

Factors affecting inferences on natural mortality and associated environmental effects in state-space age-structured assessment models: Supplementary Materials

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Referenced Figures



Fig. S1. Example simulations of environmental covariate latent processes and observations with different levels of observation error, and different assumptions about variability of the latent process.

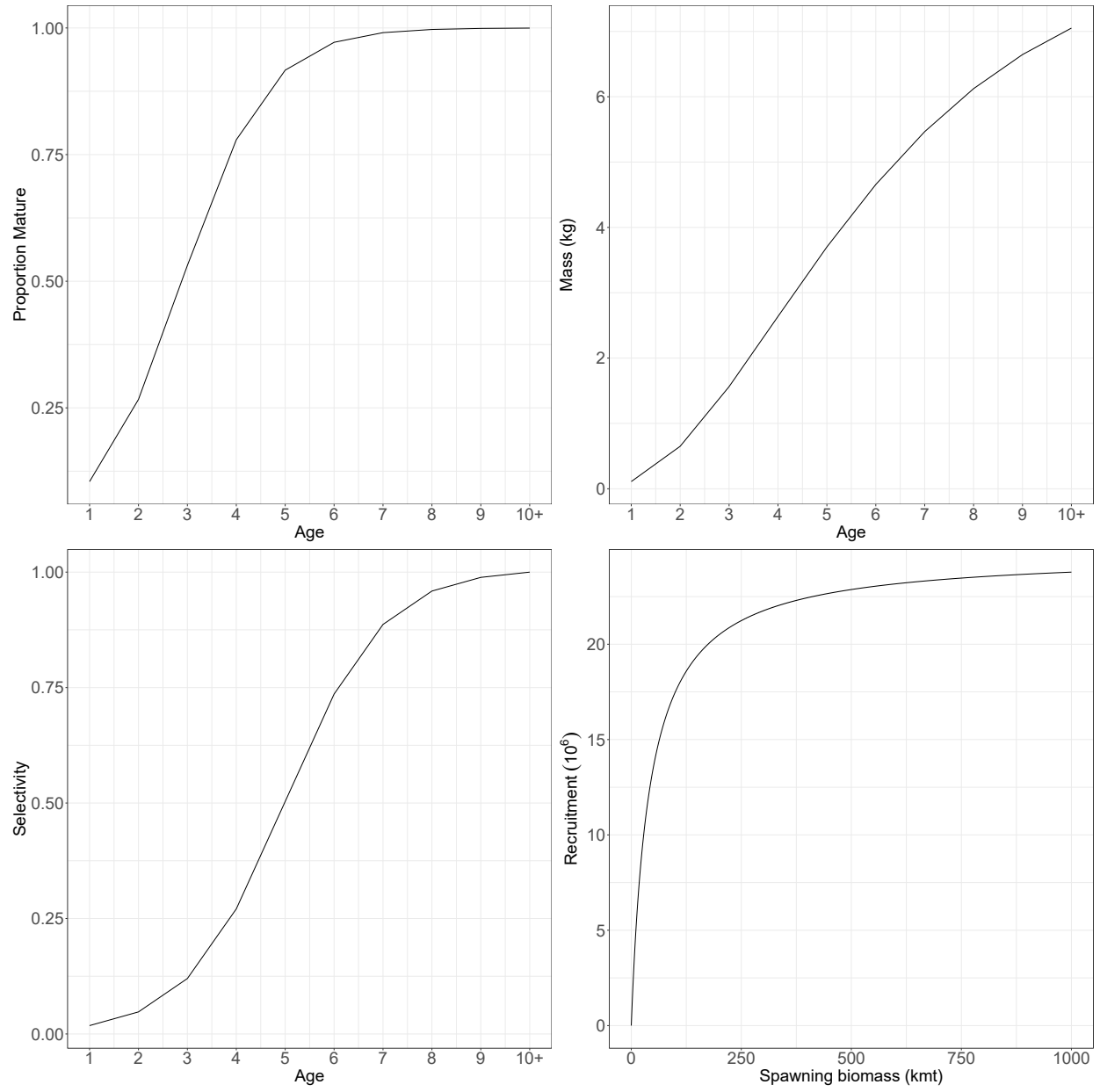


Fig. S2. The proportion mature at age, weight at age, fleet and index selectivity at age, and Beverton-Holt stock-recruit relationship assumed for the population in all operating models. For operating models with random effects on fleet selectivity, this represents the selectivity at the mean of the random effects.

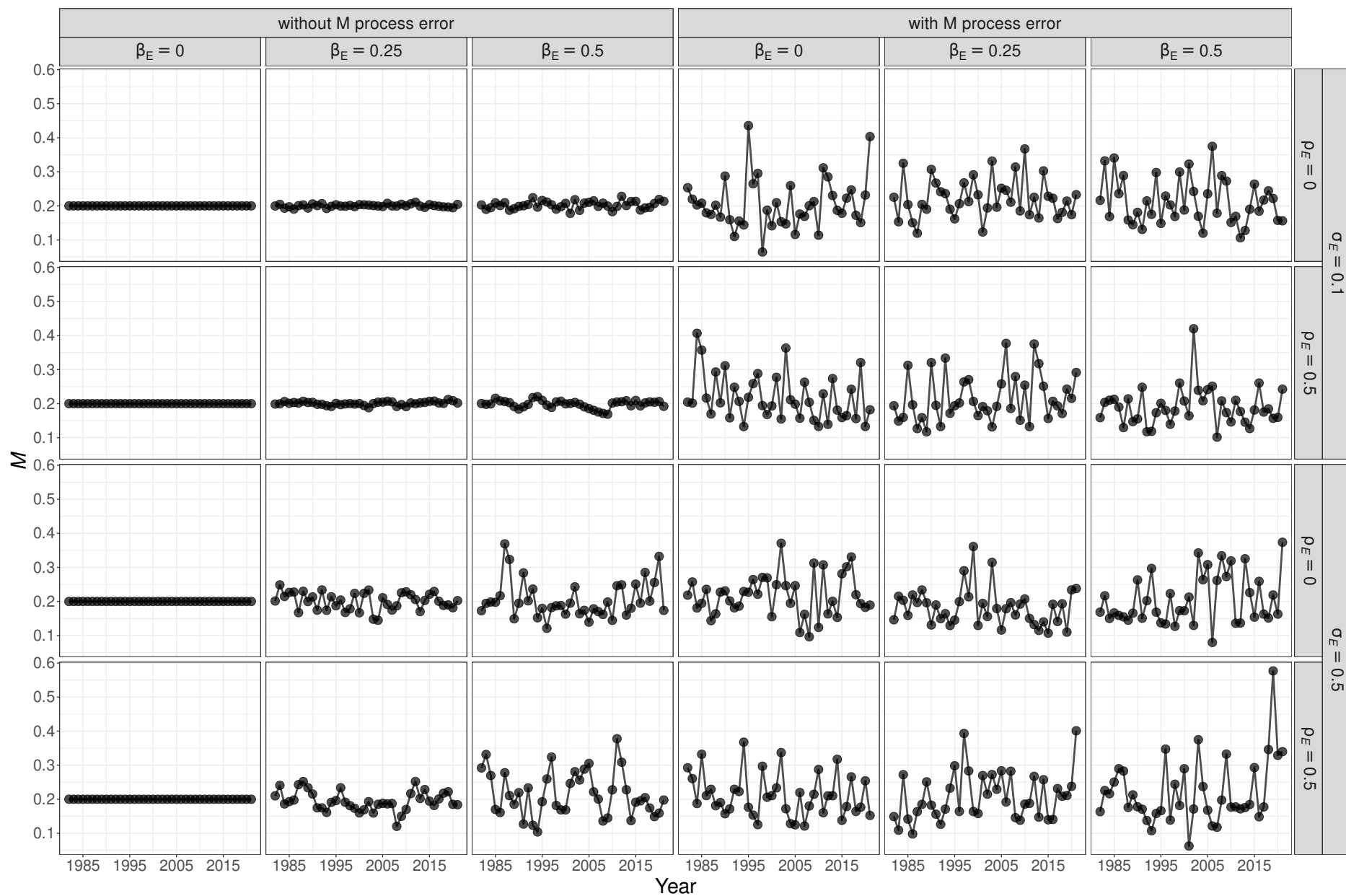


Fig. S3. Example simulations of annual M that may be a function of a temporally varying environmental covariate and autoregressive random effects. The marginal standard deviation of random effects is $\sigma_M = 0.3$.

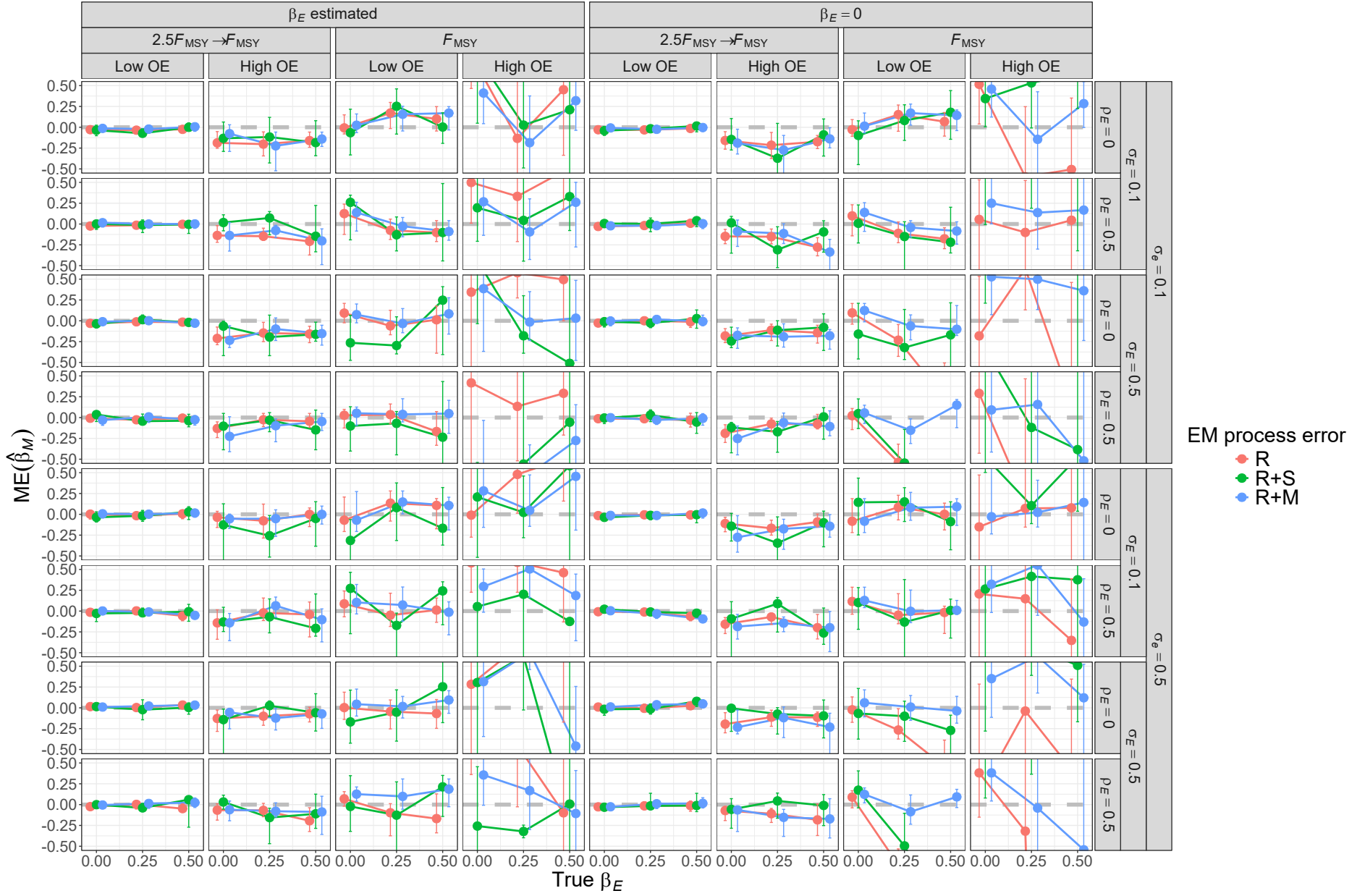


Fig. S4. For R operating models, median error (ME) of estimates of β_M from fitting estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

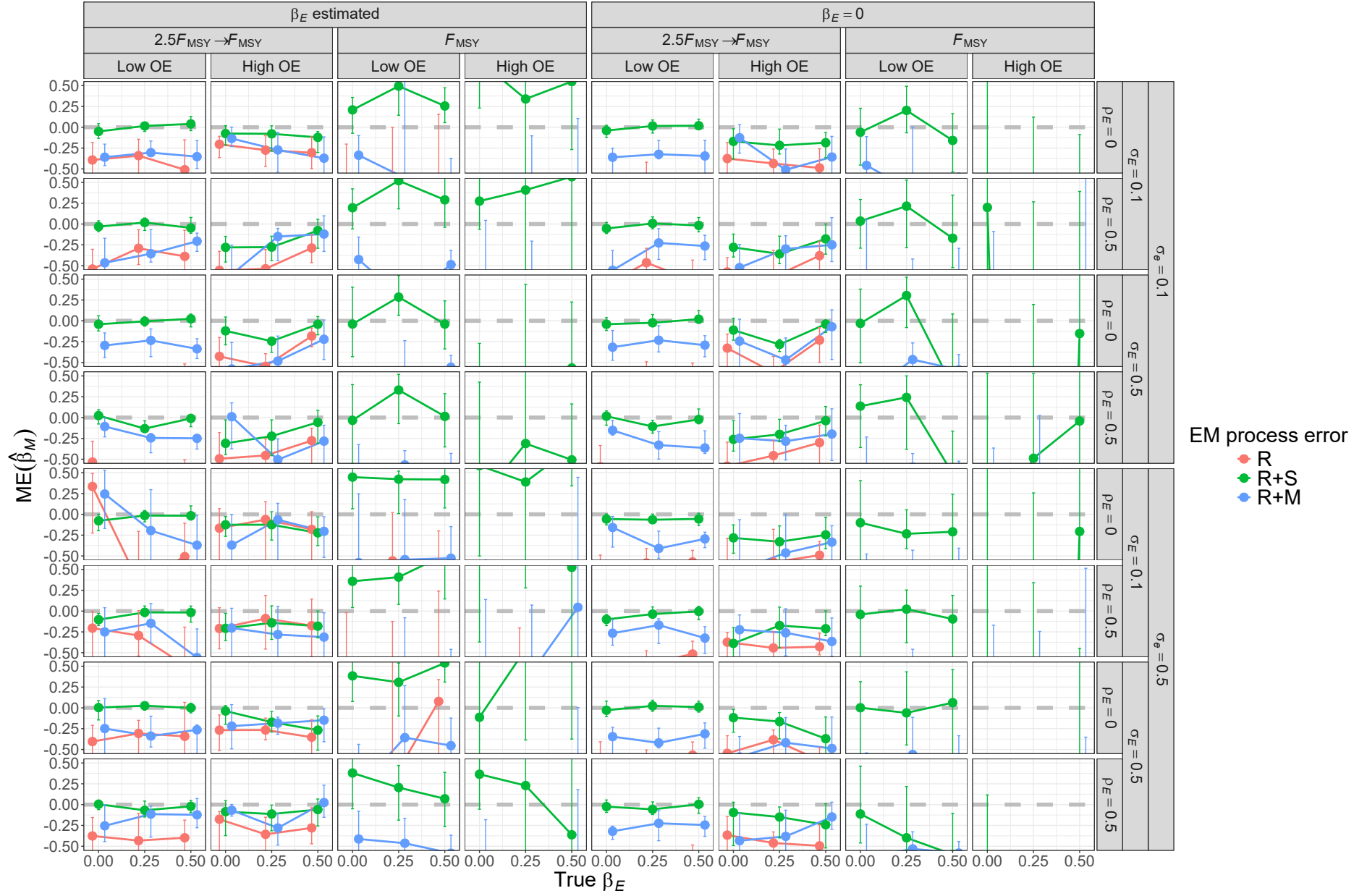


Fig. S5. For R+S operating models, median error (ME) of estimates of β_M from fitting estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

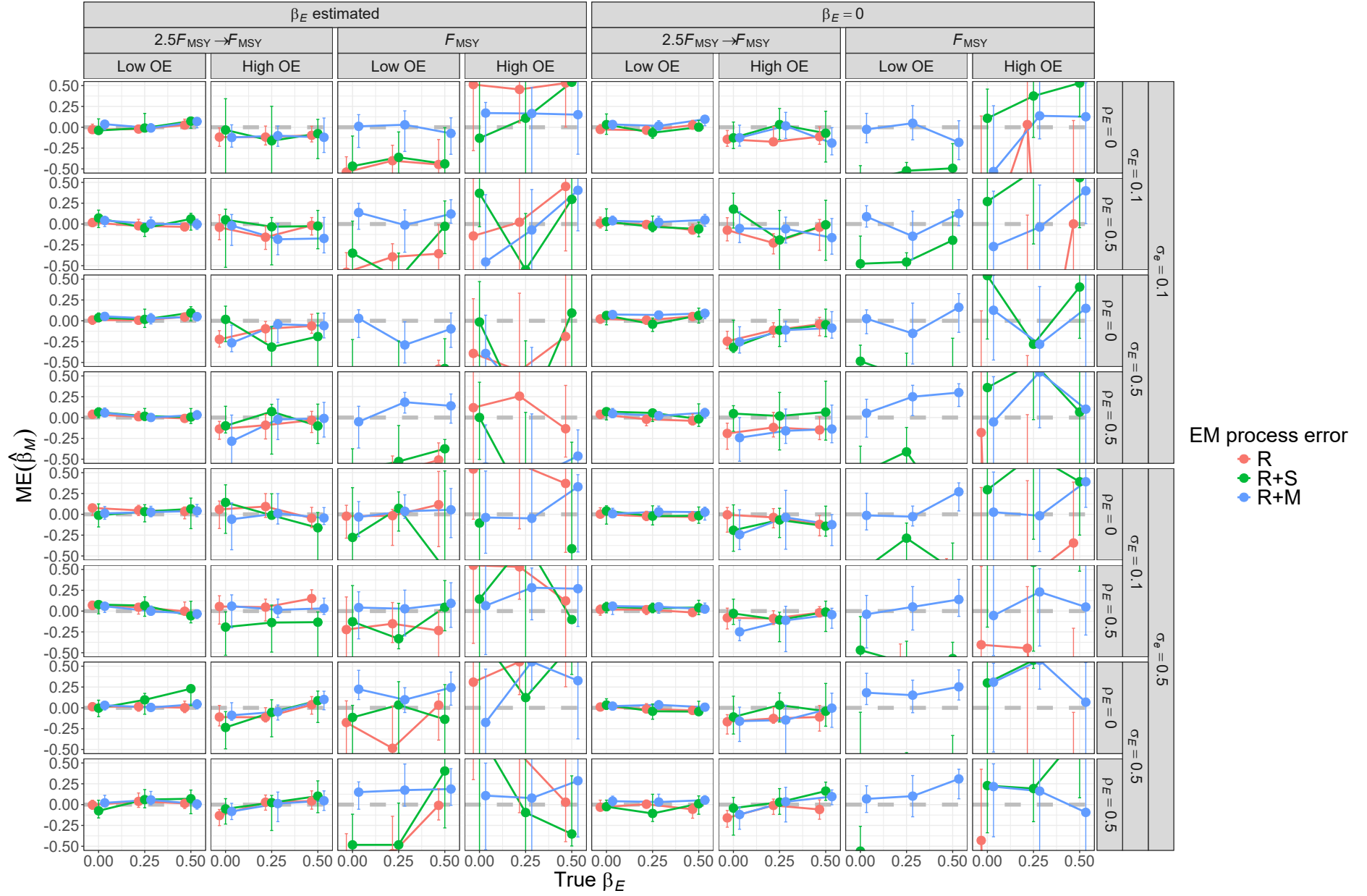


Fig. S6. For R+M operating models, median error (ME) of estimates of β_M from fitting estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

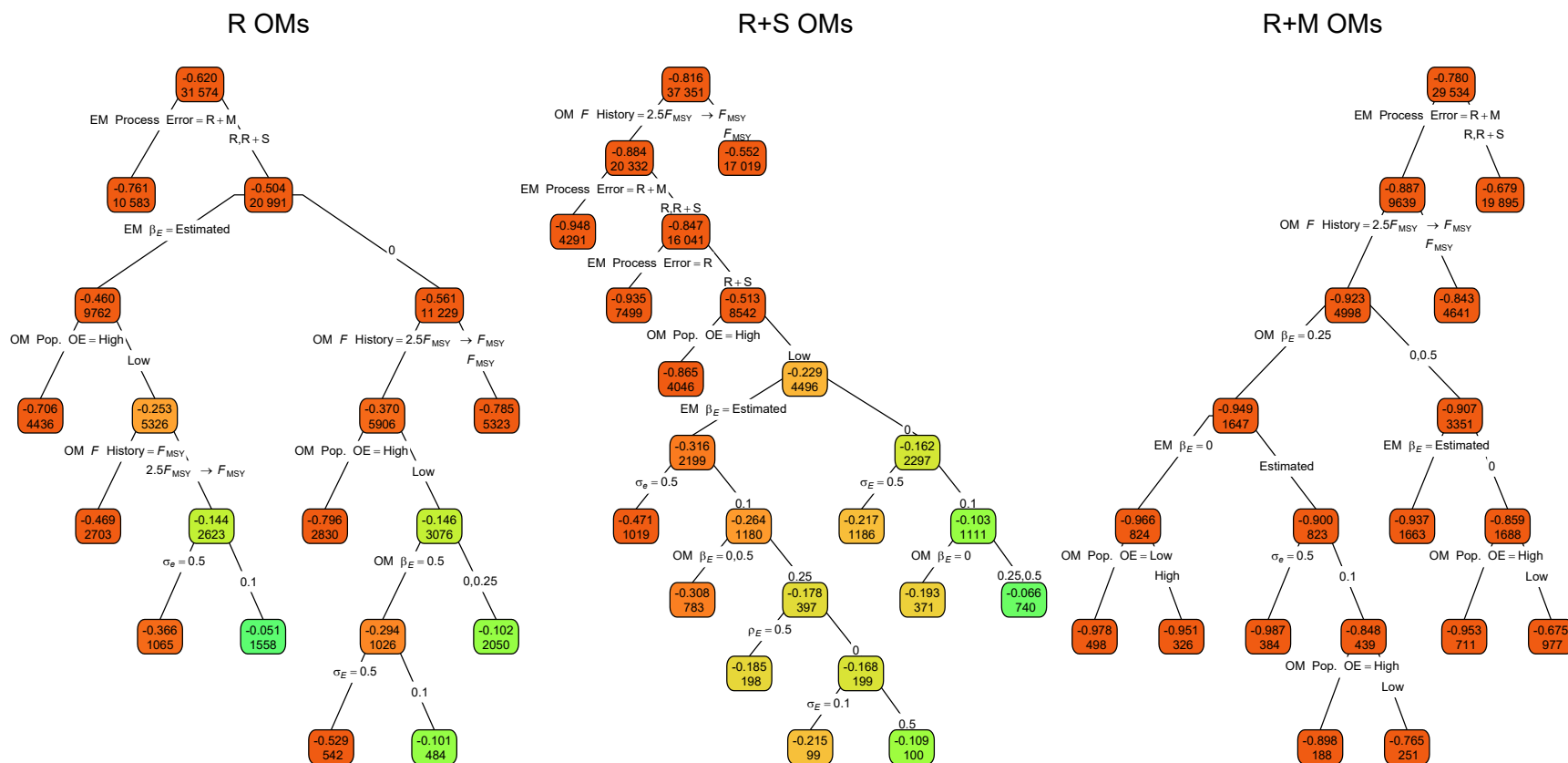


Fig. S7. Regression trees for errors of the Hessian-based standard error estimates for median M parameter ($\widehat{\text{SE}}(\hat{\beta}_M)$). Each node shows the median error for the corresponding subset. Lower or higher absolute median absolute errors of the process error assumption are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

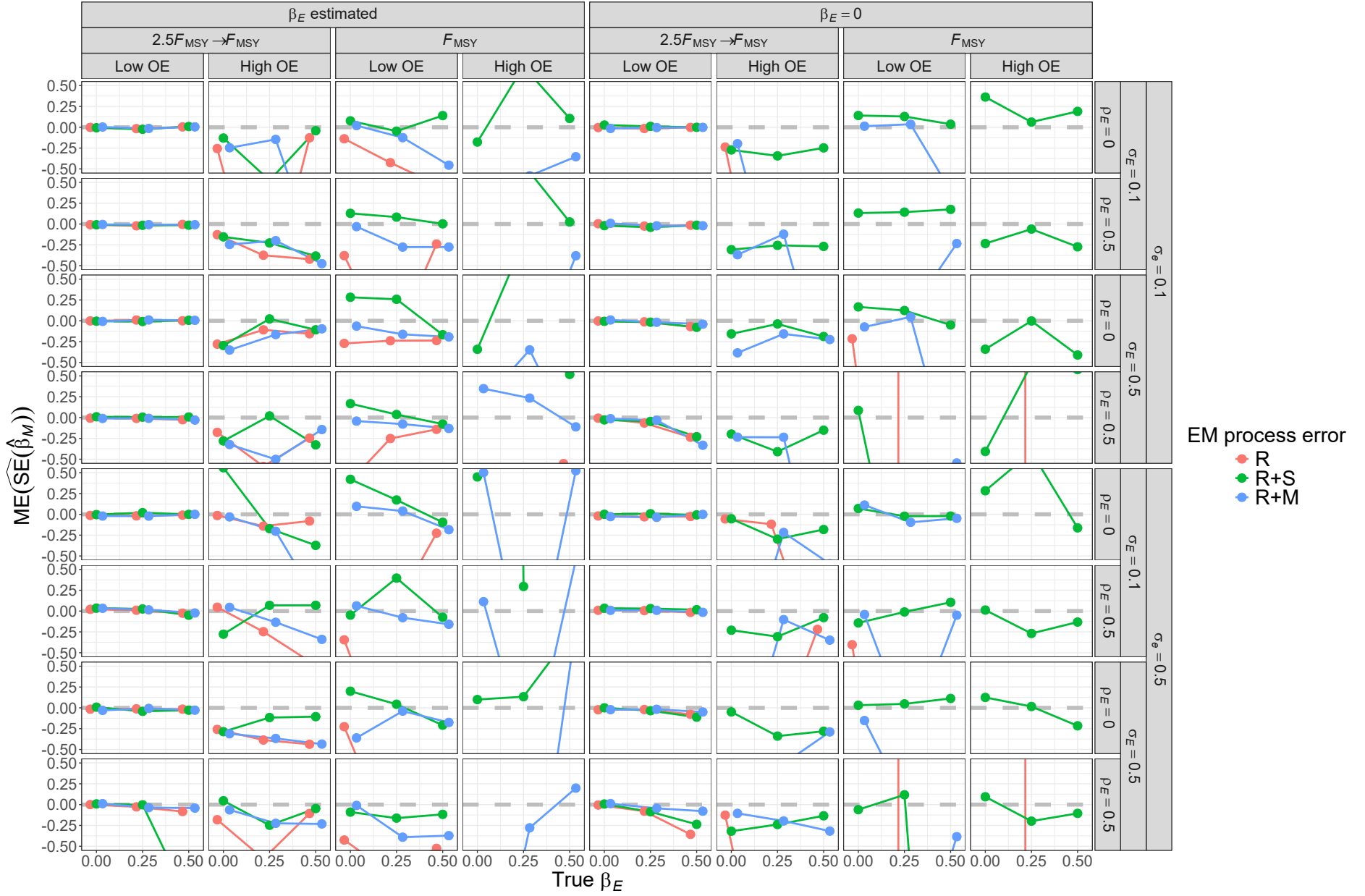


Fig. S8. For R operating models, median error (ME) of Hessian-based estimates of standard error for median M parameter ($\hat{\beta}_M$) from fitting estimating models with alternative process error assumptions and treatment of the covariate effect ($\beta_E = 0$ or estimated). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given operating model scenario.

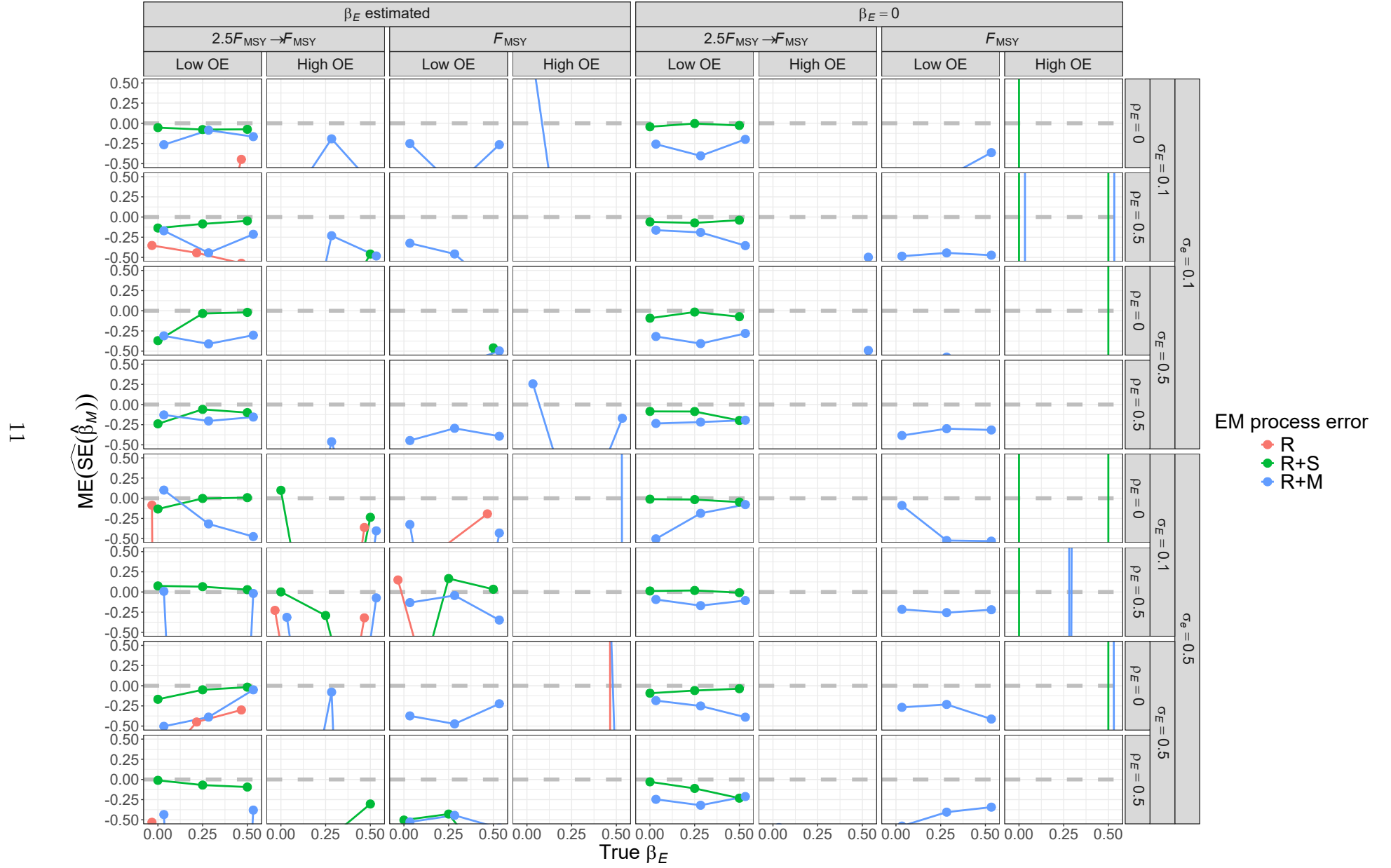


Fig. S9. For R+S operating models, median error (ME) of Hessian-based estimates of standard error for median M parameter ($\hat{\beta}_M$) from fitting estimating models with alternative process error assumptions and treatment of the covariate effect ($\beta_E = 0$ or estimated). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given operating model scenario.

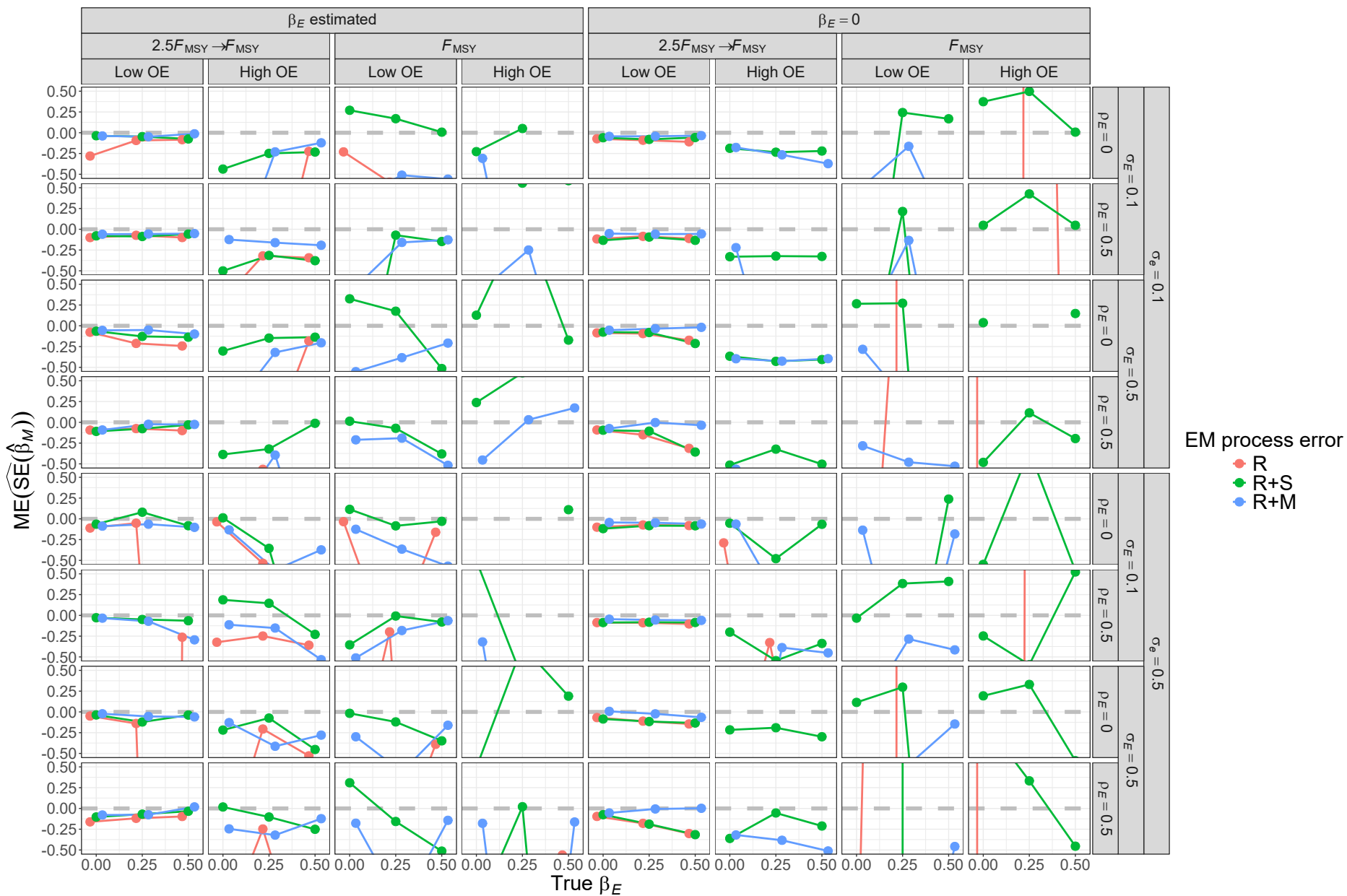


Fig. S10. For R+M operating models, median error (ME) of Hessian-based estimates of standard error for median M parameter ($\hat{\beta}_M$) from fitting estimating models with alternative process error assumptions and treatment of the covariate effect ($\beta_E = 0$ or estimated). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given operating model scenario.

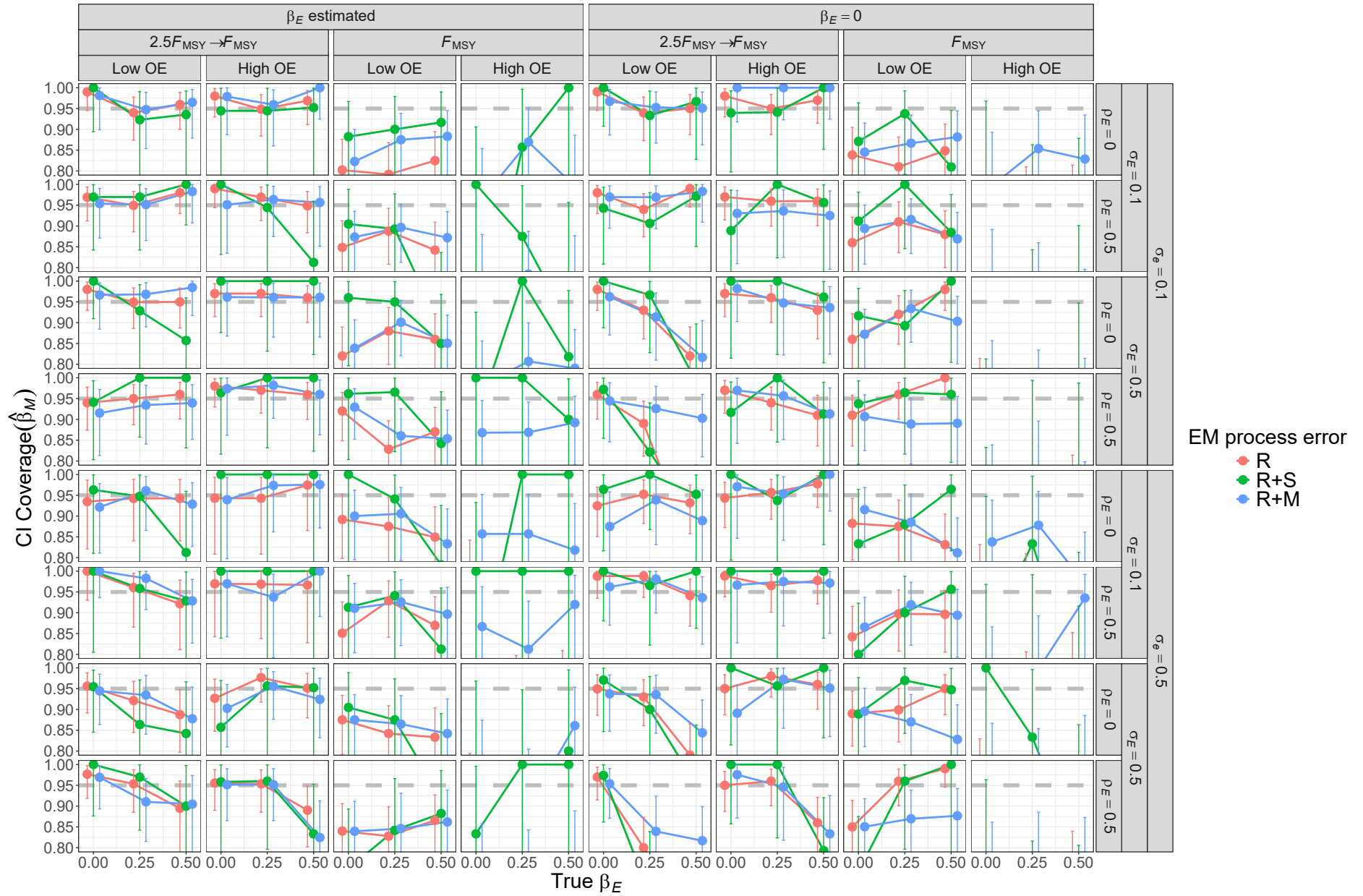


Fig. S11. For R operating models, probability of 95% confidence interval for β_M containing the true value for estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

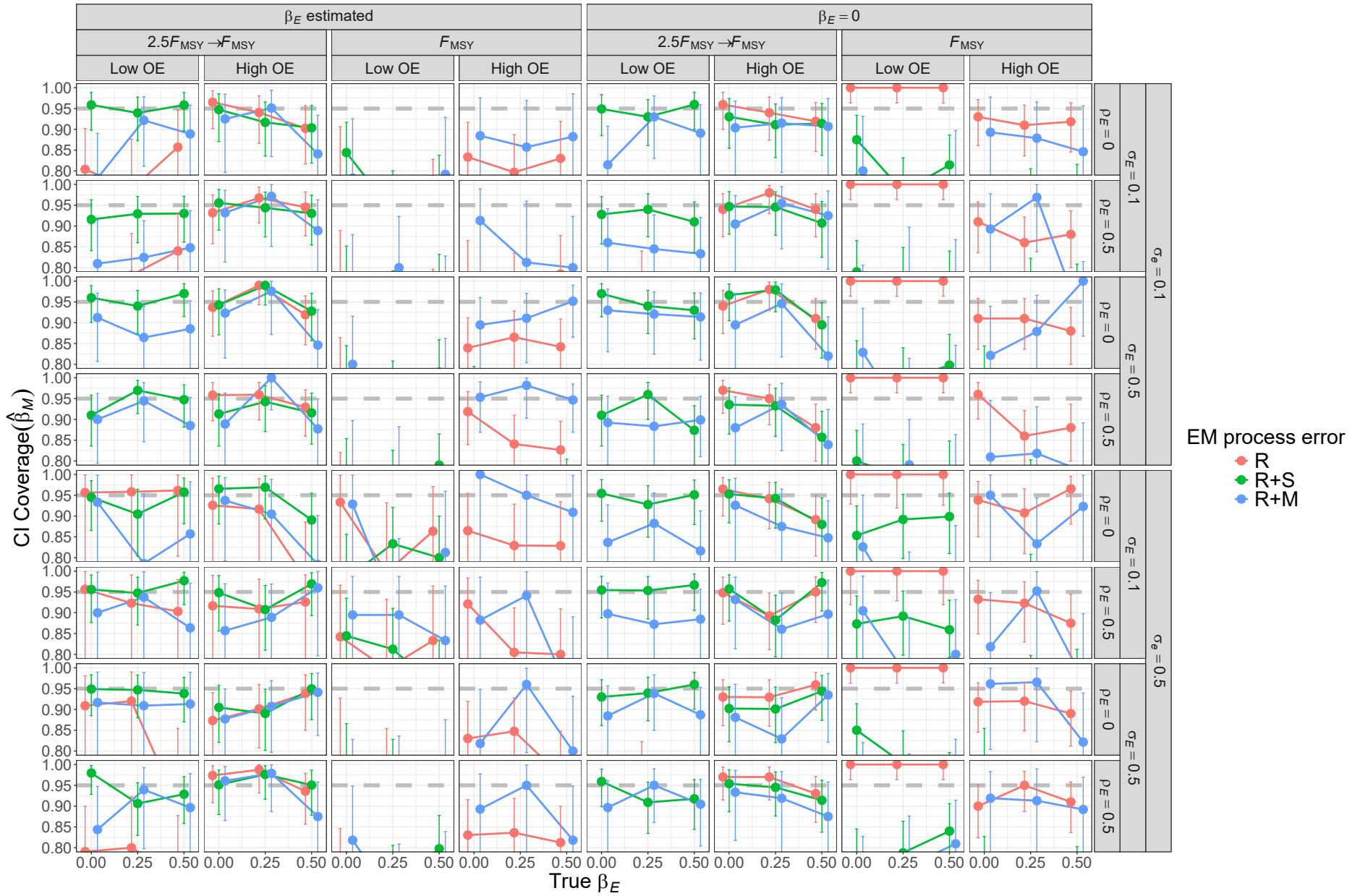


Fig. S12. For R+S operating models, probability of 95% confidence interval for β_M containing the true value for estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

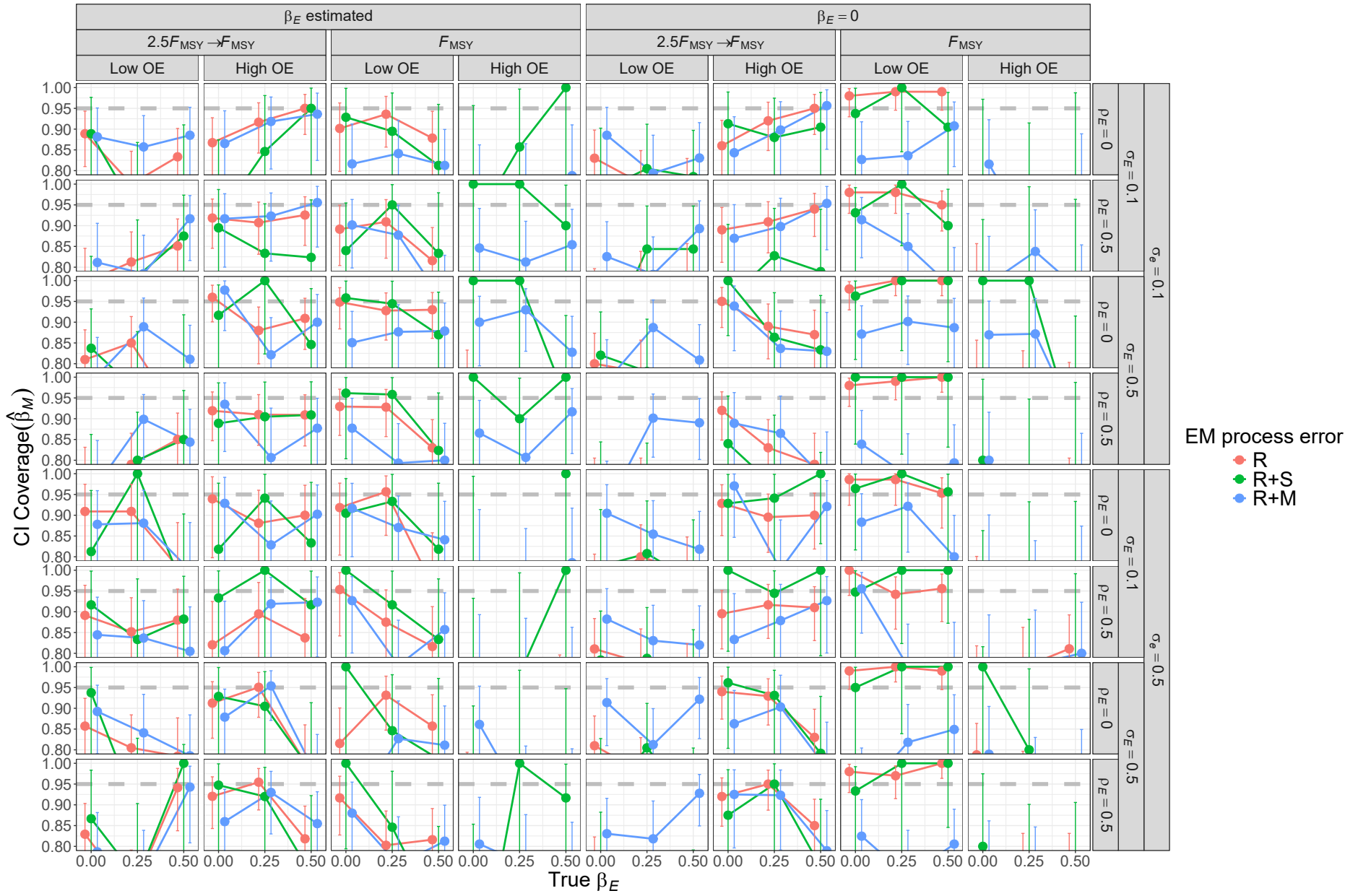


Fig. S13. For R+M operating models, probability of 95% confidence interval for β_M containing the true value for estimating models with alternative process error assumptions and treatment of covariate effect ($\beta_E = 0$ or estimated). Vertical lines represent 95% confidence intervals.

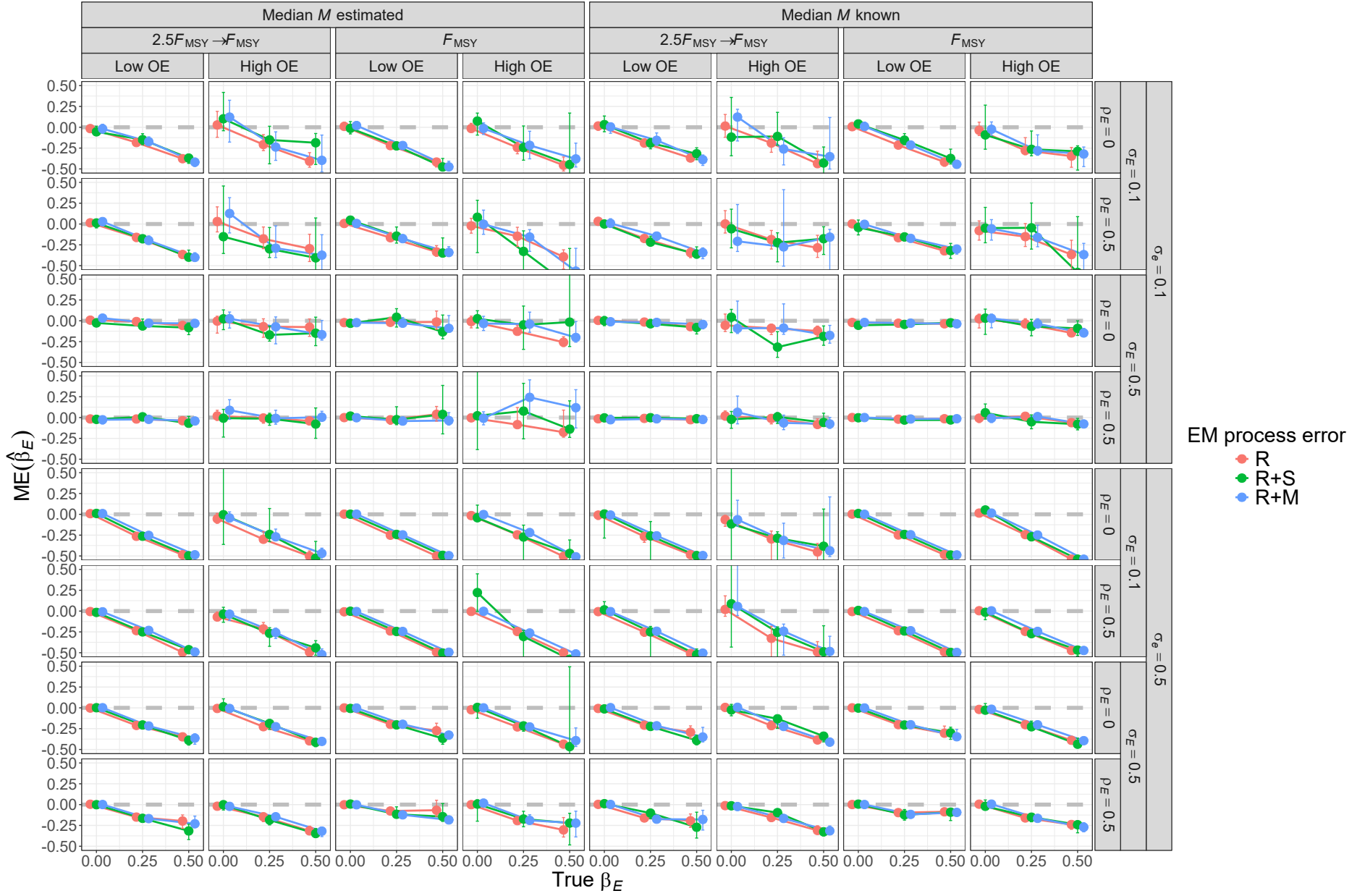


Fig. S14. For R operating models, median error (ME) of estimates of environmental effect on M (β_E) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

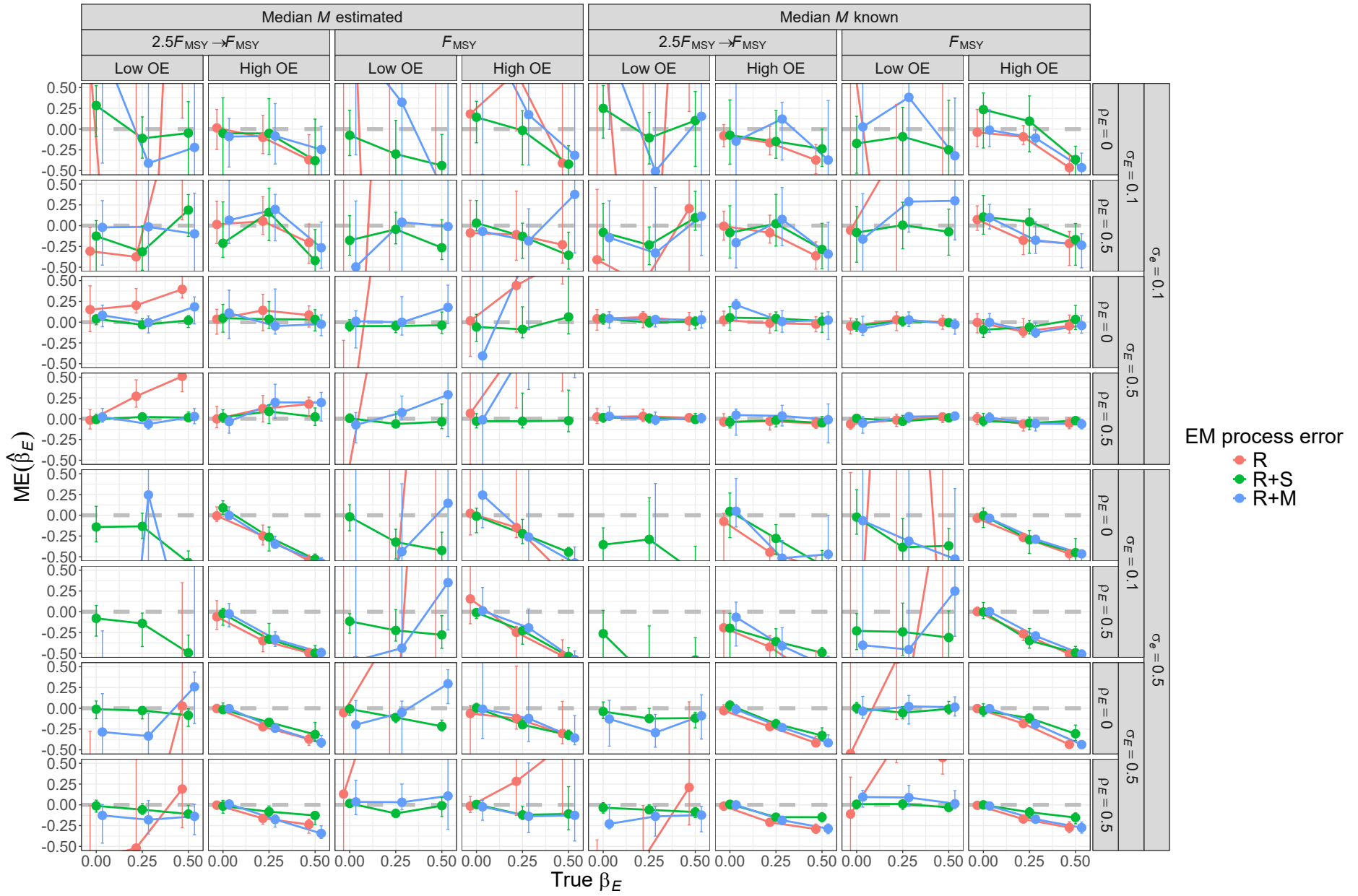


Fig. S15. For R+S operating models, median error (ME) of estimates of environmental effect on M (β_E) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

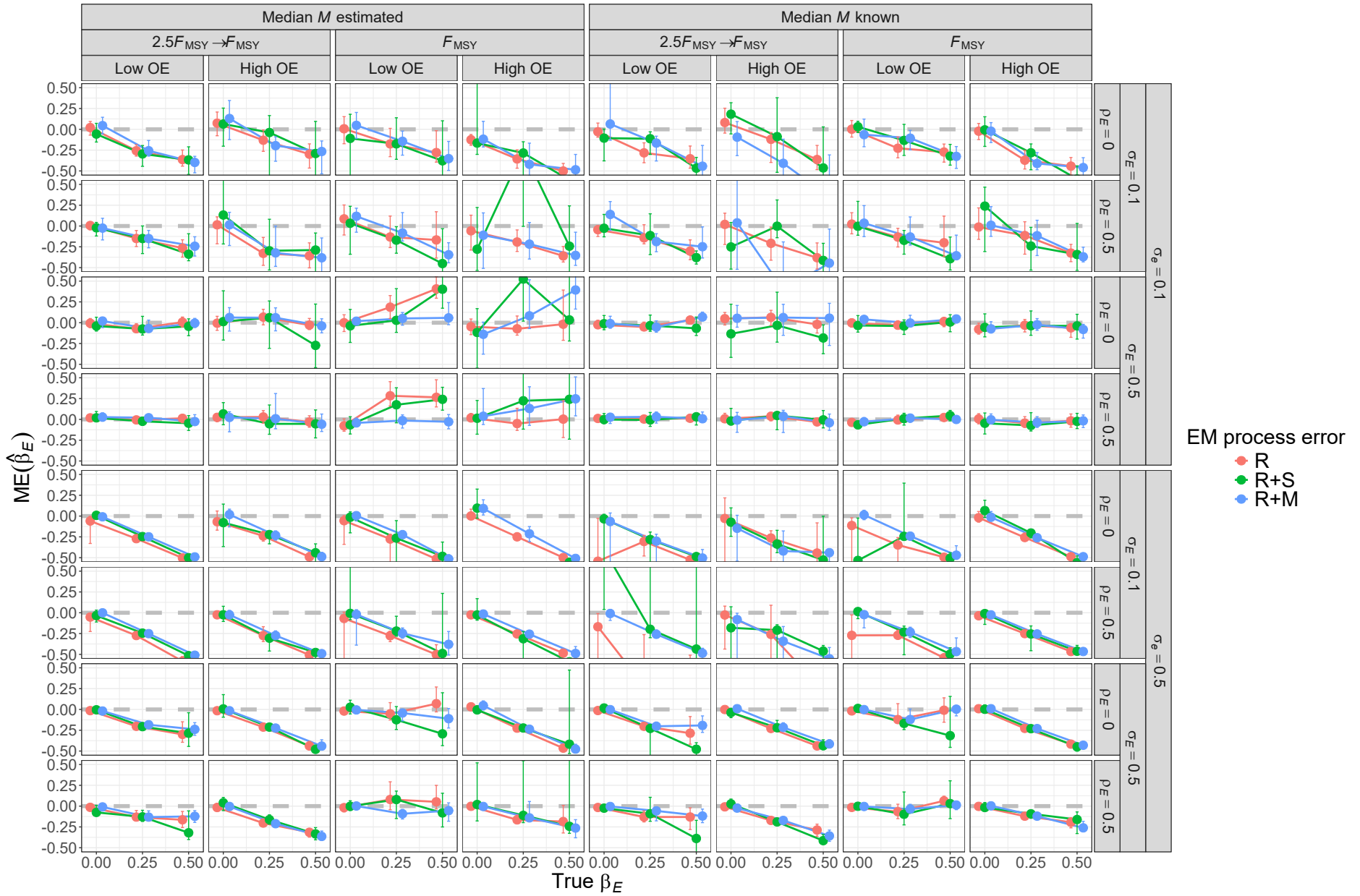


Fig. S16. For R+M operating models, median error (ME) of estimates of environmental effect on M (β_E) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

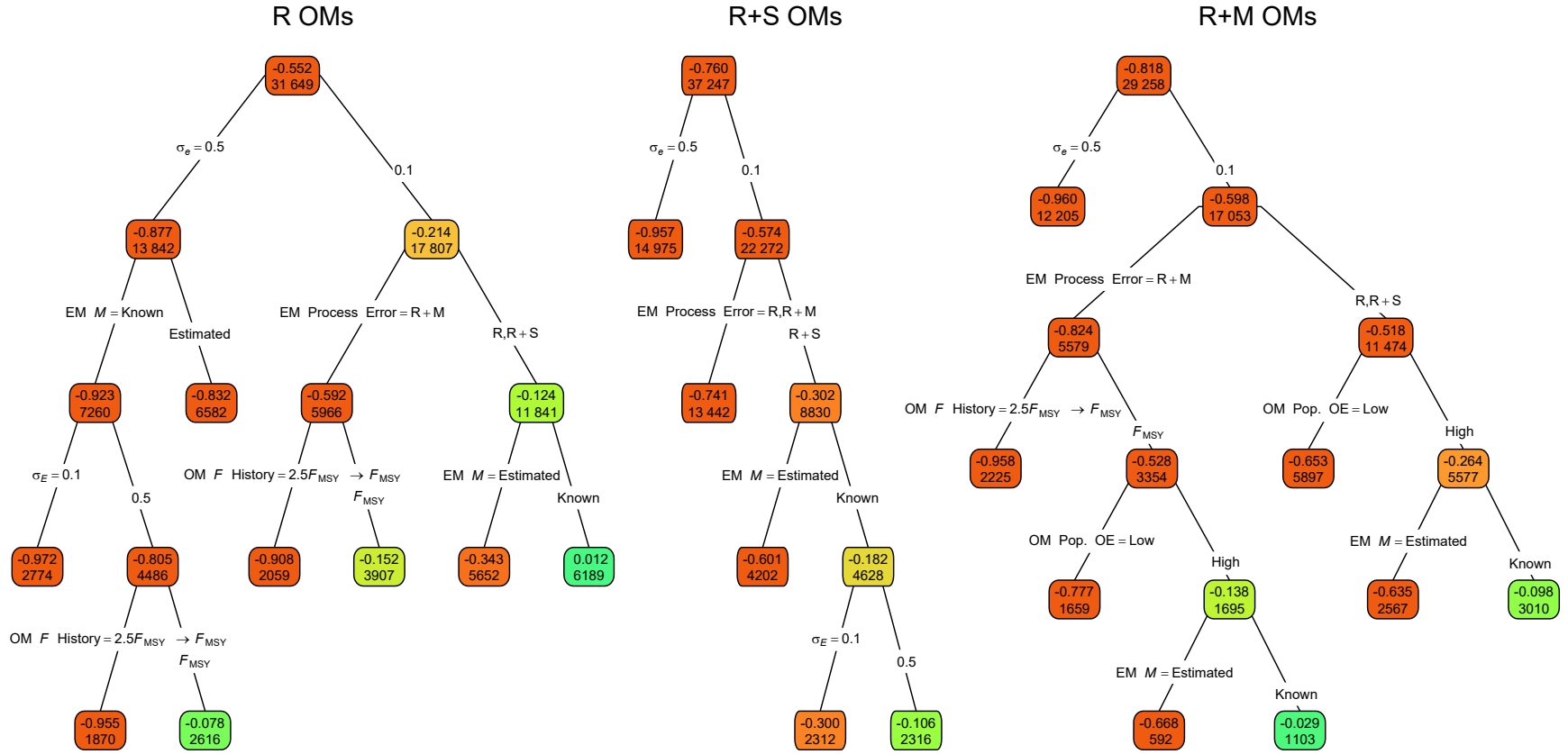


Fig. S17. Regression trees for error of the Hessian-based standard error estimates for covariate effect on M ($\widehat{SE}(\widehat{\beta}_E)$). Each node shows the median error for the corresponding subset. Lower or higher absolute median errors are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

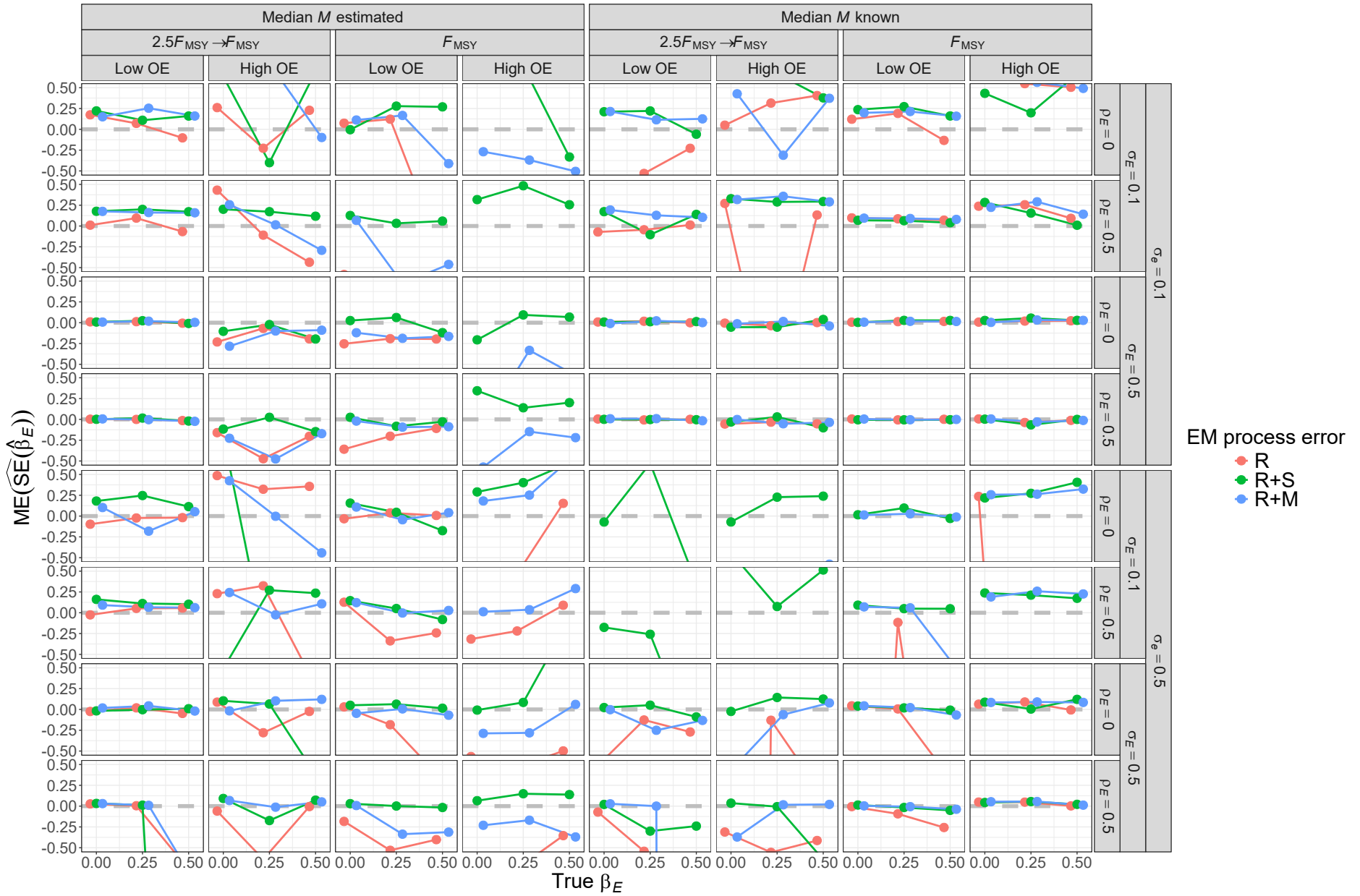


Fig. S18. For R operating models, median error (ME) of Hessian-based estimates of standard error for covariate effect on M ($\hat{\beta}_E$) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given operating model scenario.

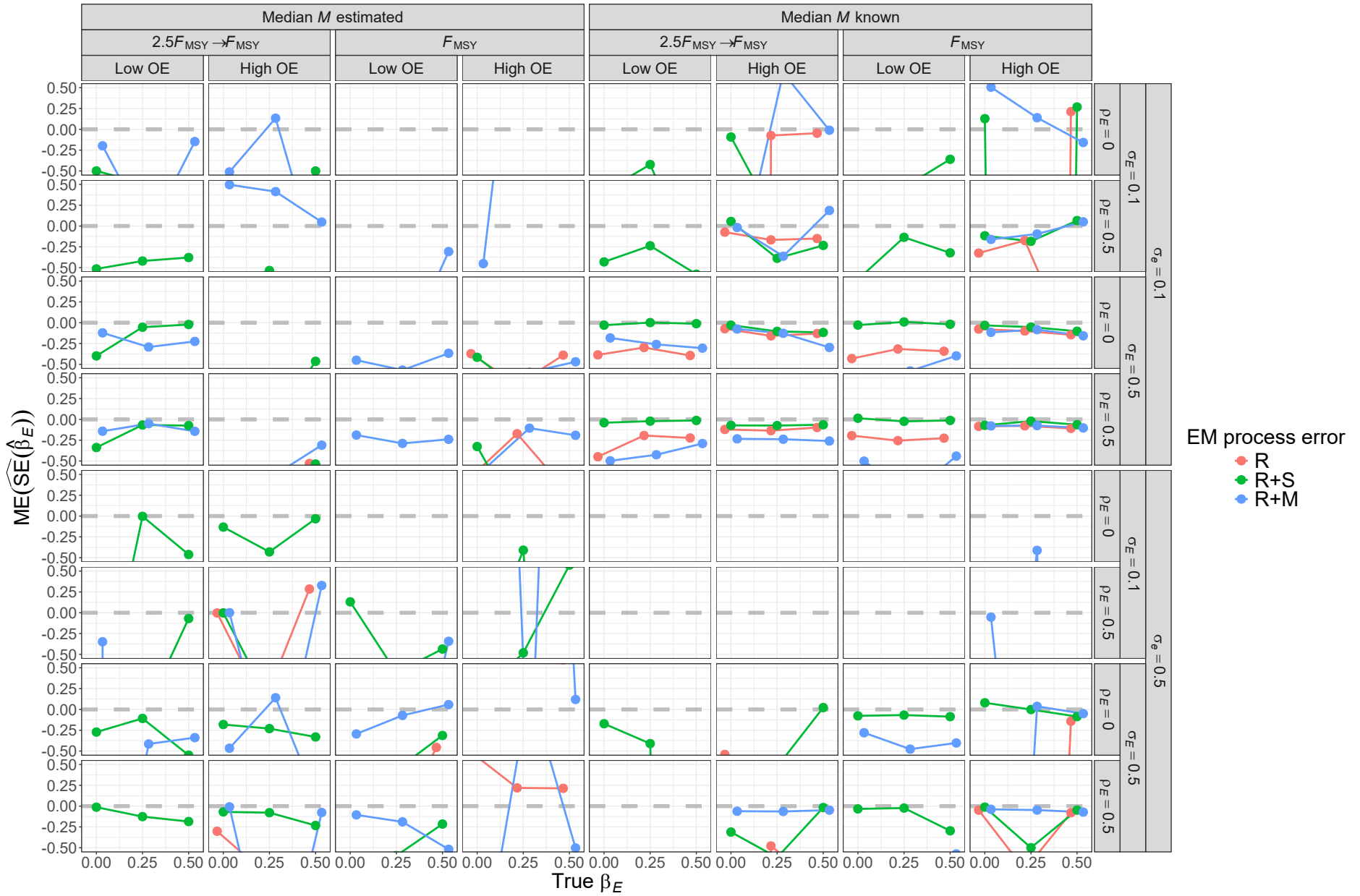


Fig. S19. For R+S operating models, median error (ME) of Hessian-based estimates of standard error for covariate effect on M (β_E) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given operating model scenario.

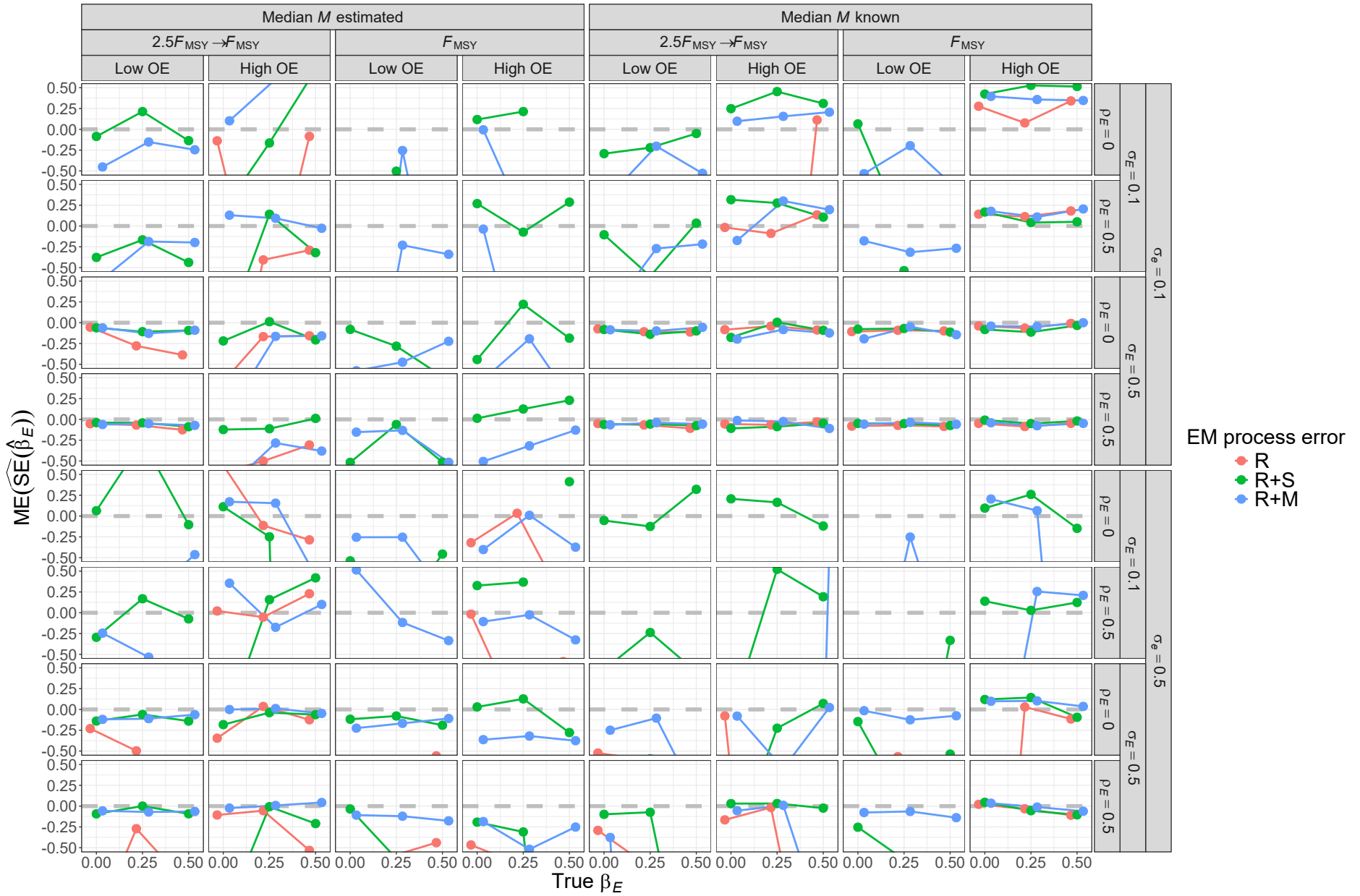


Fig. S20. For R+M operating models, median error (ME) of Hessian-based estimates of standard error for covariate effect on M (β_E) from fitting estimating models with alternative process error assumptions and treatment of median M parameter (β_M). True standard error is defined as the mean of the standard error estimates across converged fits to simulated data sets for a given operating model scenario.

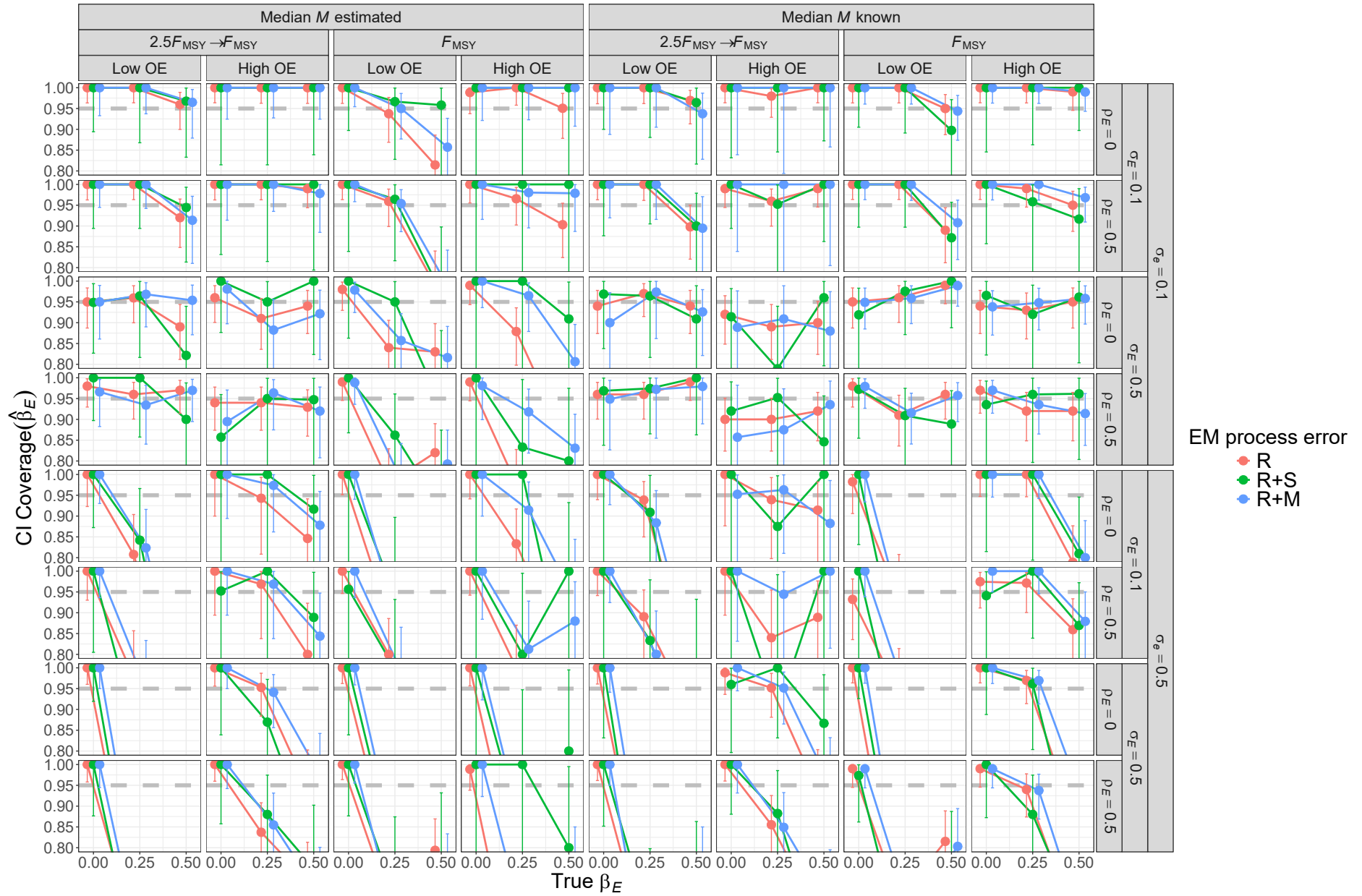


Fig. S21. For R operating models, probability of 95% confidence interval for β_E containing the true value for estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

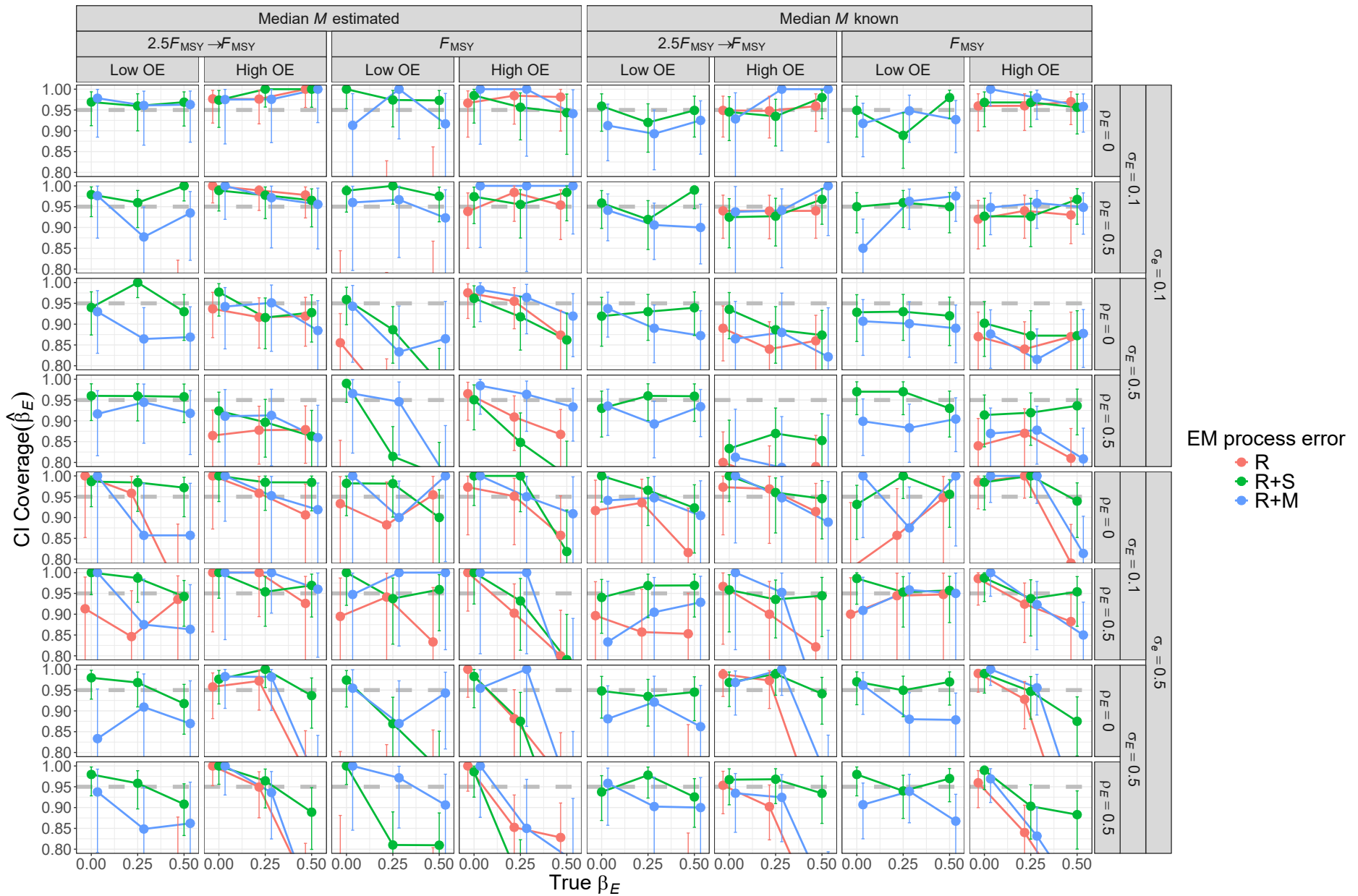


Fig. S22. For R+S operating models, probability of 95% confidence interval for β_E containing the true value for estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

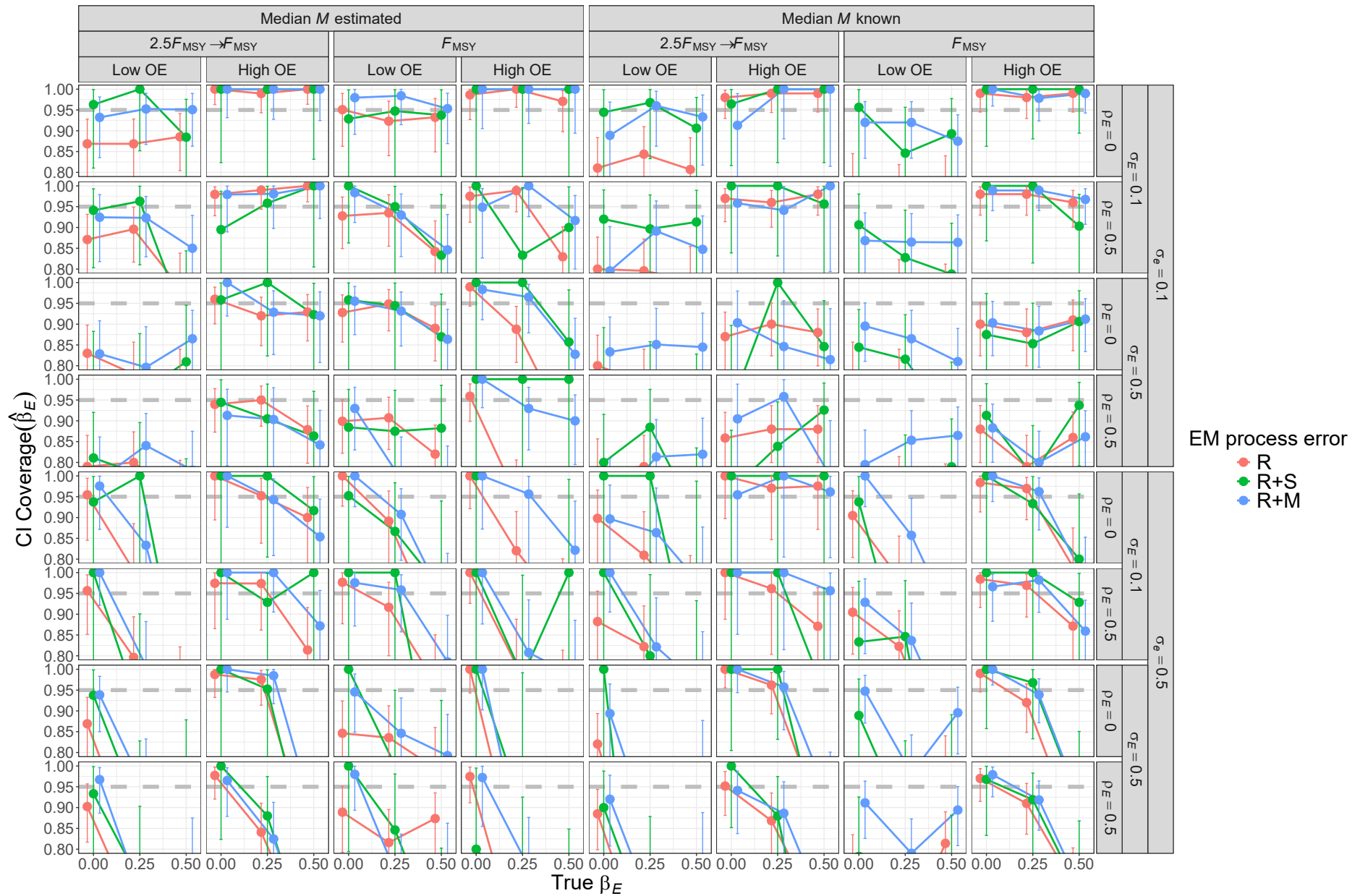


Fig. S23. For R+M operating models, probability of 95% confidence interval for β_E containing the true value for estimating models with alternative process error assumptions and treatment of median M parameter (β_M). Vertical lines represent 95% confidence intervals.

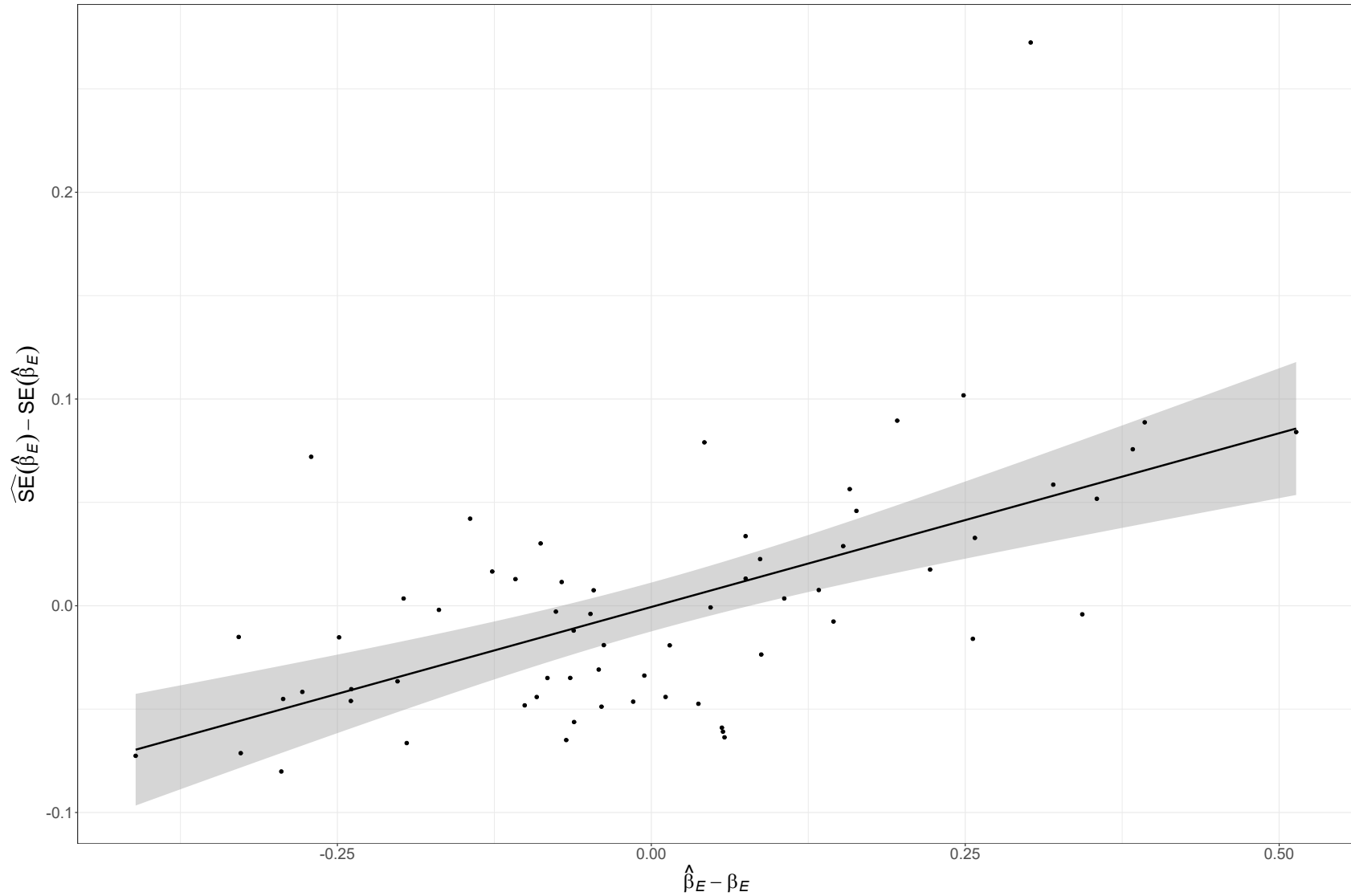


Fig. S24. Positive correlation of covariate effect estimates and Hessian-based standard error estimates for estimating model that also estimates the median M parameter (β_M) and has correct R+M process error assumption fitted to simulated data from the operating model with R+M process errors, temporal contrast in fishing pressure, low observation uncertainty for both population (*LowOE*) and covariate observations ($\sigma_e = 0.1$), high and uncorrelated temporal variability in the true covariate ($\sigma_E = 0.5$ and $\rho_E = 0$), and the strongest covariate effect on M ($\beta_E = 0.5$).

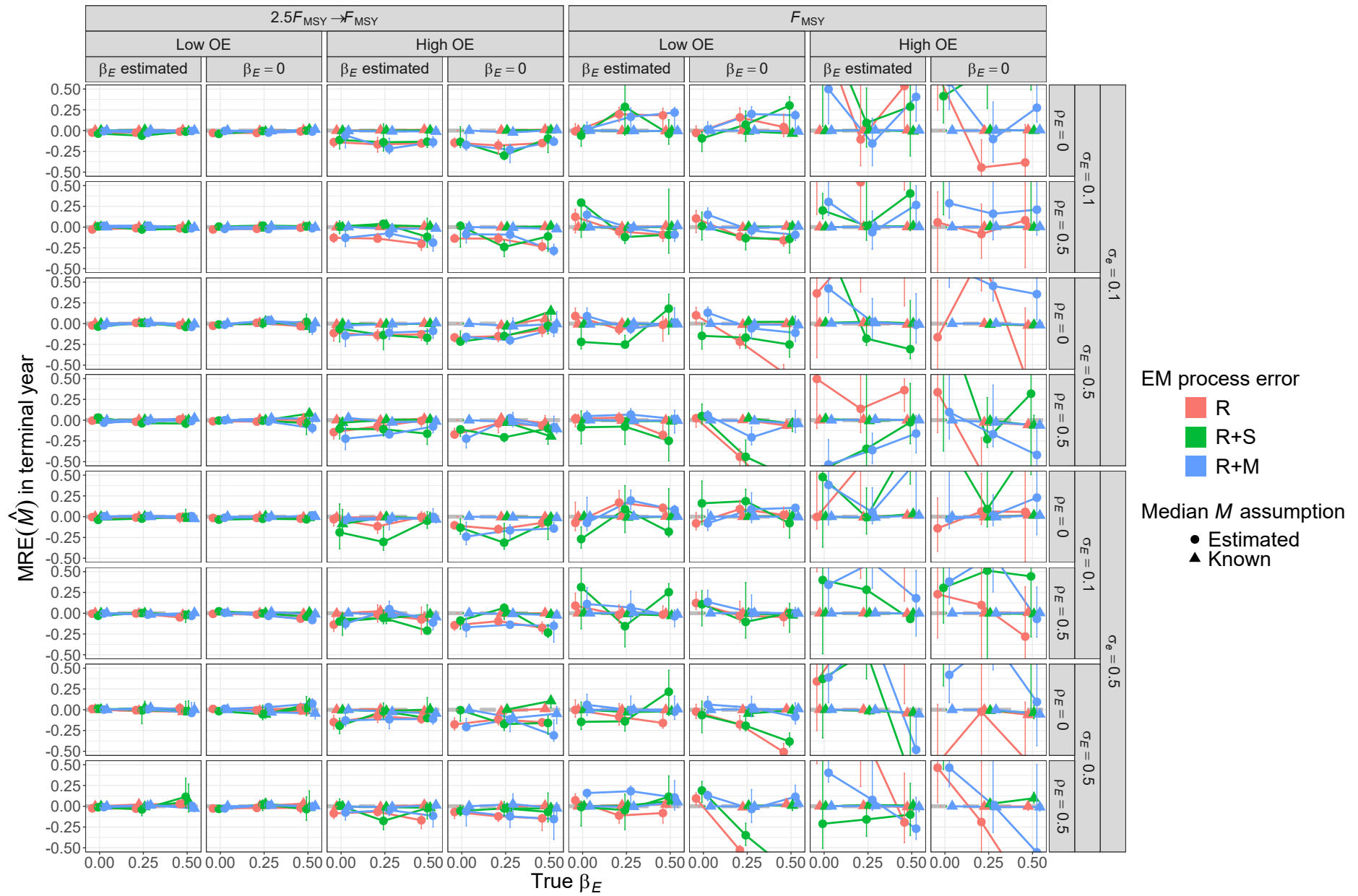


Fig. S25. For R operating models, median relative error (MRE) of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

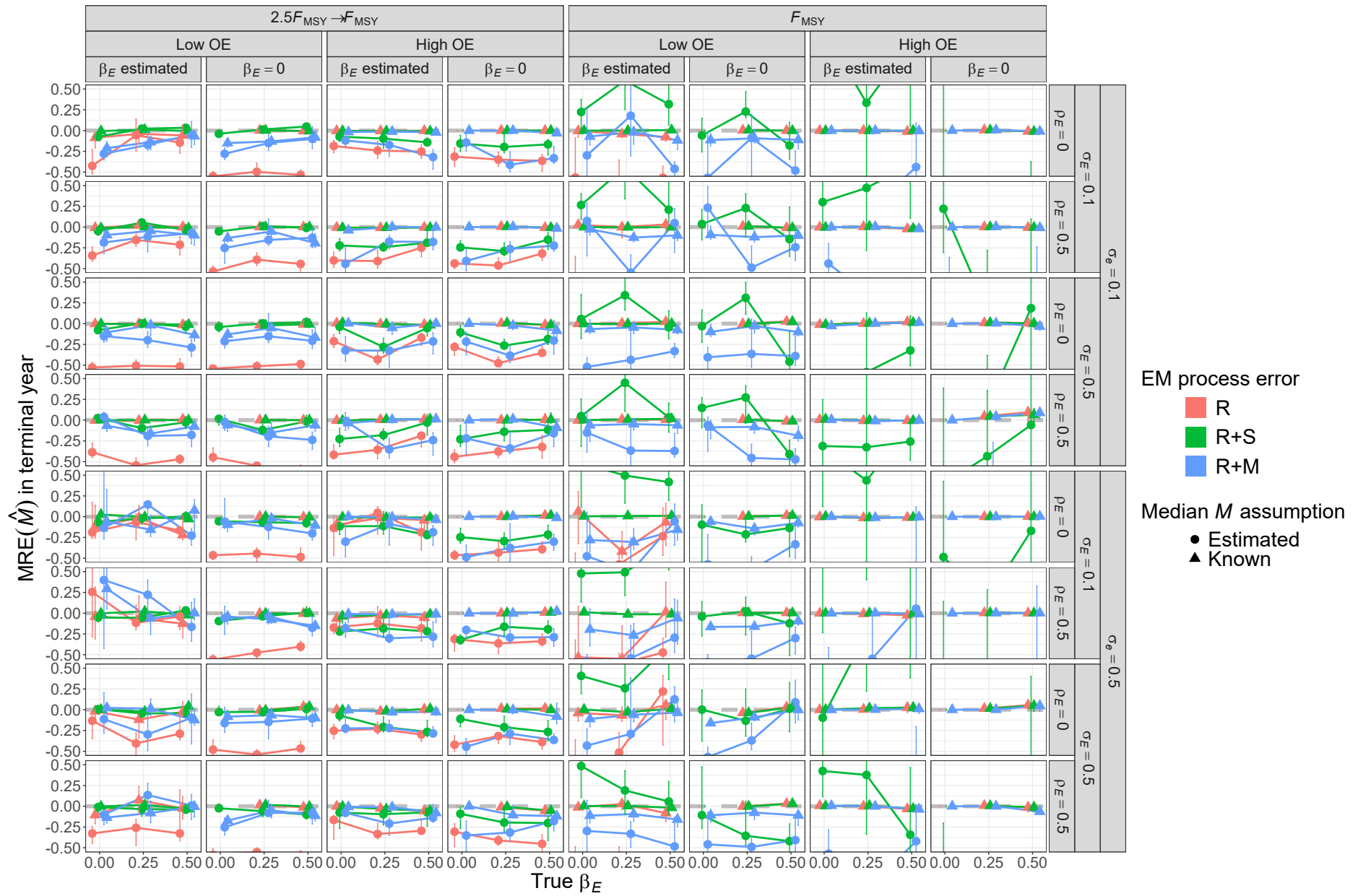


Fig. S26. For R+S operating models, median relative error (MRE) of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

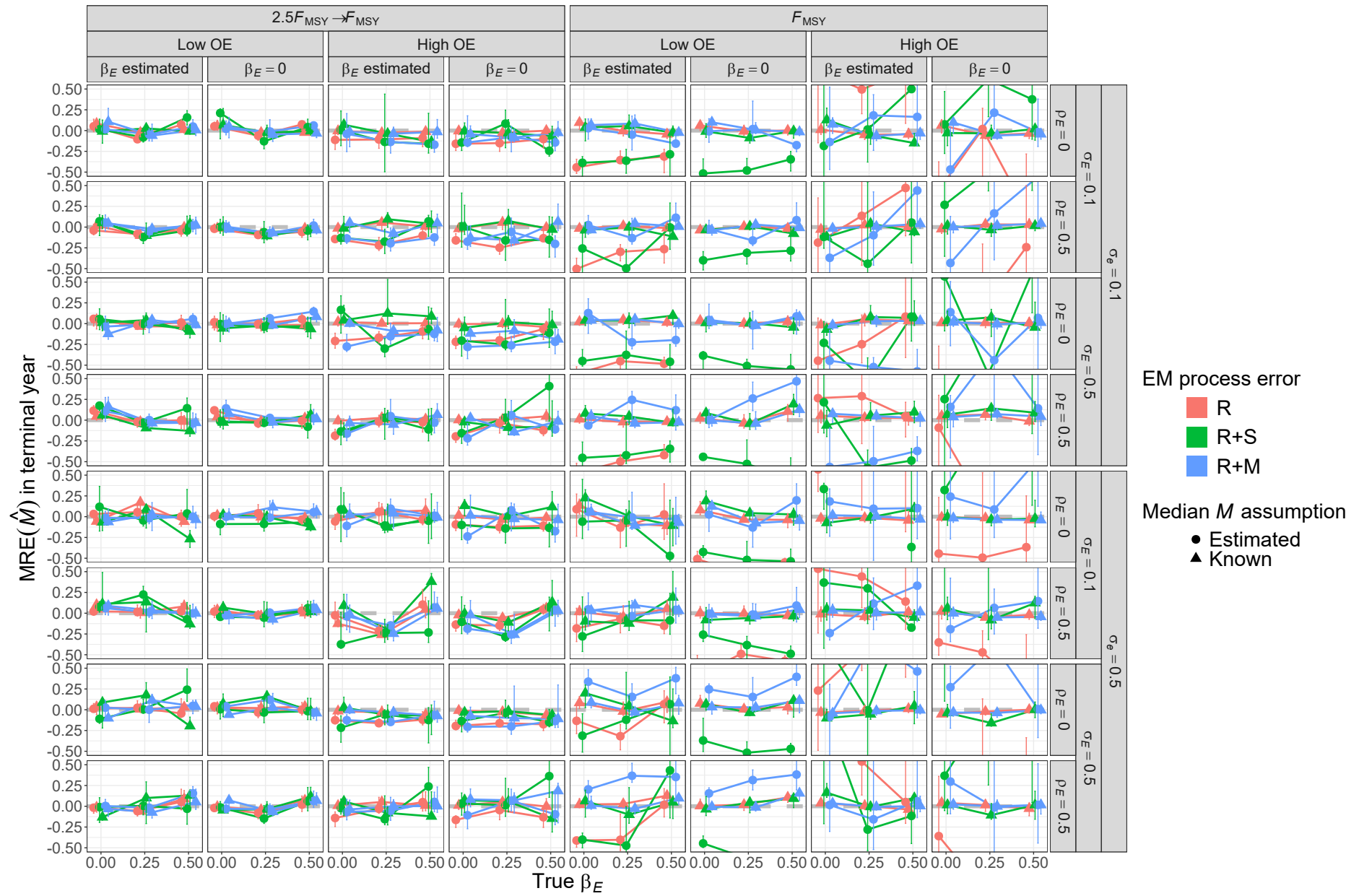


Fig. S27. For R+M operating models, median relative error (MRE) of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

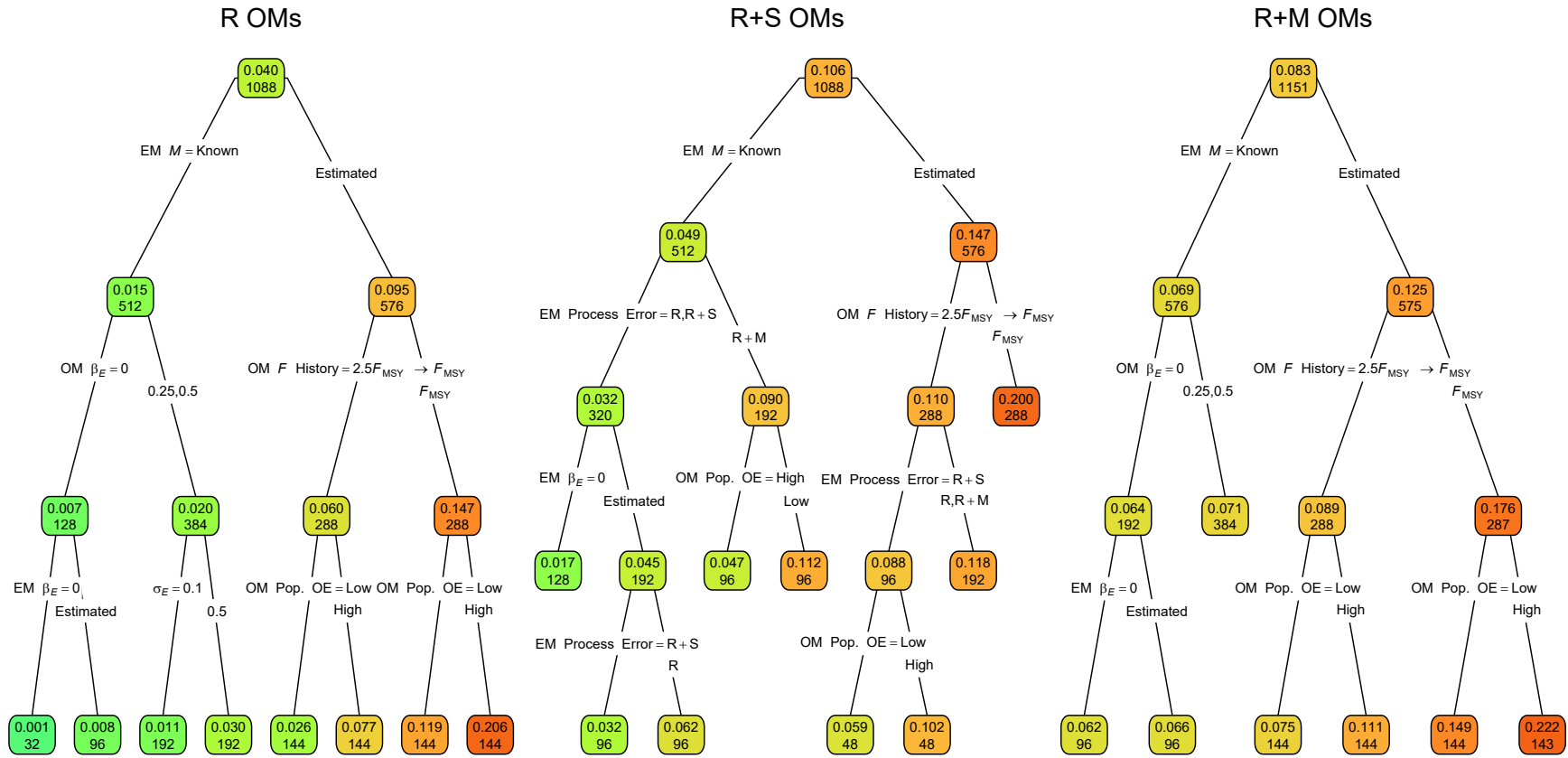


Fig. S28. Regression trees for the log-transformed RMSE of terminal year M . Each node shows the median RMSE (top) and number of observations (bottom) for the corresponding subset. Lower or higher median RMSE are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

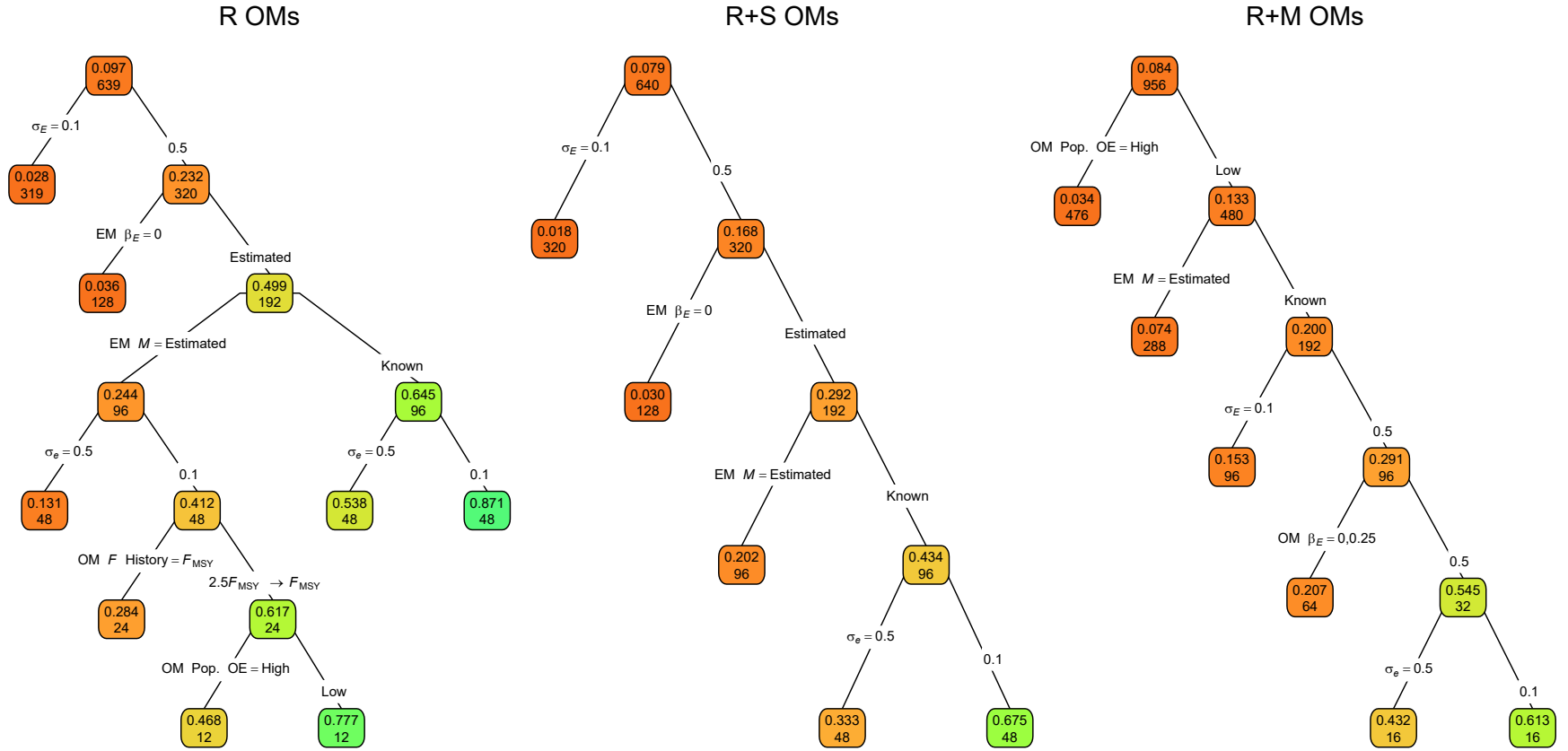


Fig. S29. Regression trees for the correlation of terminal year M estimates and true values. Each node shows the median correlation for the corresponding subset. Lower or higher median correlations are indicated by more red or green polygons, respectively. See Figure 1 caption and Table S1 for further details.

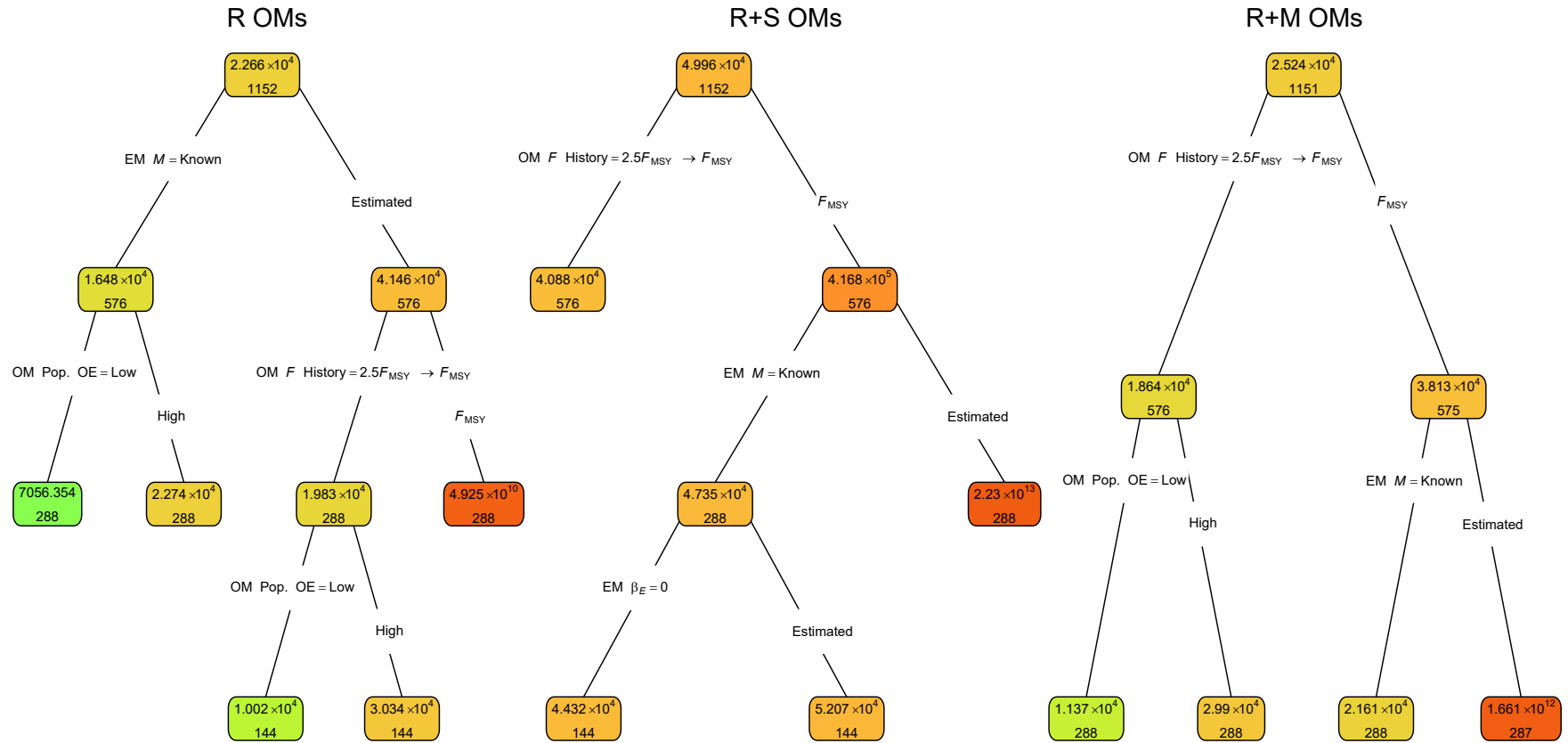


Fig. S30. Regression trees for the log-transformed RMSE of terminal year SSB. Each node shows the median RMSE (top) and number of observations (bottom) for the corresponding subset. Lower or higher median RMSE are indicated by more red or green polygons, respectively.

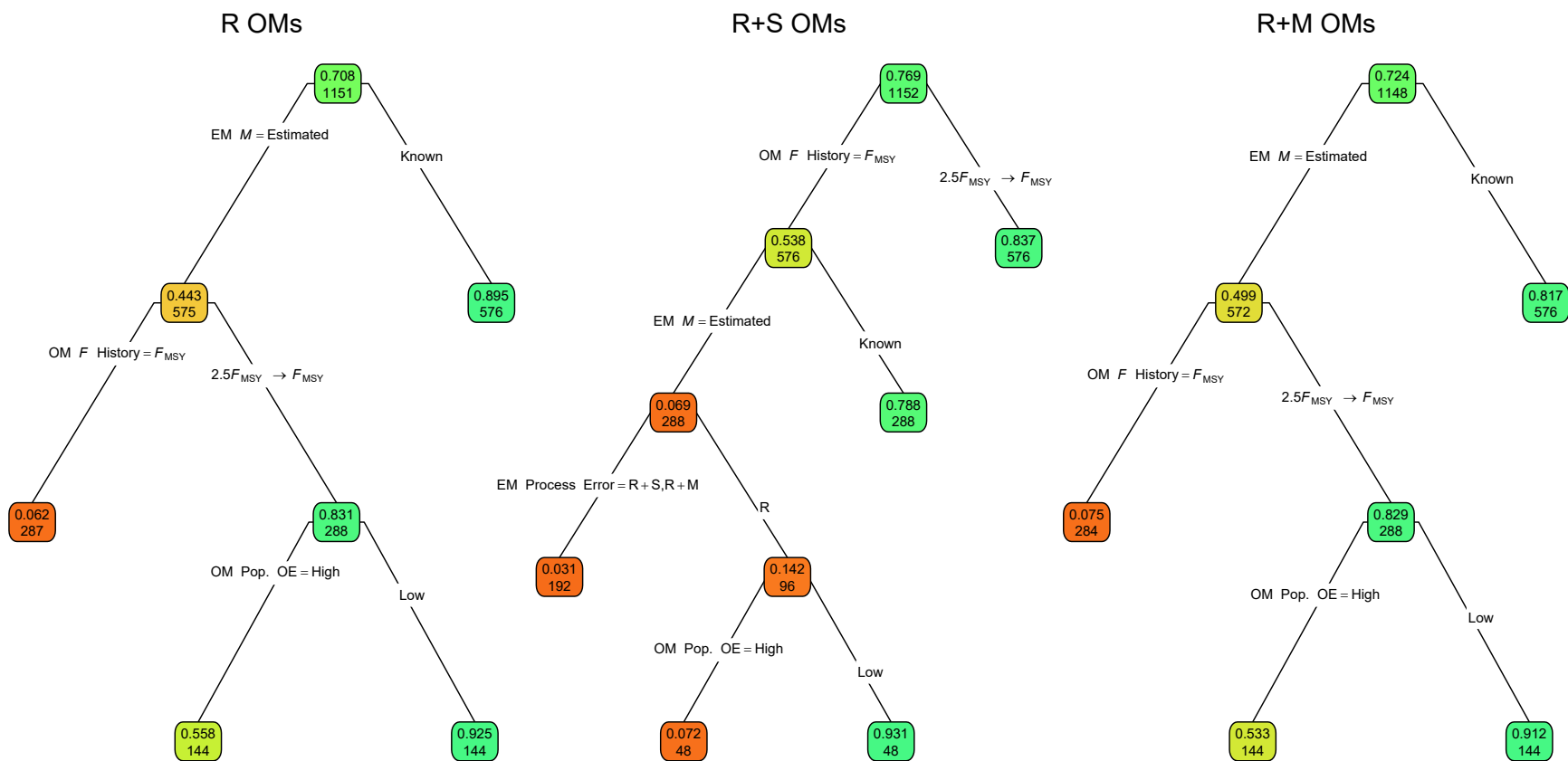


Fig. S31. Regression trees for the correlation of terminal year SSB estimates and true values. Each node shows the median correlation for the corresponding subset. Lower or higher median correlations are indicated by more red or green polygons, respectively. See Figure 1 caption and Table S1 for further details.

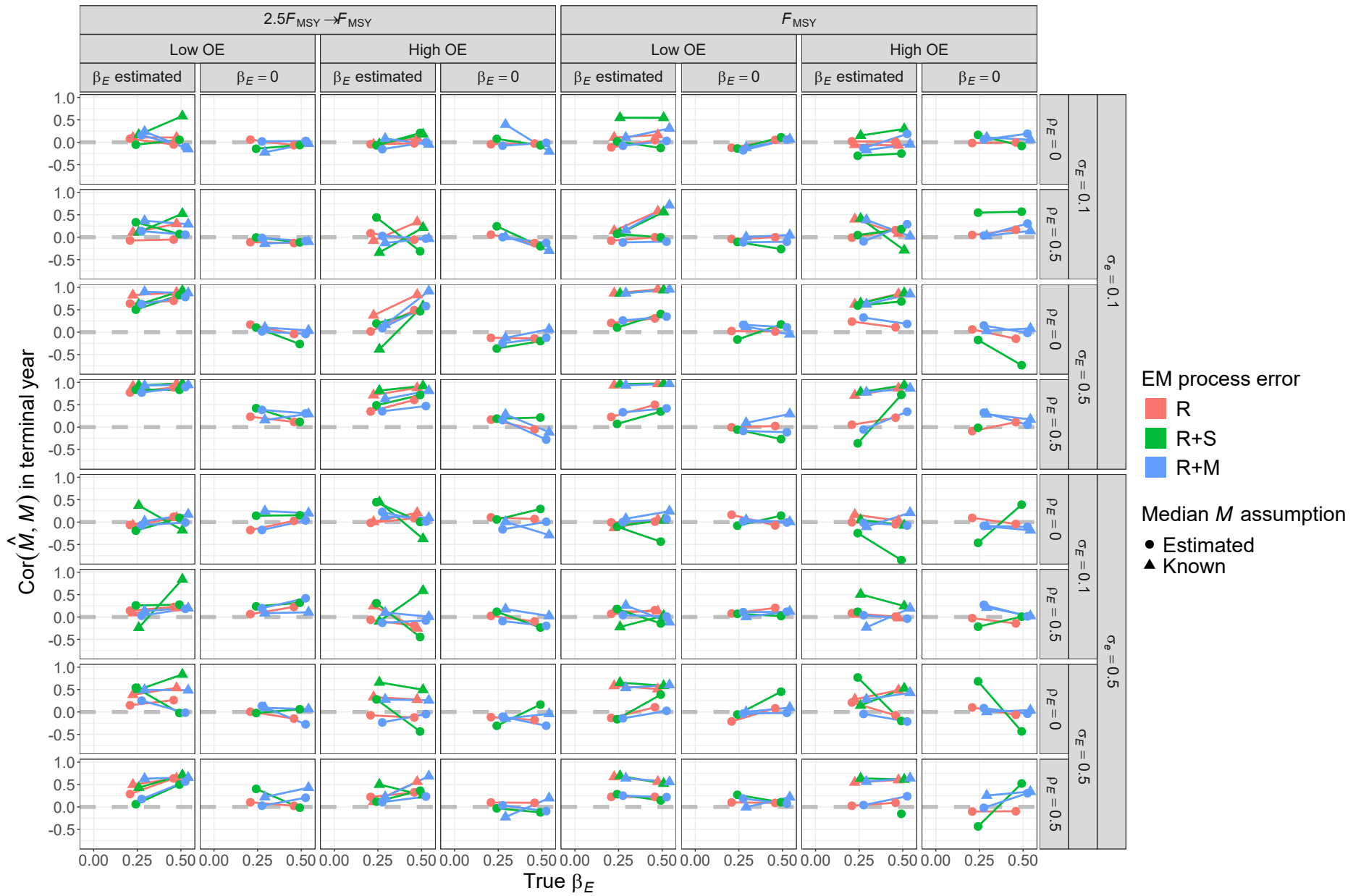


Fig. S32. For R operating models, correlation of estimates and true values of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

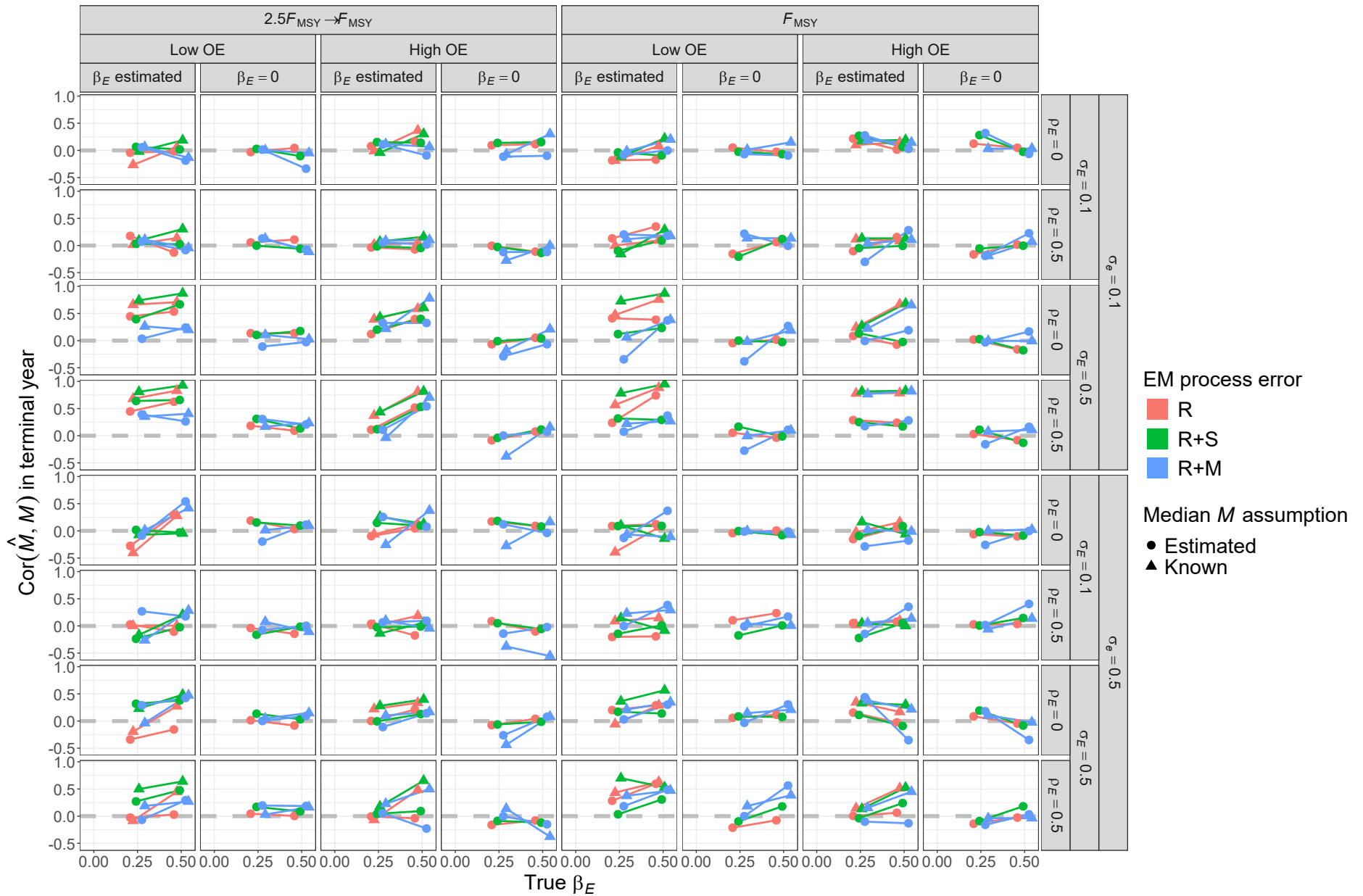


Fig. S33. For R+S operating models, correlation of estimates and true values of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

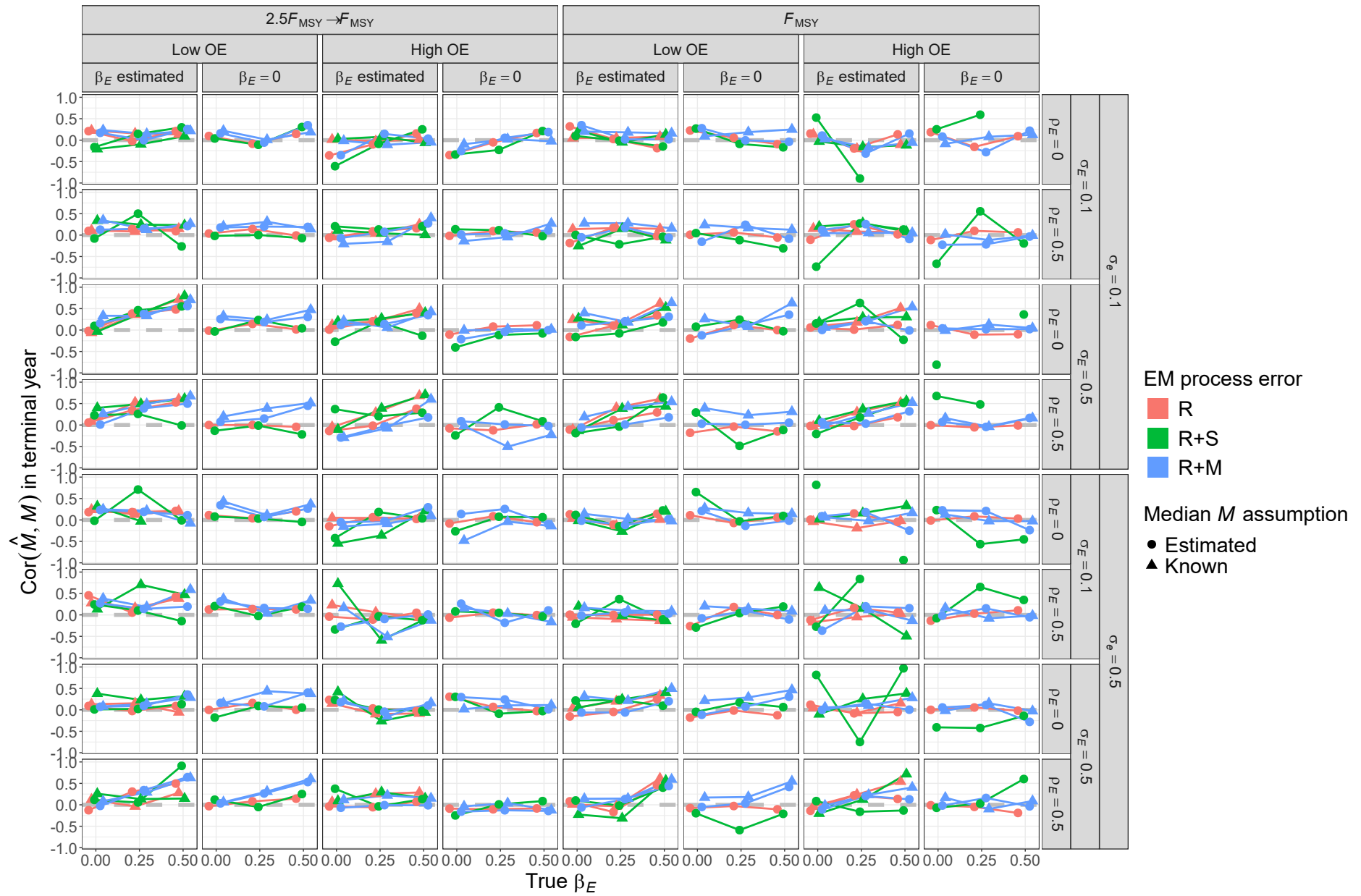


Fig. S34. For R+M operating models, correlation of estimates and true values of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

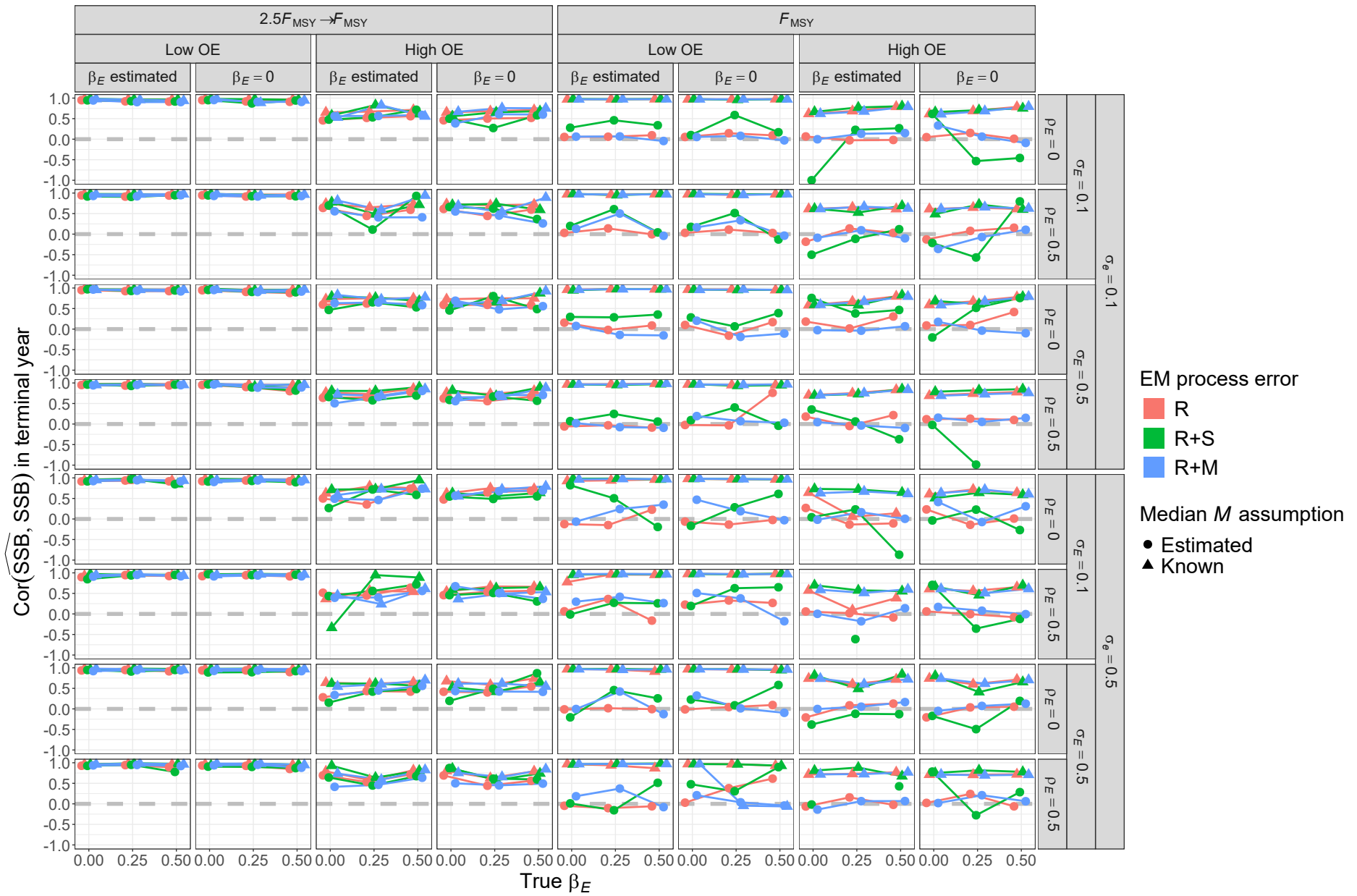


Fig. S35. For R operating models, correlation of estimates and true values of terminal year SSB for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

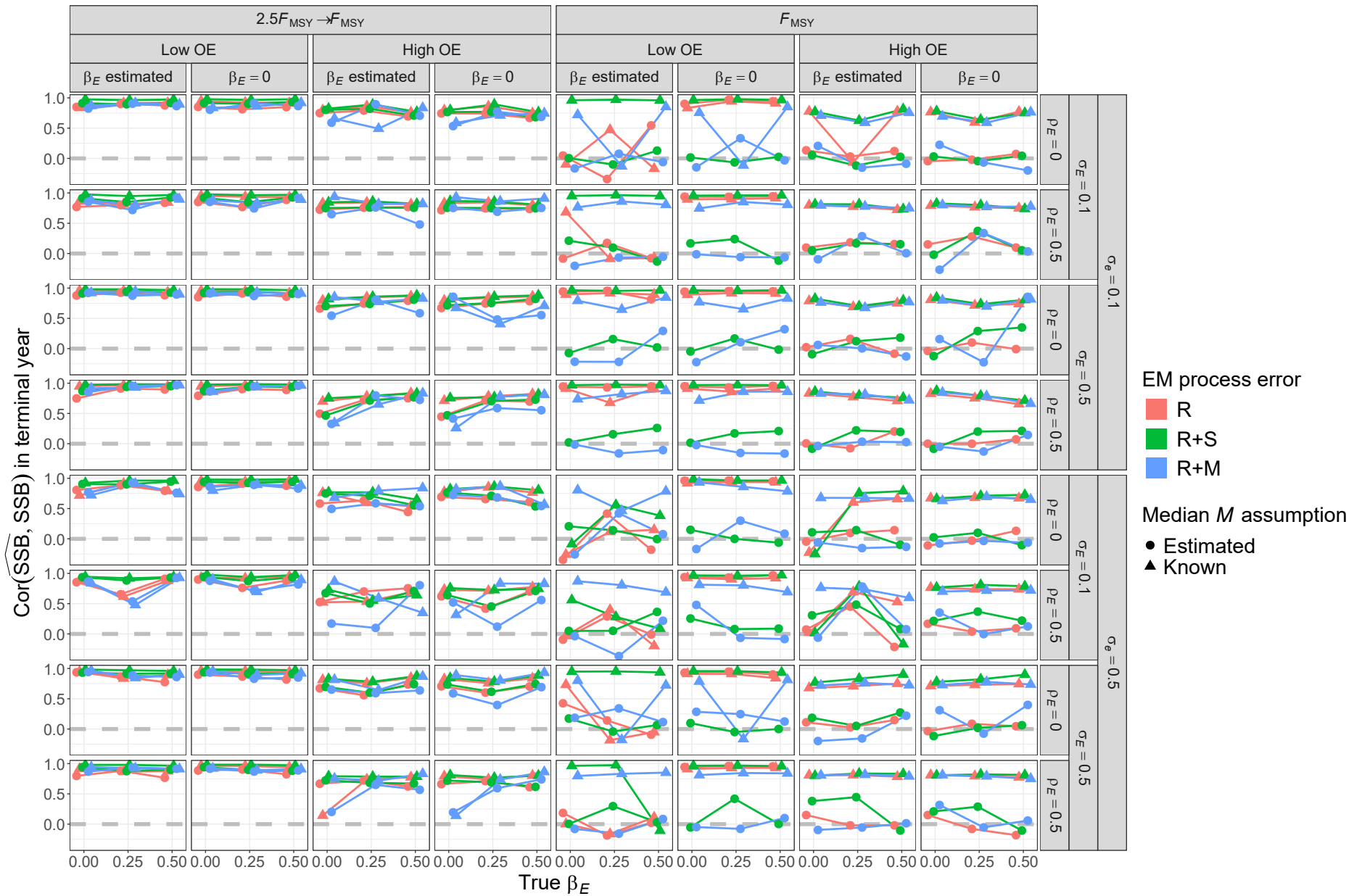


Fig. S36. For R+S operating models, correlation of estimates and true values of terminal year SSB for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

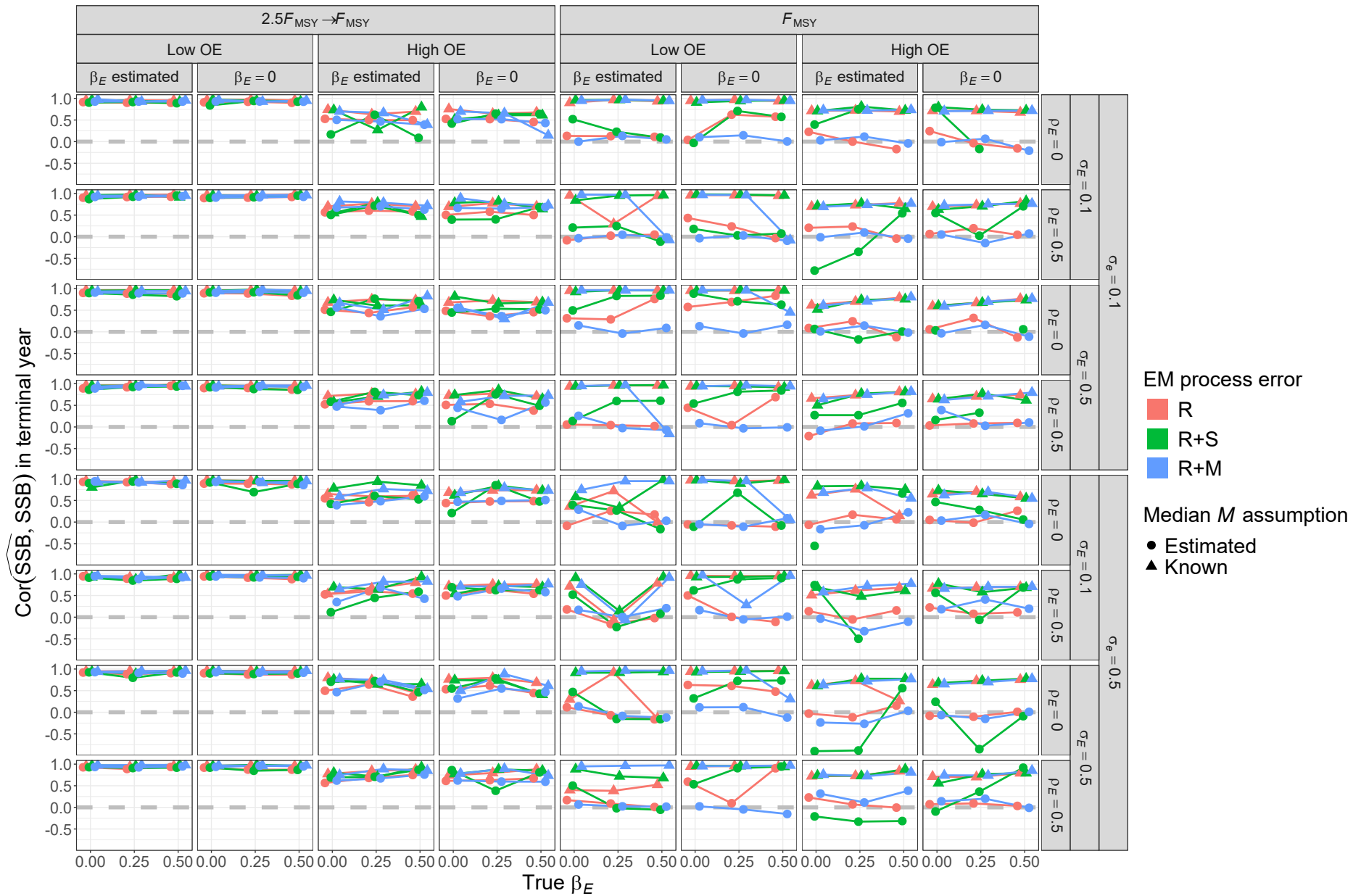


Fig. S37. For R+M operating models, correlation of estimates and true values of terminal year SSB for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

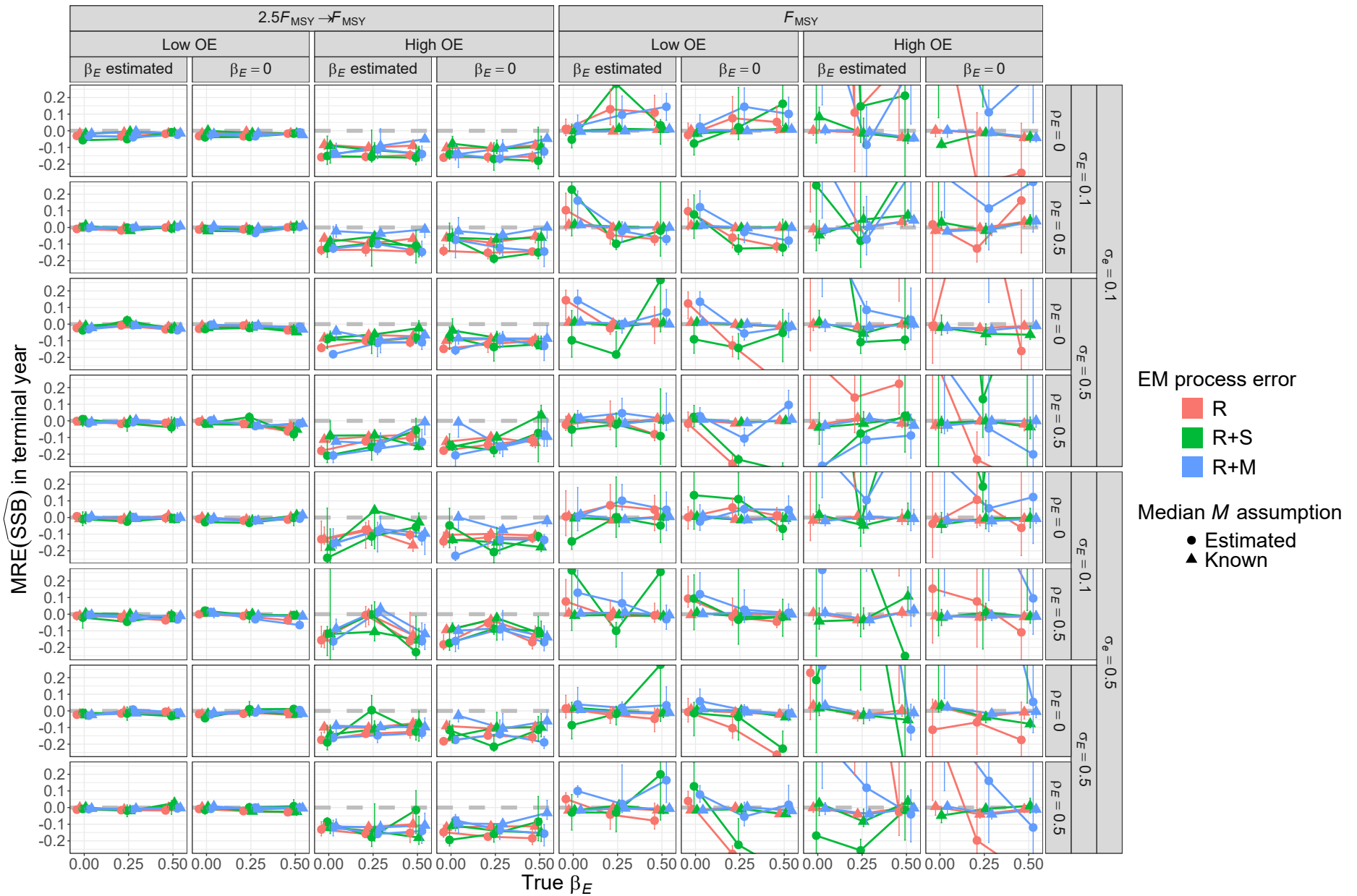


Fig. S38. For R operating models, median relative error (MRE) of estimates of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

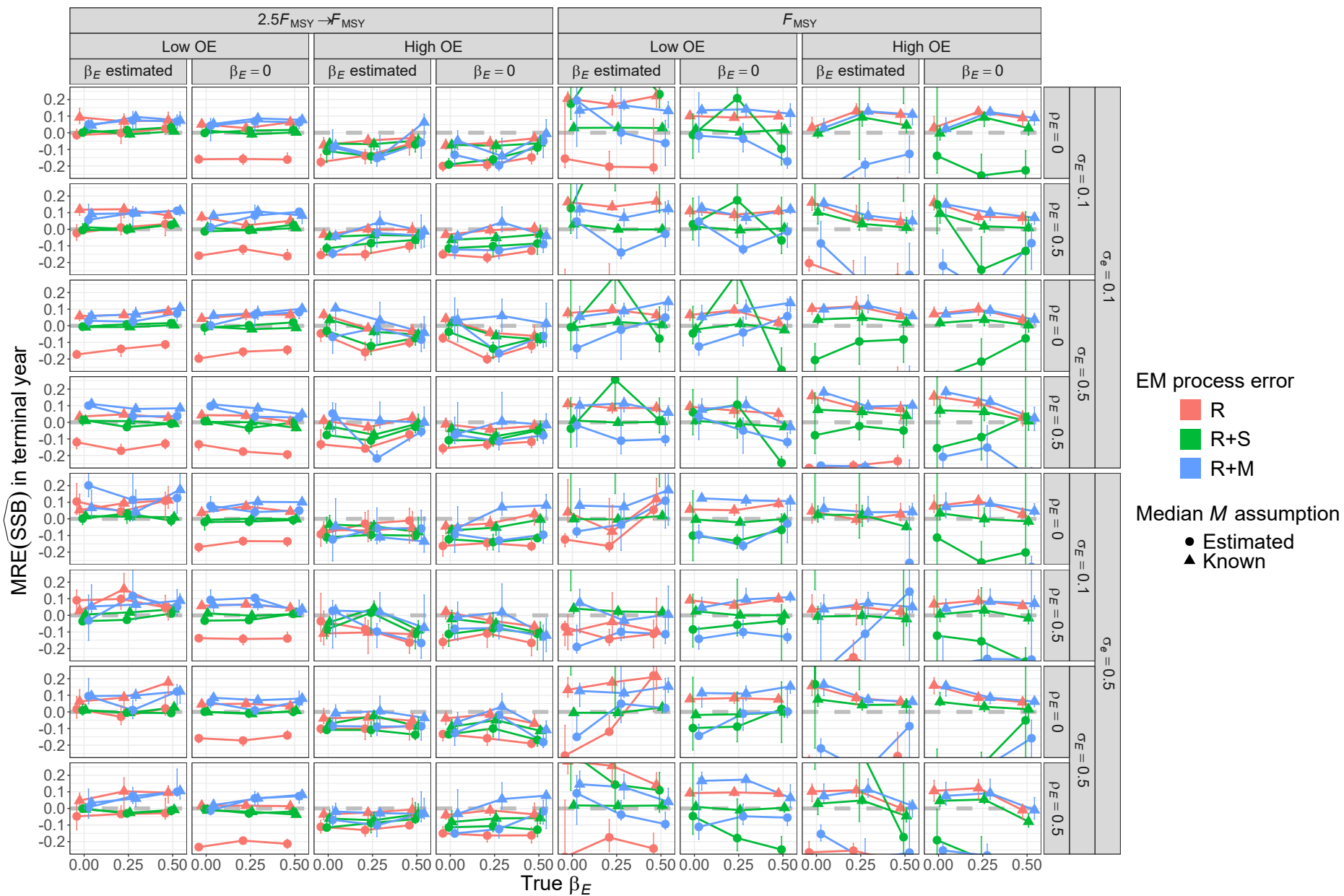


Fig. S39. For R+S operating models, median relative error (MRE) of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

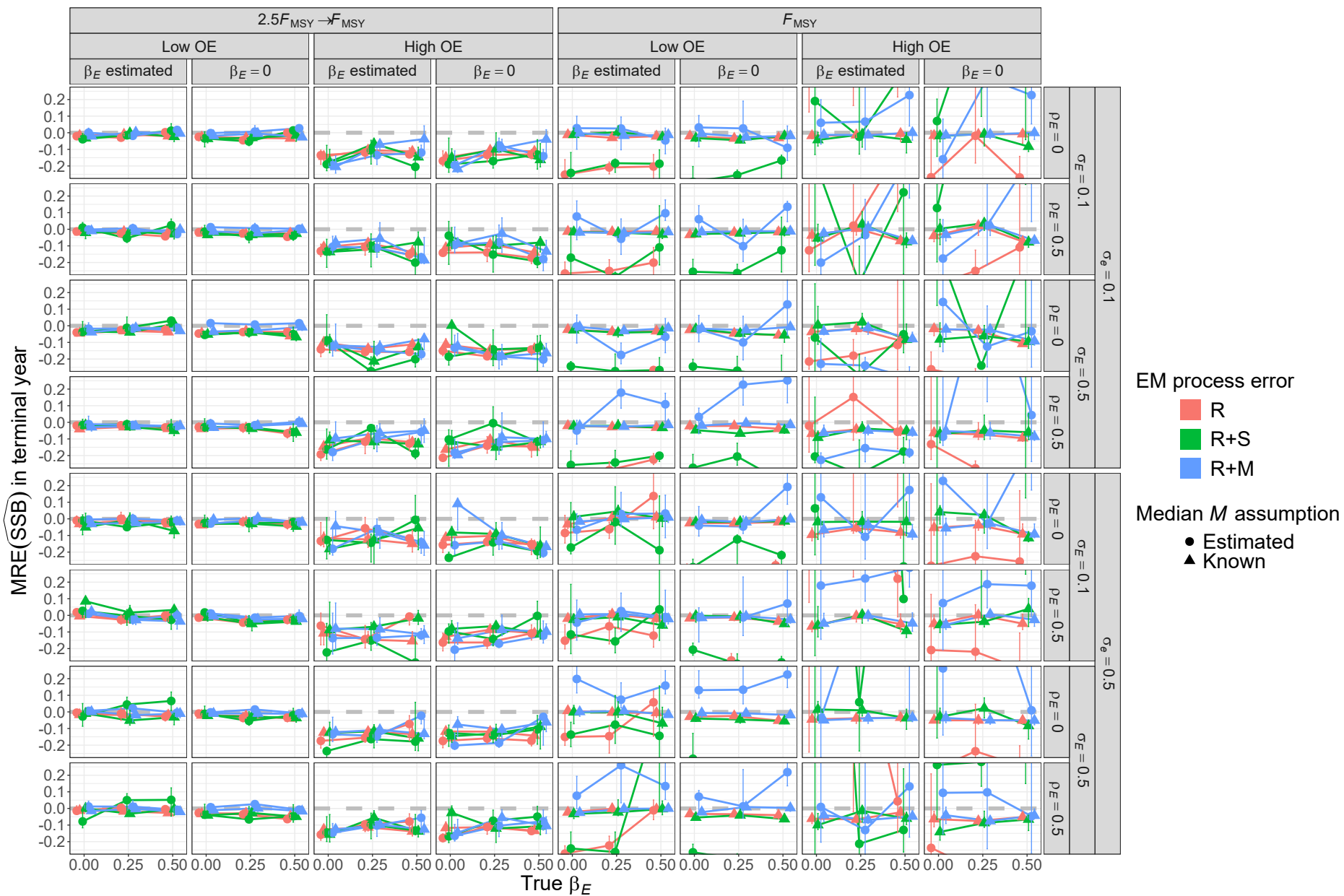


Fig. S40. For R+M operating models, median relative error (MRE) of estimates of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

Referenced Tables

Table S1. Operating (OM) and estimating model (EM) attributes used in analyses of deviance reduction and classification and regression trees. R, R+S, and R+M, denote models (OM or EM) with process error on recruitment only, recruitment and apparent survival, and recruitment and natural mortality, respectively. SD denotes standard deviation and F_{MSY} is the fishing mortality rate that achieves maximum sustainable yield.

Factor	Levels
OM Process error	R, R+S, R+M
OM Fishing History	$F_{\text{MSY}}, 2.5F_{\text{MSY}} \rightarrow F_{\text{MSY}}$
OM Covariate effect size (β_E)	0, 0.25, 0.5
OM Population Observation Error	Low, High
OM Covariate Observation Error	Low ($\sigma_e = 0.1$), High ($\sigma_e = 0.5$)
OM Covariate Process Error SD (σ_E)	0.1, 0.5
OM Covariate Process Error Correlation (ρ_E)	0, 0.5
EM Process Error	R, R+S, R+M
EM Covariate Effect	None ($\beta_E = 0$), β_E estimated
EM Median Natural Mortality Rate Parameter (β_M)	Known, Estimated

Percent deviance reduction results

Estimating model convergence

For fitted logistic regression models to indicators of convergence, the estimating model process error assumption provided the largest percent reduction in deviance (9-25%) of all operating model and estimating model attributes for all three operating model process error types (Table S2). Including second order interactions of operating model and estimating model factors also provided further reduction of residual deviance (between 7 and 11%), indicating successful convergence depended on a combination of operating model and estimating model attributes.

AIC performance of process error source selection

For all OM process error types, the level of error in both the population and covariate observations were the attributes that resulted in the largest percent reductions in deviance for multinomial regression fits to indicators of AIC selection of the process error configuration (Table S3). For R+S and R+M OMs, level of error in population observations provided deviance reduction (19% and 13%, respectively) much larger than any other OM and EM factors whereas error in covariate observations provided the largest reduction (8%) for R OMs. Lesser deviance reduction (2-3%) occurred for the true covariate effect size and whether the EM made the correct assumption about including a covariate effect for R OMs and error in covariate observations for R+M OMs. Including second and third order interactions, provided the largest reductions in residual deviance for R OMs (24%) and little reduction for R+S and R+M OMs (4% and 6%, respectively).

AIC performance of covariate effect selection

The factors explaining the largest reduction in deviance for logistic regression models fitted to indicators of AIC selection of the correct covariate effect assumption depended on the magnitude of the true effect assumed in the OM (Table S4). When OMs had no covariate effect ($\beta_E = 0$) the OM process error type provided the largest reduction in deviance (4%). When OMs assumed a covariate effect ($\beta_E > 0$), level of error in population observations and magnitude of temporal variability in the true covariate provided the largest reductions in deviance (4% and 8-21%, respectively). The reduction in deviance provided by the magnitude of covariate temporal variability also increased with larger true covariate effect. Including second and third order interactions provided a modest reduction in deviance (5-7%).

Median natural mortality rate bias

Across all OM process error types, regression models fit to errors in $\hat{\beta}_M$ showed largest reductions in deviance from the type of OM fishing pressure (4-17%), EM process error

assumption (2-12%), and the EM covariate effect assumption (3-7%, Table S5). The level of observation error also provided a similar reduction in deviance for R OMs (3%). Relatively large reductions in deviance also occurred with second and third order interactions (7-16%).

Median natural mortality rate standard error bias

The EM process error assumption provided the largest reduction in deviance for regression models fitted to errors in $\widehat{SE}(\hat{\beta}_M)$ from R (3%) and R+M (7%) OM simulations (Table S6). The type of OM fishing pressure provided the largest reduction (13%) for R+S OMs and a lesser reduction in deviance for R+M OMs (4%). The EM covariate effect assumption also provided similar reductions in deviance for R+S (9%) and R+M (4%) OMs.

Median natural mortality rate confidence interval bias

For logistic regressions of indicators of CIs including β_M , factors that provided the largest reductions in deviance were the OM fishing history and level of error in population observations for all OM process error types (0.3-6%) (Table S7). Including second and third order interactions provided the largest relative increase in deviance reduction for R+S (8-11%) and R+M OMs (5-8%) and lesser increases for R OMs (3-5%).

Covariate effect bias

For regression models fit to errors in estimation of β_E , we found very little reduction in deviance for any of the OM and EM factors (Table S8). For the normal model in these regressions, deviance is the sum of squares and there were extremely large estimates of β_E (and errors) for many simulations which resulted in extremely large sums of squares and relative differences from the null model were therefore small (<1%). Nevertheless, observation error of the covariate provided either the largest or second largest reductions in deviance of any of the single OM and EM attributes for OM process error configuration types. OM fishing history provided reductions of similar scale for R and R+S OMs. Other factors providing similar reductions were true covariate effect size for R OMs and level of error in population observations and EM process error assumption for R+S OMs. Including second and third order interactions also provided further reductions in residual deviance similar in scale to the individual factors.

Covariate effect standard error bias

Factors providing the largest reductions in deviance for estimation error of $\widehat{SE}(\hat{\beta}_E)$ were generally similar to those that were most important for estimation of β_E , but deviance reductions were one or two orders of magnitude larger (Table S9). Level of error in covariate observations and EM process error assumption was important for all OM process error configurations (6-13% and 3-4%, respectively). Level of error in population observations provided

relatively large reductions in deviance for R+S and R+M OMs (3% and 8%, respectively). The OM fishing pressure history provided a lesser reduction in deviance for all OM process error types (2-4%). Like estimation of β_E , additional large reductions in deviance also occurred when second and third order interactions of OM and EM attributes were included in the regression models (11-31%).

Covariate effect confidence interval bias

For logistic regressions of indicators of CIs including the true value of β_E , factors that provided the largest reductions in deviance were true covariate effect size for R and R+M OMs (11% and 3%, respectively) and EM process error configuration for R+S OMs (7%) (Table S10). Covariate observation error also provided relatively large reduction in deviance for R OMs (8%). Lesser reductions of 1-2% occurred for level of error in population observations and covariate temporal variability for all OM process error types. Including second and third order interactions also provided relatively large further reductions in deviance (5-9%).

Terminal year natural morality rate bias

Regressions of estimation errors show largest reductions in deviance from whether the EM treats β_M as known (Table S11). For R and R+S OMs, annual M was constant, and therefore, terminal M was known when the true β_M was assumed known and there will be no error in any of those EMs. Beyond the β_M assumption, the OM fishing history, level of error in population observations, and EM assumptions for covariate effect and process error all provided notable deviance reductions.

Terminal year natural morality rate accuracy

Like bias of terminal year M , the EM assumption for β_M provided the largest reduction in deviance (22-45%) for accuracy as measured by RMSE for all OM process error types (Table S12). Including second and third order interactions also provided large further reductions in deviance (21-38%).

Using correlation to measure accuracy of terminal year M estimation provides a different perspective. Treatment of β_M provided relatively large reductions in deviance for all OM process error types (3-12%), but level of variability in the covariate was equally or more important (3-15%) as was whether the covariate effect was included in the EM (12%) for R and R+S OMs (Table S13). Including second and third order interactions also provided relatively large further reductions in deviance (15 to 24%).

Terminal year spawning stock biomass bias

Regression model fits showed largest reductions in deviance for OM fishing history (2-3%), error in population observations (1% for R and R+M OMs), and EM β_M assumption (2%;

Table S14). The type of EM process error assumed also provided a deviance reduction similar in scale for R+S OMs (1%). Including second and third order interactions of OM and EM attributes provided further deviance reductions of 5-10%.

Terminal year spawning stock biomass accuracy

Largest reductions in deviance were provided by whether β_M was known or estimated (20-21%) and whether there was temporal contrast in fishing pressure (20-33%) across all OM process error types (Table S15). Lesser reductions were also provided by the EM process error assumption for R and R+M OMs. Inclusion of second and third order interactions of OM and EM attributes provided further reductions in deviance of 26-40%.

Using correlation to measure accuracy, temporal contrast in fishing pressure and EM treatment of β_M provided relatively important reductions in deviance like RMSE, but level of error in population observations was important rather than EM process error type (Table S16). Including second and third order interactions also provided relatively large reductions in deviance (6 to 20%).

Table S2. Percent reduction in deviance for logistic regression models fit to indicators of convergence (providing Hessian-based standard errors). Table rows give each operating model (OM) and estimating model (EM) factor included individually, combined, and with second and third order interactions. Table columns categorize operating models (OMs) with process error on recruitment only (R), recruitment and apparent survival (R+S), or recruitment and natural mortality (R+M). See Table S1 for further details.

Factor	R	R+S	R+M
OM F history	0.86	<0.01	0.24
OM Obs. Error	0.56	0.03	0.26
OM σ_e	0.66	2.99	0.93
OM σ_E	0.53	2.22	0.62
OM ρ_E	<0.01	0.02	<0.01
OM β_E	0.02	<0.01	<0.01
EM Process Error	24.53	9.33	23.06
EM β_E assumption	0.16	2.96	0.51
EM M Assumption	0.18	2.83	0.40
All factors	28.92	22.67	27.34
+ All Two Way	36.15	33.54	34.03
+ All Three Way	37.95	36.58	35.57

Table S3. Percent reduction in deviance for multinomial logistic regression models fit to indicators of process error assumption in the estimating models with lowest AIC. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.23	0.06	0.30
OM Obs. Error	0.54	18.58	13.43
OM σ_e	8.41	0.05	2.23
OM σ_E	0.17	0.08	0.59
OM ρ_E	1.29	0.03	0.06
OM β_E	3.42	0.04	0.38
EM M Assumption	0.35	0.34	0.48
EM β_E Assumption Correct	2.35	0.22	0.78
All factors	21.97	19.29	19.07
+ All Two Way	38.25	21.13	24.73
+ All Three Way	46.26	22.85	26.97

Further detailed results

Table S4. Percent reduction in deviance for logistic regression models fit to indicators of correct estimating model (EM) covariate effect assumption (no effect for operating models assuming $\beta_E = 0$ or estimated effect for operating models assuming $\beta_E > 0$) with lowest AIC. Table columns categorize operating models with different true values for β_E . See Tables S2 and S1 for further details.

Factor	$\beta_E = 0$	$\beta_E = 0.25$	$\beta_E = 0.5$
OM F history	<0.01	0.09	0.18
OM Obs. Error	1.19	4.06	4.64
OM Process Error	4.11	0.71	0.25
OM σ_e	0.66	2.00	1.97
OM σ_E	0.35	7.97	20.66
OM ρ_E	0.21	1.23	1.48
EM M Assumption	0.23	0.15	0.15
EM Process Error Assumption Correct	1.95	0.39	0.06
All factors	6.95	17.65	33.44
+ All Two Way	10.58	23.21	38.36
+ All Three Way	11.83	24.95	39.79

Table S5. Percent reduction in deviance for linear regression models fit to errors in estimation of the median natural mortality rate parameter ($\hat{\beta}_M$). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	4.03	16.52	6.51
OM Obs. Error	2.74	0.54	0.64
OM σ_e	0.01	<0.01	0.01
OM σ_E	0.18	0.02	0.15
OM ρ_E	0.07	0.05	<0.01
OM β_E	0.37	0.04	0.10
EM Process Error	2.31	11.99	2.65
EM β_E assumption	2.83	7.20	3.25
All factors	12.14	33.71	12.81
+ All Two Way	21.74	46.46	19.84
+ All Three Way	26.38	49.41	22.24

Table S6. Percent reduction in deviance for linear regression models fit to errors in estimation of the Hessian-based standard error of the median M parameter ($\widehat{\text{SE}}(\hat{\beta}_M)$). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.55	12.91	3.65
OM Obs. Error	0.98	0.01	0.07
OM σ_e	<0.01	0.03	0.11
OM σ_E	0.13	0.03	0.19
OM ρ_E	<0.01	0.02	<0.01
OM β_E	0.12	0.07	0.23
EM Process Error	3.16	8.79	6.83
EM β_E assumption	0.18	8.96	3.81
All factors	5.32	28.39	14.71
+ All Two Way	11.80	44.67	22.70
+ All Three Way	17.83	51.54	29.46

Table S7. Percent reduction in deviance for logistic regression models fit to indicators of whether CIs for estimates of the median natural mortality rate parameter ($\hat{\beta}_M$) includes the true value. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	6.08	1.46	0.14
OM Obs. Error	1.48	0.41	0.29
OM σ_e	0.01	0.06	0.01
OM σ_E	0.13	0.02	0.04
OM ρ_E	<0.01	<0.01	0.02
OM β_E	0.08	0.11	0.09
EM Process Error	0.53	0.08	0.04
EM β_E assumption	<0.01	0.19	0.01
All factors	8.46	2.32	0.62
+ All Two Way	11.97	10.57	5.99
+ All Three Way	13.35	13.51	8.44

Table S8. Percent reduction in deviance for linear regression models fit to errors in estimation of the covariate effect on natural mortality rate ($\hat{\beta}_E$). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.03	0.06	0.02
OM Obs. Error	<0.01	0.06	0.02
OM σ_e	0.04	0.08	0.06
OM σ_E	<0.01	0.02	0.01
OM ρ_E	<0.01	0.01	<0.01
OM β_E	0.05	<0.01	<0.01
EM Process Error	0.01	0.06	0.01
EM M assumption	0.02	0.02	<0.01
All factors	0.18	0.37	0.14
+ All Two Way	0.50	1.11	0.35
+ All Three Way	1.38	1.88	0.66

Table S9. Percent reduction in deviance for linear regression models fit to errors in estimation of the Hessian-based standard error of the covariate effect on M ($\widehat{\text{SE}}(\hat{\beta}_E)$). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	1.84	2.68	4.00
OM Obs. Error	0.06	2.74	8.04
OM σ_e	6.27	7.28	12.83
OM σ_E	0.48	1.05	0.46
OM ρ_E	0.17	0.01	0.02
OM β_E	0.37	0.01	0.05
EM Process Error	2.90	3.89	2.98
EM M assumption	0.15	0.34	0.47
All factors	13.01	20.95	29.24
+ All Two Way	28.58	32.25	42.77
+ All Three Way	44.44	42.83	51.84

Table S10. Percent reduction in deviance for logistic regression models fit to indicators of whether CIs for covariate effect estimates includes the true value. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.24	0.05	0.01
OM Obs. Error	1.36	1.91	1.78
OM σ_e	8.26	0.24	0.92
OM σ_E	0.92	1.70	1.56
OM ρ_E	0.03	0.05	0.06
OM β_E	11.10	0.68	2.72
EM Process Error	0.01	6.51	0.27
EM M assumption	0.25	0.29	0.06
All factors	23.61	12.16	7.78
+ All Two Way	30.17	18.08	12.63
+ All Three Way	32.58	20.62	14.87

Table S11. Percent reduction in deviance for linear regression models fit to errors in terminal year natural mortality rate (M) as measured by Eq. 2. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
Convergence	<0.01	0.18	0.01
OM F history	0.66	2.69	1.00
OM Obs. Error	0.29	0.43	0.16
OM σ_e	<0.01	<0.01	<0.01
OM σ_E	0.02	<0.01	<0.01
OM ρ_E	<0.01	0.01	<0.01
OM β_E	0.03	0.01	0.01
EM Process Error	0.22	1.42	0.43
EM β_E assumption	0.35	1.07	0.39
EM M assumption	0.84	5.10	1.22
All factors	2.56	10.83	3.37
+ All Two Way	5.41	18.75	6.38
+ All Three Way	7.93	22.49	8.76

Table S12. Percent reduction in deviance for linear regression models fit to log-transformed RMSE of terminal year natural mortality (M). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.26	2.21	14.65
OM Obs. Error	1.65	0.71	4.68
OM σ_e	0.02	0.16	0.01
OM σ_E	1.06	1.23	0.84
OM ρ_E	0.02	0.01	<0.01
OM β_E	3.73	1.94	1.14
EM Process Error	2.04	3.85	0.87
$EM\beta_E$ assumption	0.19	0.59	0.07
EM M assumption	22.28	45.15	44.70
All factors	32.83	56.71	66.99
+ All Two Way	57.65	84.17	87.96
+ All Three Way	74.93	94.57	91.94

Table S13. Percent reduction in deviance for linear regression models fit to correlation of terminal year natural mortality (M) estimates and true values. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	0.05	0.02	0.37
OM Obs. Error	2.00	0.89	3.73
OM σ_e	2.30	2.56	0.37
OM σ_E	14.91	13.78	3.03
OM ρ_E	1.98	0.33	0.14
OM β_E	0.41	3.76	3.87
EM Process Error	0.43	1.52	0.62
$EM\beta_E$ assumption	12.72	12.30	1.56
EM M assumption	12.34	7.16	3.03
All factors	42.11	38.31	15.76
+ All Two Way	63.38	62.40	31.99
+ All Three Way	72.79	75.98	46.62

Table S14. Percent reduction in deviance for linear regression models fit to errors in terminal year spawning stock biomass (SSB) as measured by Eq. 2. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
Convergence	0.02	0.13	<0.01
OM F history	2.08	2.74	1.75
OM Obs. Error	1.09	0.18	1.19
OM σ_e	0.01	<0.01	<0.01
OM σ_E	<0.01	<0.01	<0.01
OM ρ_E	<0.01	<0.01	<0.01
OM β_E	<0.01	0.01	<0.01
EM Process Error	0.52	1.05	0.57
EM β_E assumption	<0.01	0.05	0.01
EM M assumption	1.76	2.12	1.51
All factors	5.97	6.58	5.21
+ All Two Way	12.81	13.00	11.97
+ All Three Way	16.41	15.57	15.73

Table S15. Percent reduction in deviance for linear regression models fit to log-transformed RMSE of terminal year spawning stock biomass (SSB). See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	19.88	32.67	20.92
OM Obs. Error	2.21	0.11	1.48
OM σ_e	0.05	0.16	0.03
OM σ_E	0.08	0.16	0.02
OM ρ_E	0.03	<0.01	0.02
OM β_E	0.05	0.03	0.14
EM Process Error	5.72	0.13	5.73
$EM\beta_E$ assumption	0.03	0.79	0.05
EM M assumption	20.05	21.11	19.43
All factors	48.09	55.17	47.82
+ All Two Way	79.09	80.98	78.40
+ All Three Way	88.30	89.49	87.67

Table S16. Percent reduction in deviance for linear regression models fit to correlation of terminal year spawning stock biomass (SSB) estimates and true values. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	19.27	25.70	19.59
OM Obs. Error	12.53	3.60	9.46
OM σ_e	0.08	0.44	0.09
OM σ_E	0.10	0.44	0.11
OM ρ_E	<0.01	<0.01	0.09
OM β_E	0.09	0.05	<0.01
EM Process Error	0.07	0.92	0.71
EM β_E assumption	<0.01	1.32	0.38
EM M assumption	28.94	19.74	25.36
All factors	61.19	52.21	56.09
+ All Two Way	81.48	72.16	74.98
+ All Three Way	85.23	83.22	80.48

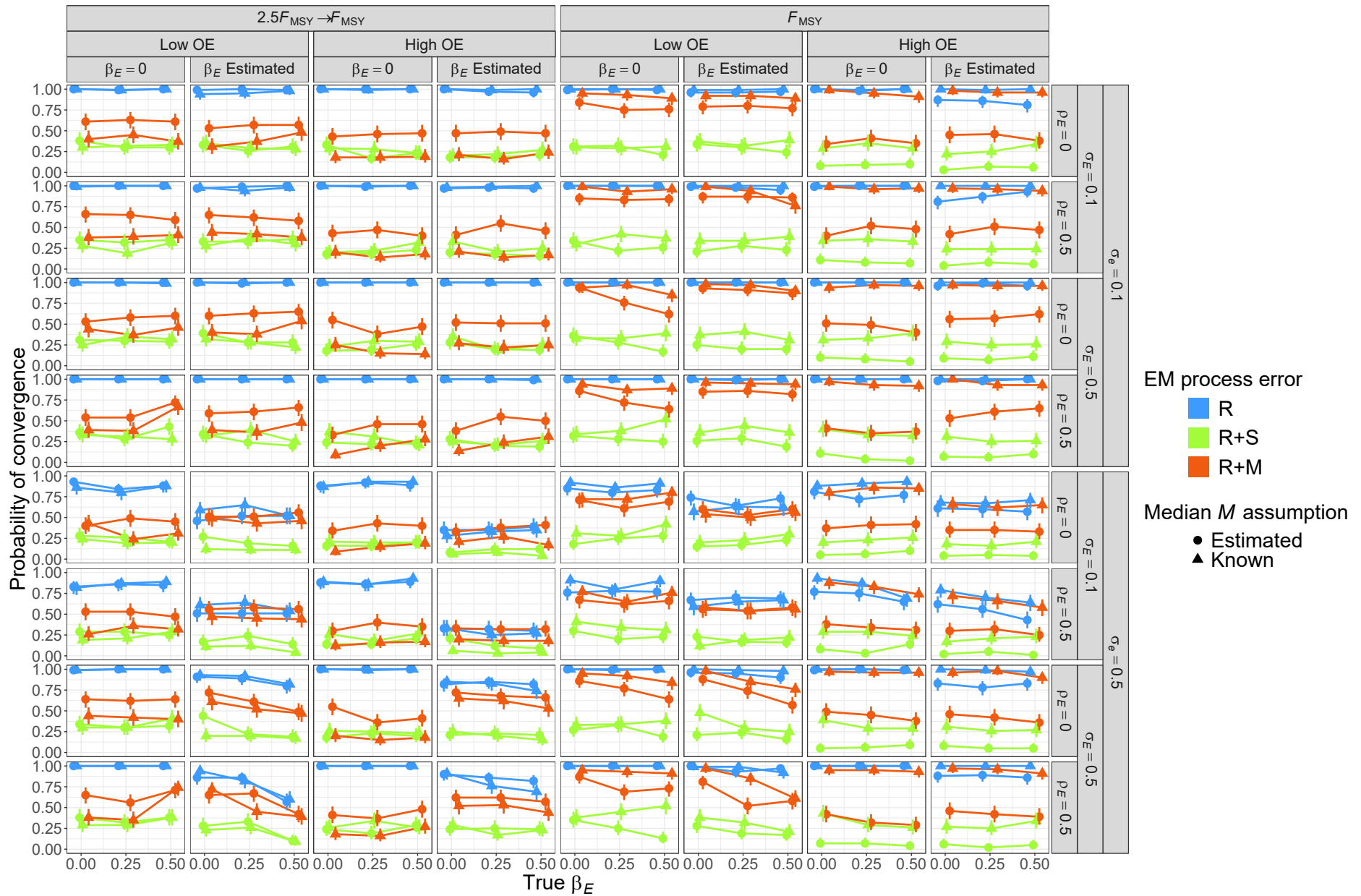


Fig. S41. Estimated probability of fits providing Hessian-based standard errors for estimating models assuming alternative process error, that estimate or assume known median M , and that estimate or assume no covariate effect on median M when fitted to R operating models and three levels of true covariate effect on median M (x axis). Vertical lines represent 95% confidence intervals.

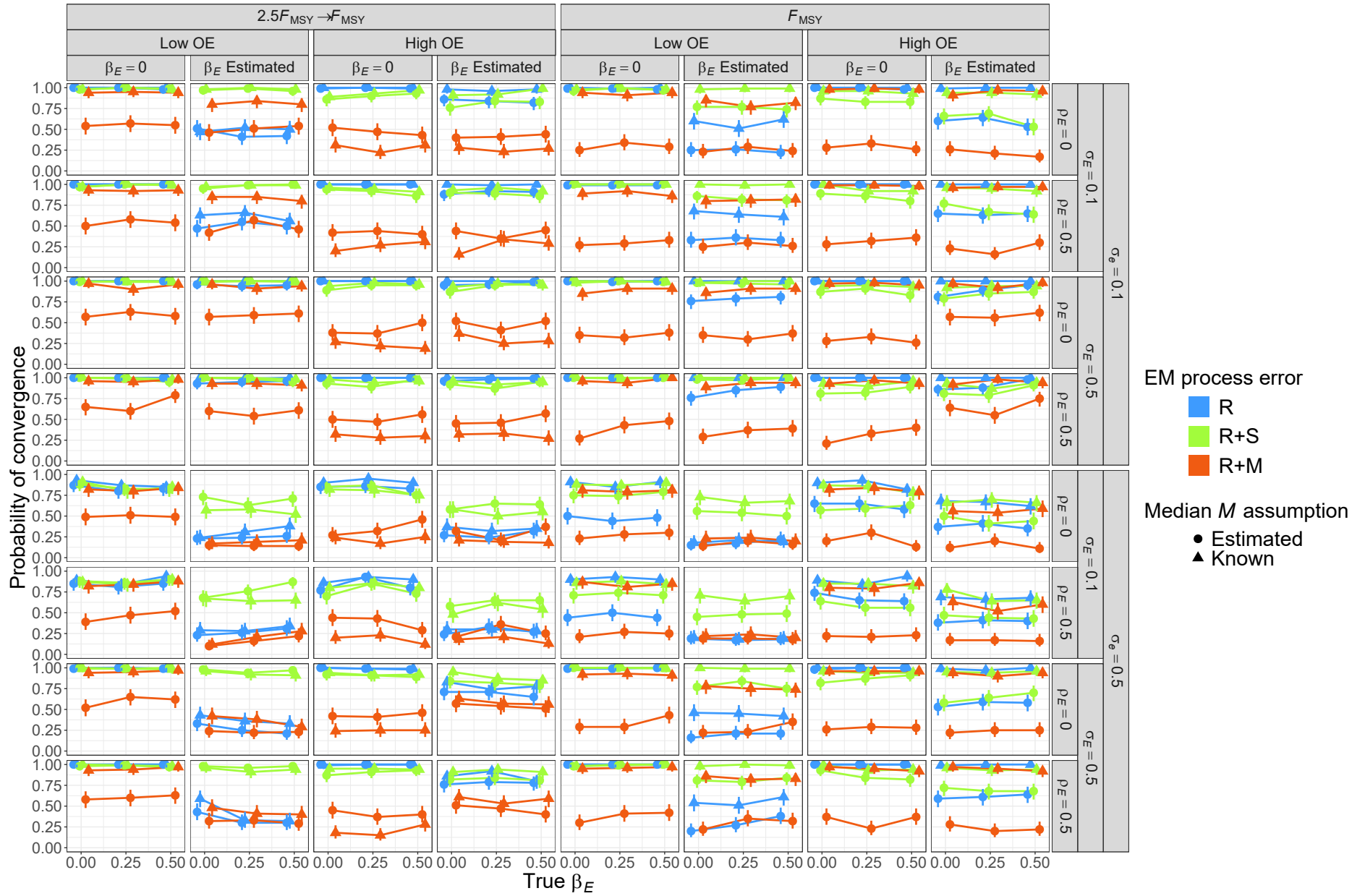


Fig. S42. Estimated probability of fits providing Hessian-based standard errors for estimating models assuming alternative process error, that estimate or assume known median M , and that estimate or assume no covariate effect on median M when fitted to R+S operating models and three levels of true covariate effect on median M (x axis). Vertical lines represent 95% confidence intervals.

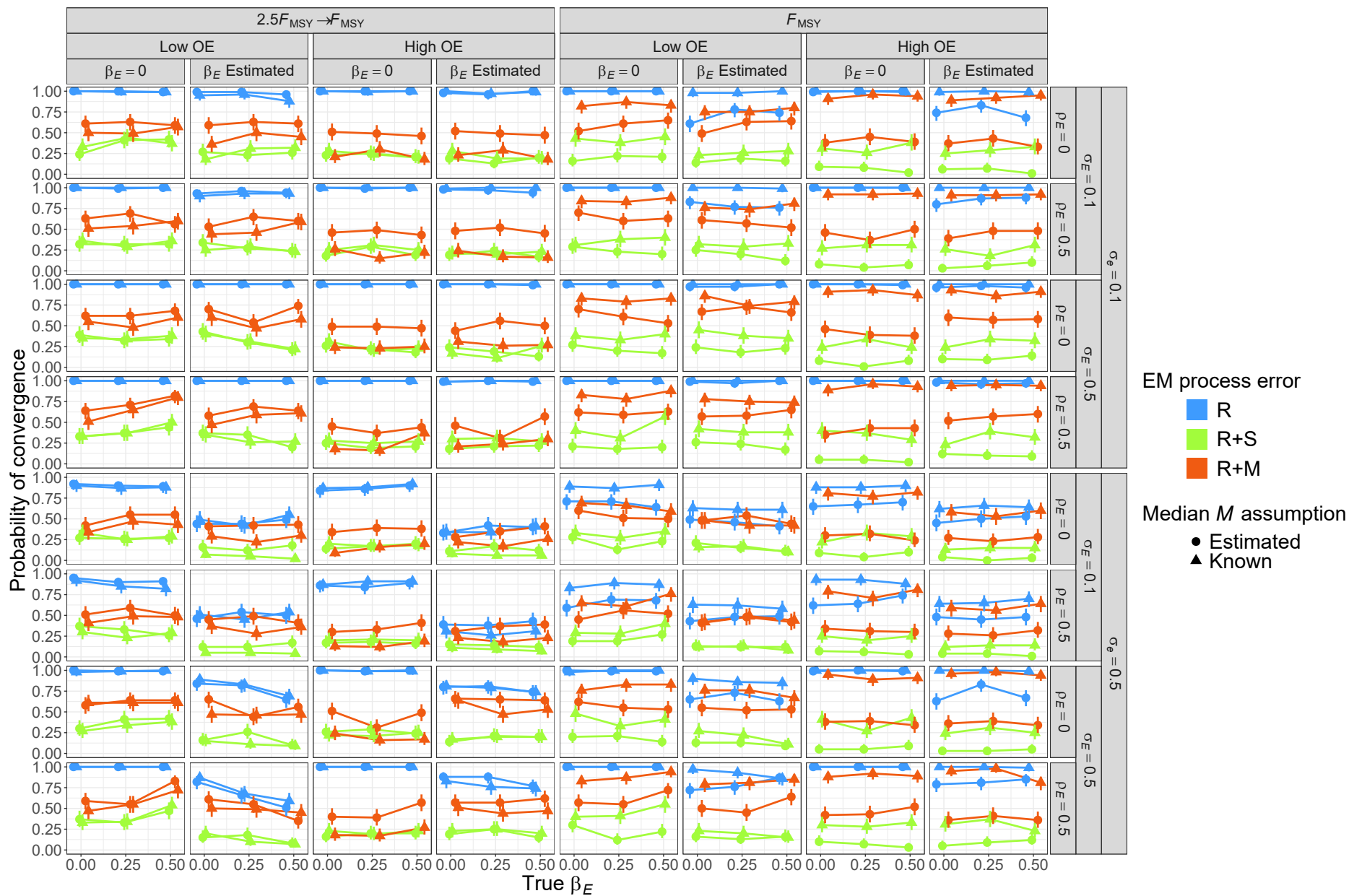


Fig. S43. Estimated probability of fits providing Hessian-based standard errors for estimating models assuming alternative process error, that estimate or assume known median M , and that estimate or assume no covariate effect on median M when fitted to R+M operating models and three levels of true covariate effect on median M (x axis). Vertical lines represent 95% confidence intervals.

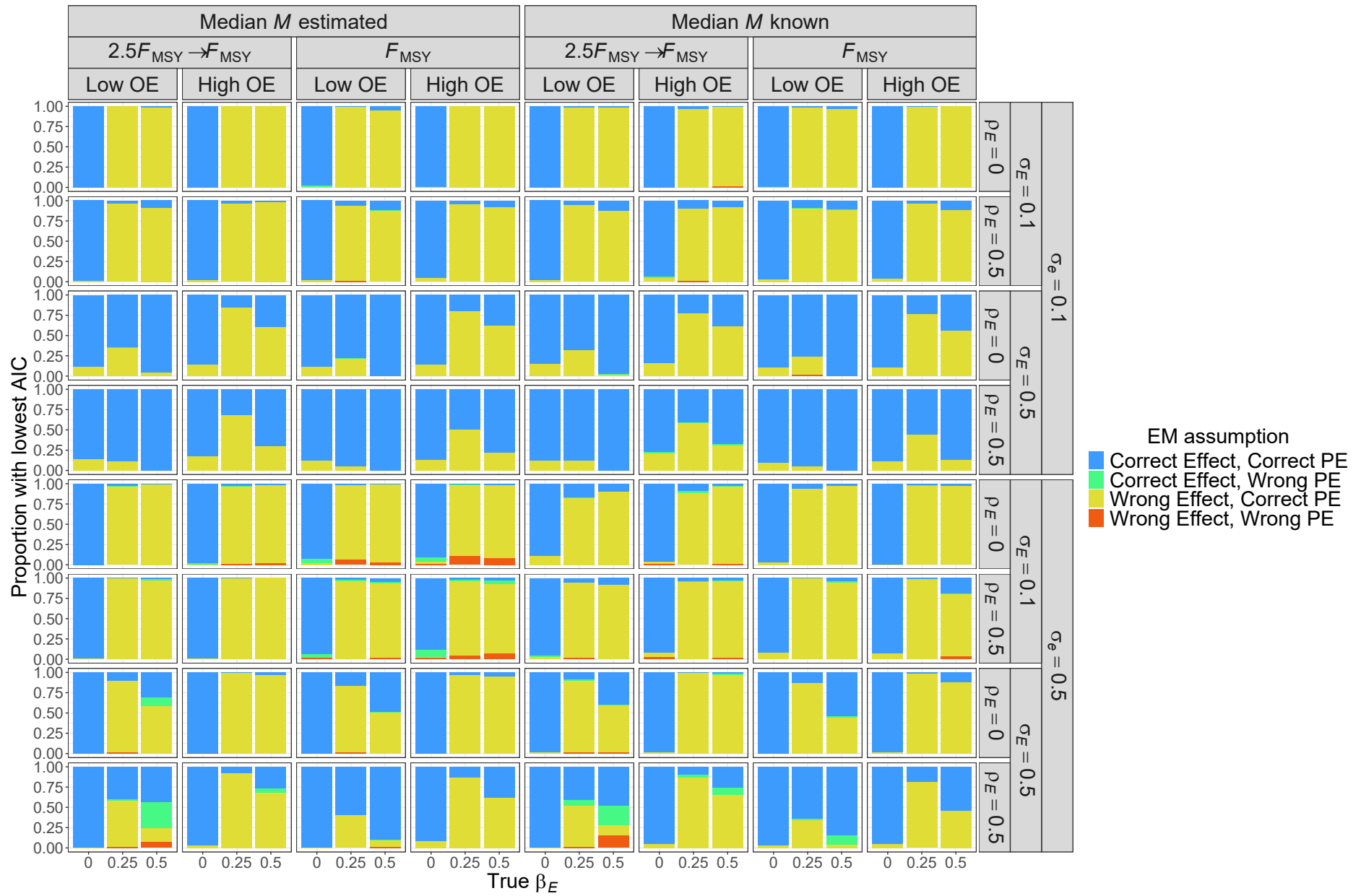


Fig. S44. Proportion of simulated data sets for R operating models where the estimating model type (treatment of environmental covariate and assumed process error type) had the lowest AIC.

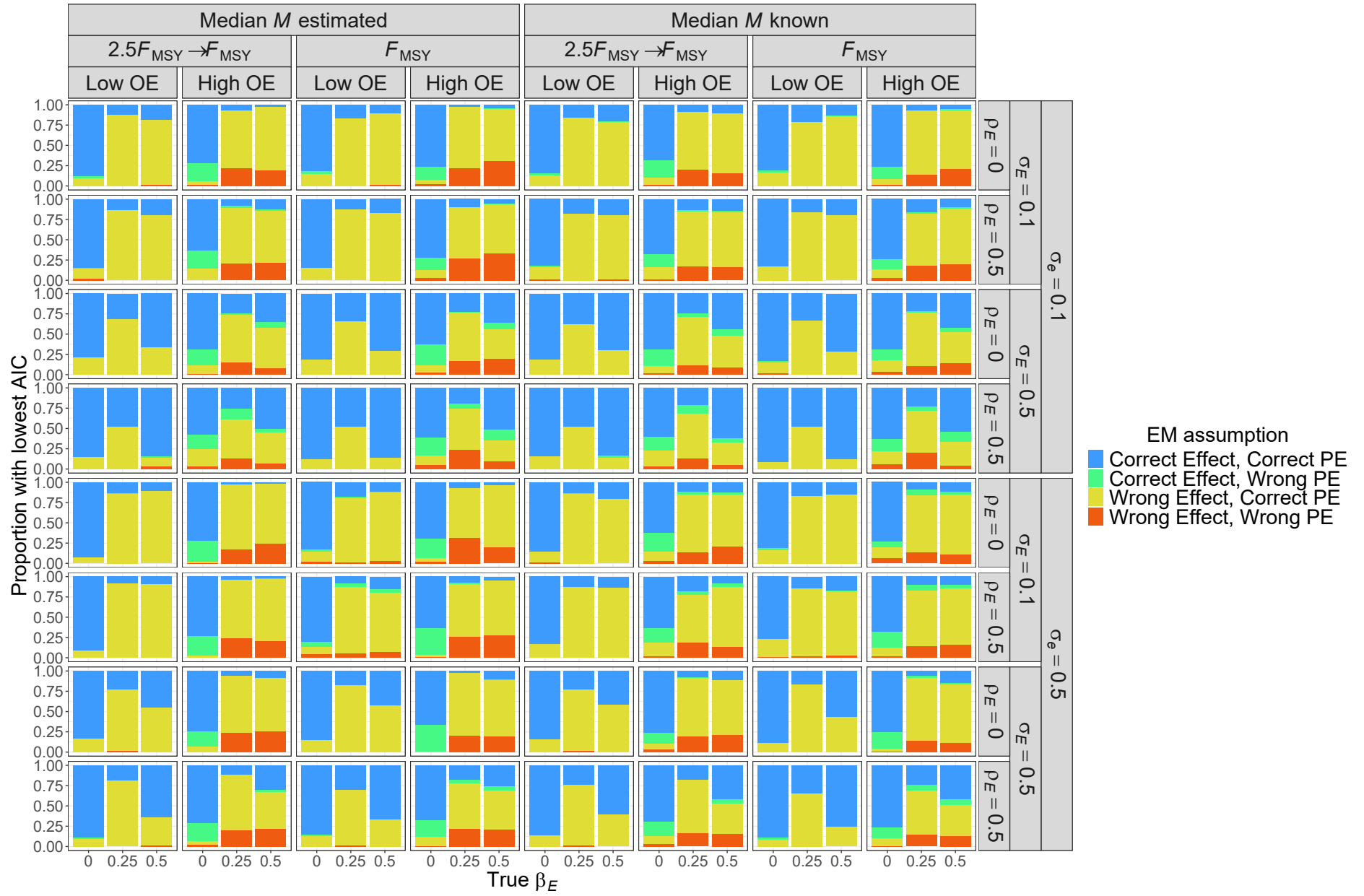


Fig. S45. Proportion of simulated data sets for R+S operating models where the estimating model type (treatment of environmental covariate and assumed process error type) had the lowest AIC.

Fig. S46. Proportion of simulated data sets for R+M operating models where the estimating model type (treatment of environmental covariate and assumed process error type) had the lowest AIC.

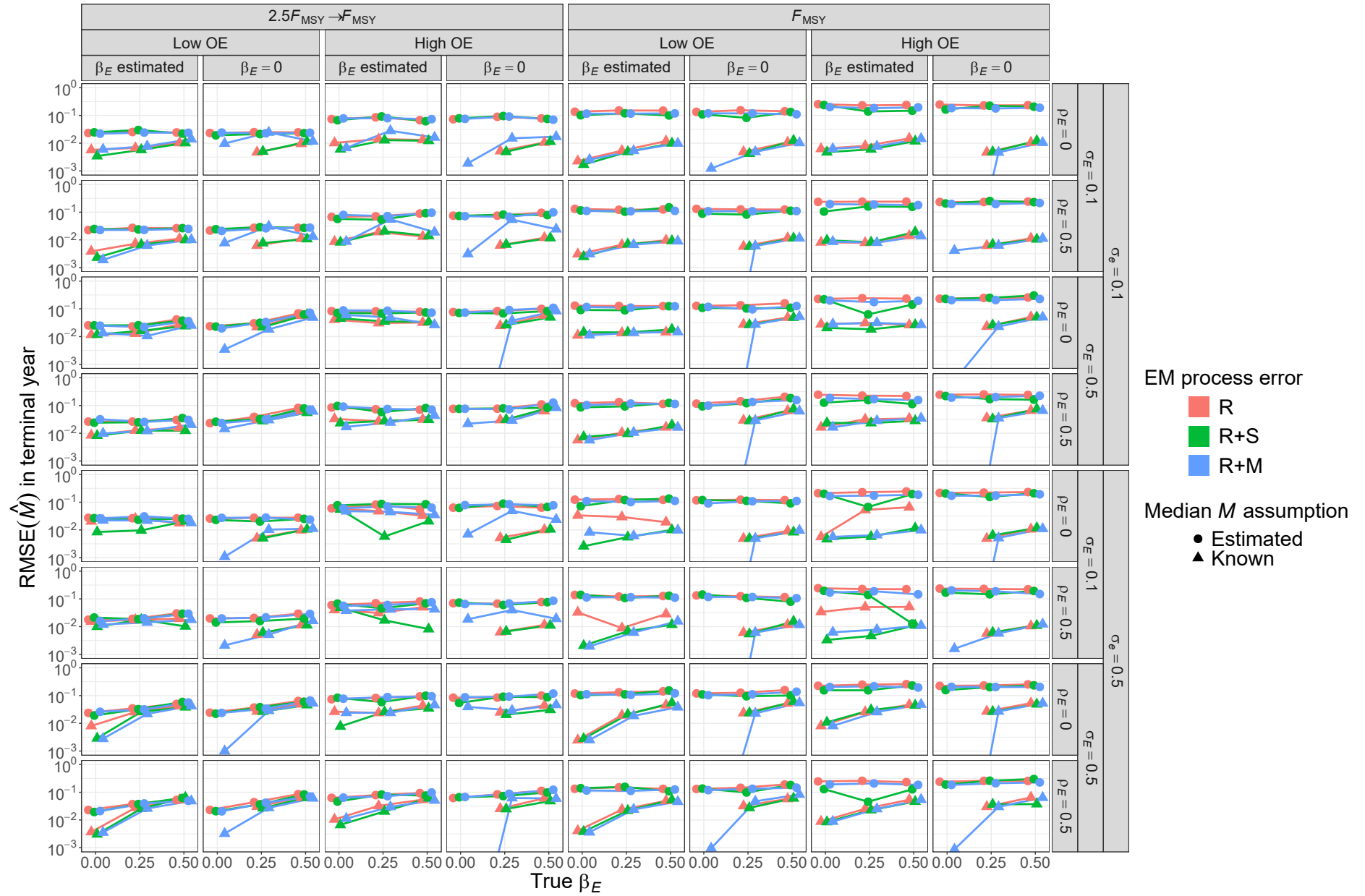


Fig. S47. For R operating models, RMSE of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

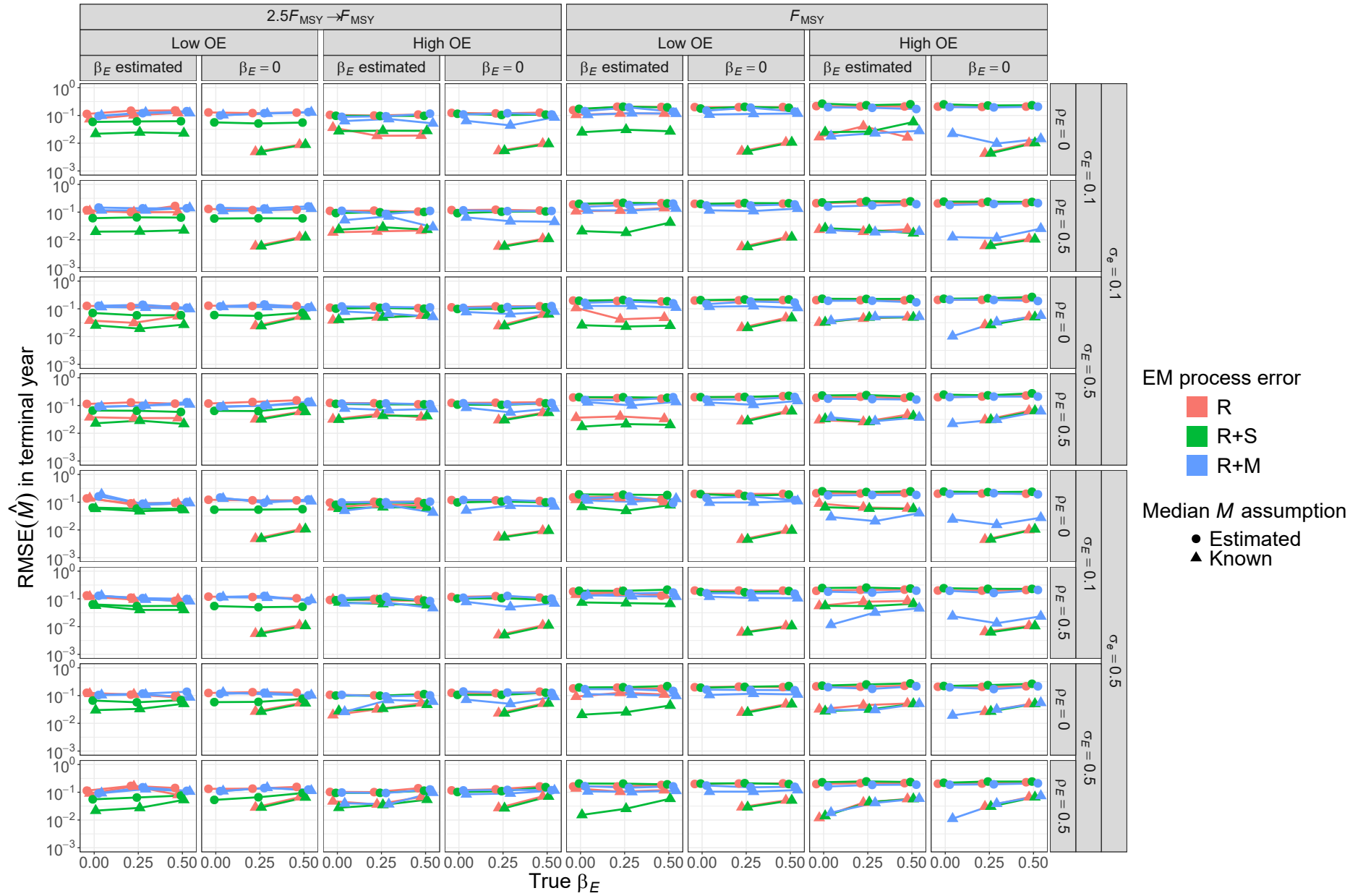


Fig. S48. For R+S operating models, RMSE of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

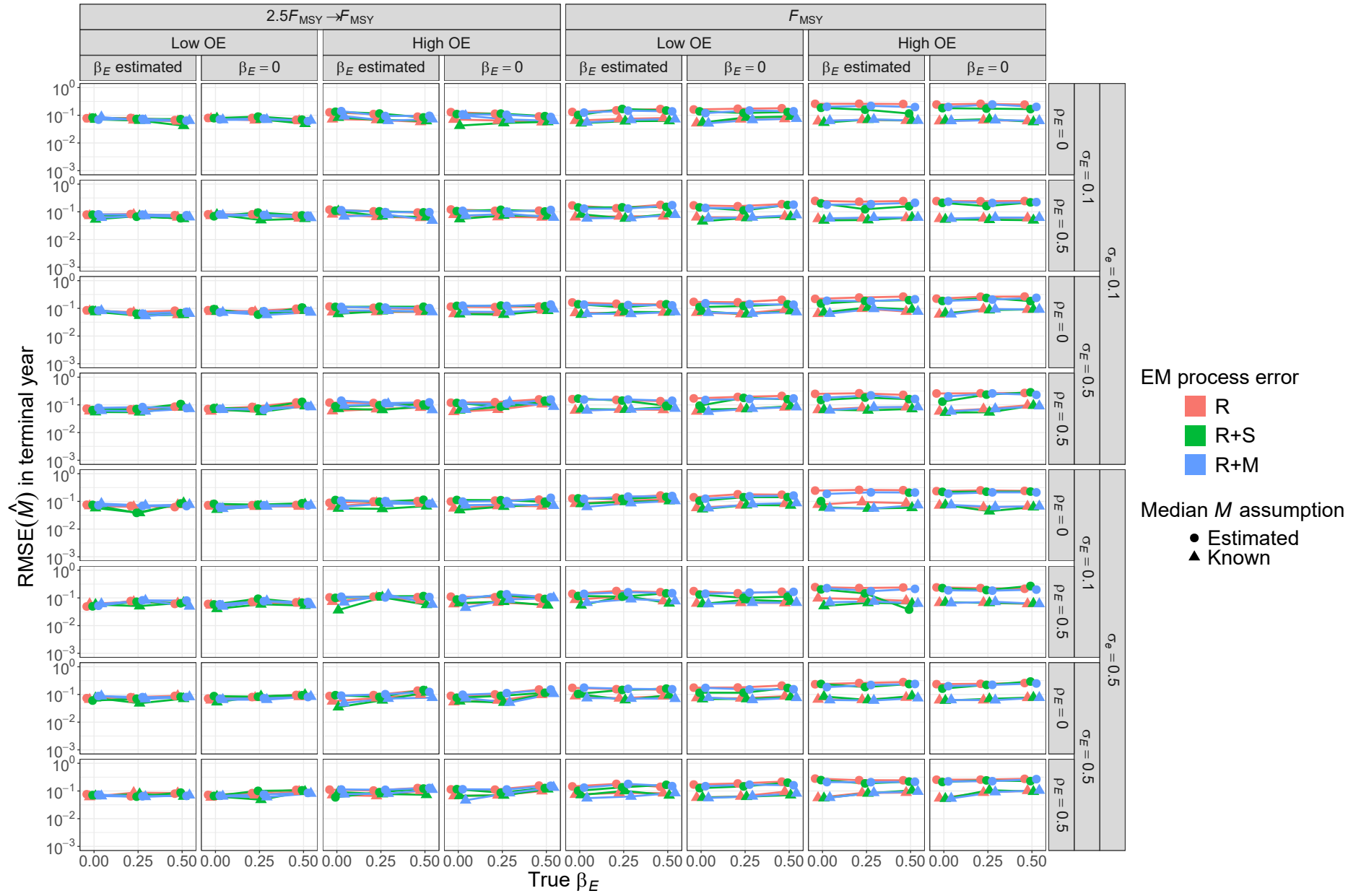


Fig. S49. For R+M operating models, RMSE of estimates of terminal year M for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

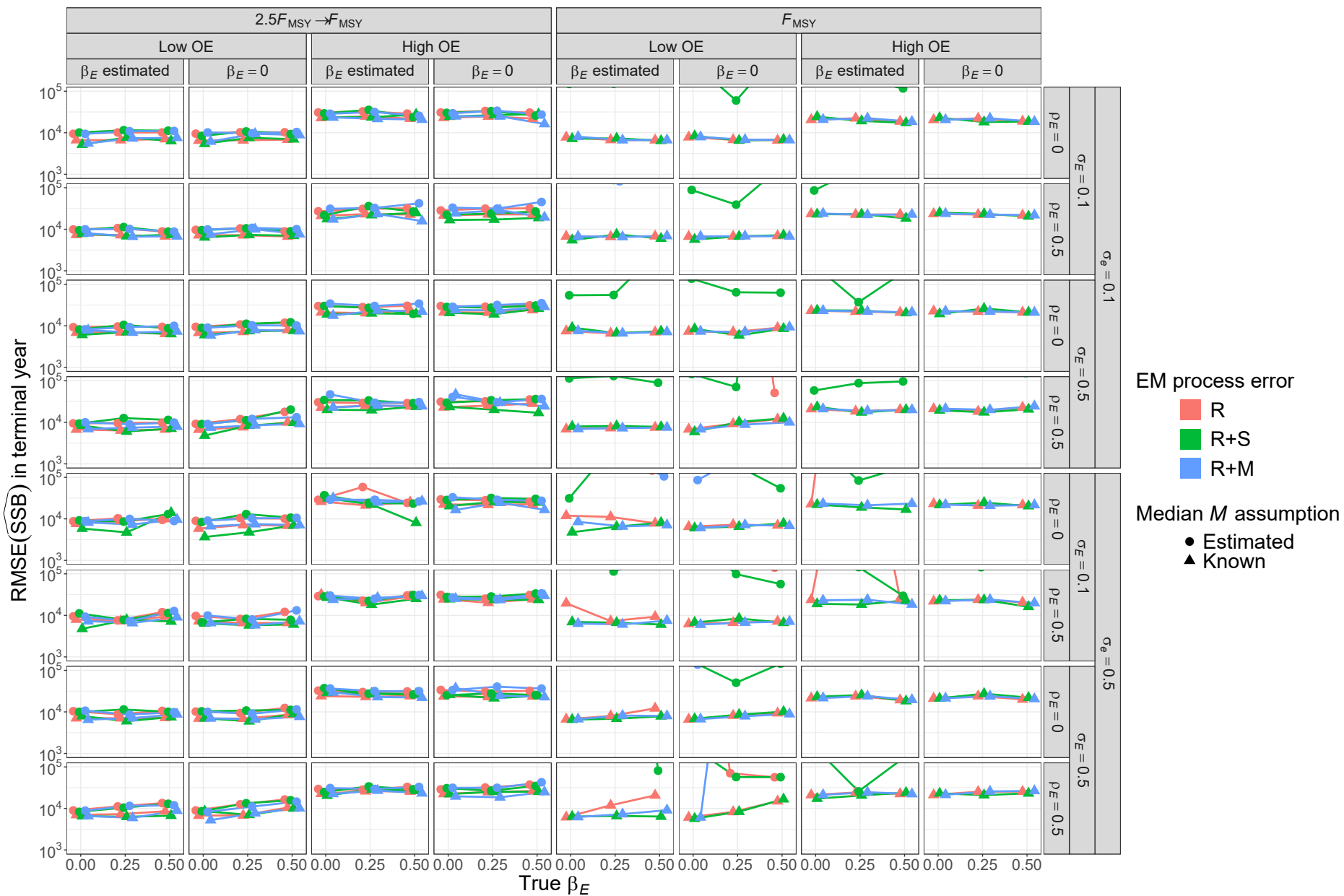


Fig. S50. For R operating models, RMSE of estimates of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

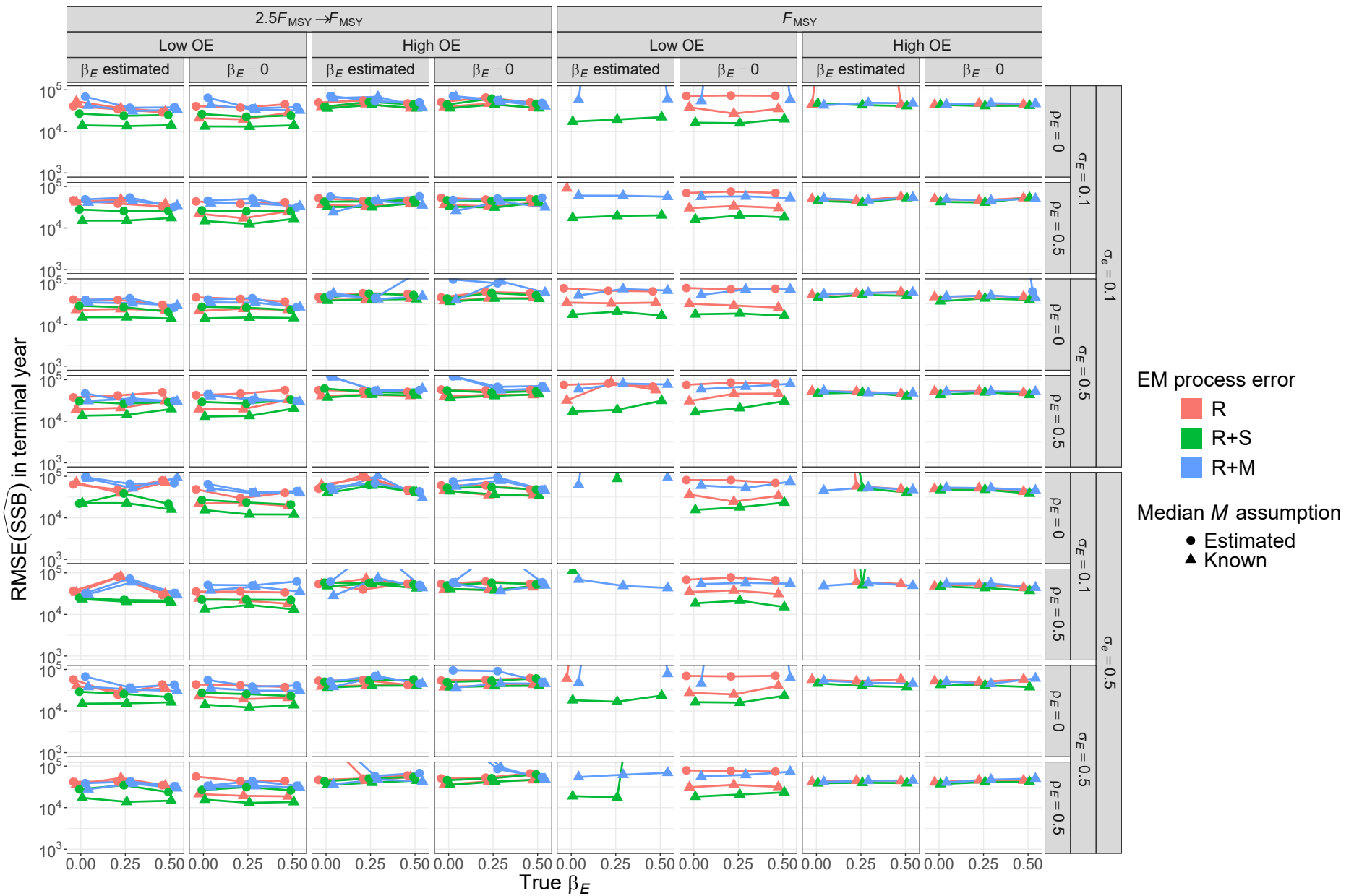


Fig. S51. For R+S operating models, RMSE of estimates of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

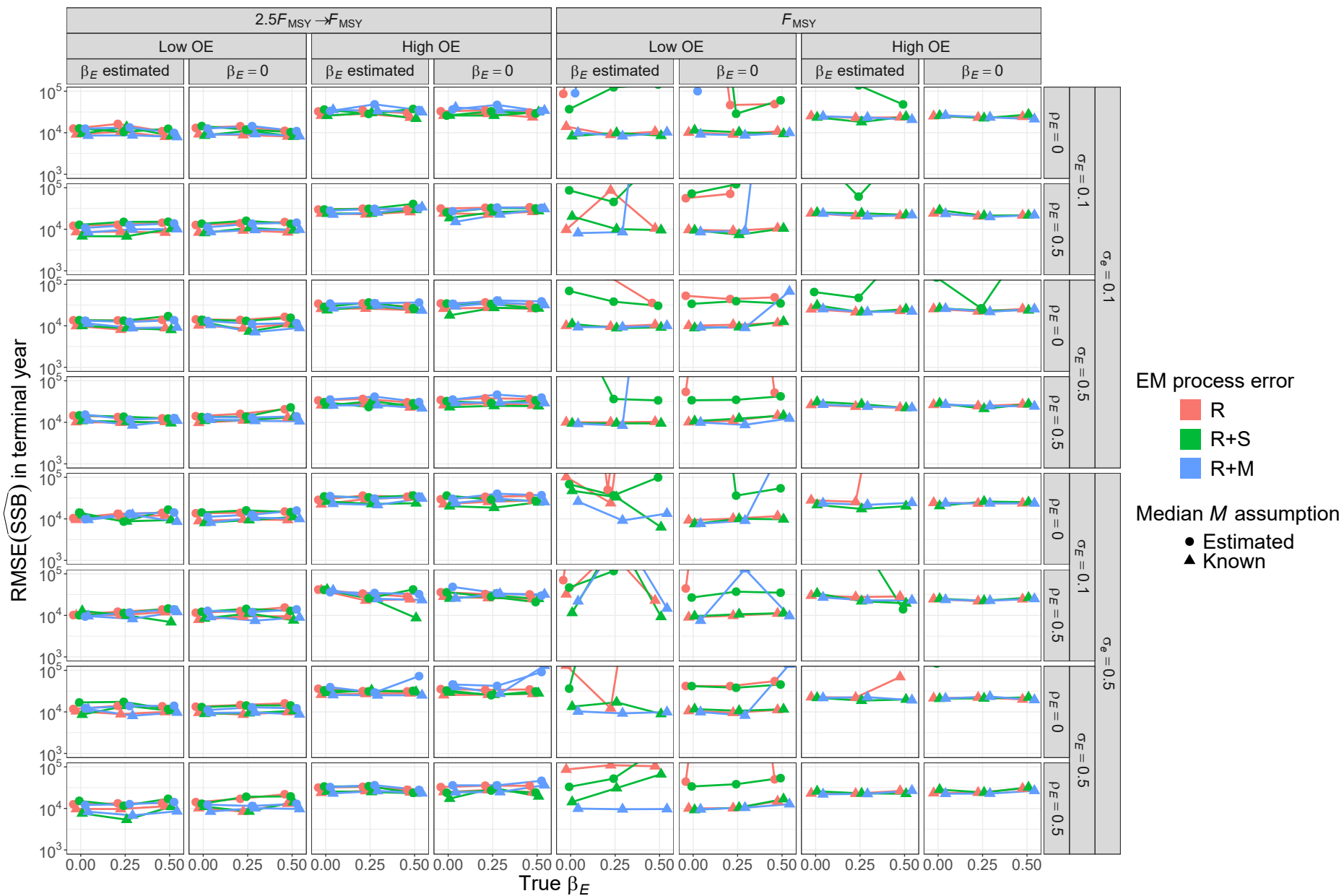


Fig. S52. For R+M operating models, RMSE of estimates of SSB in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

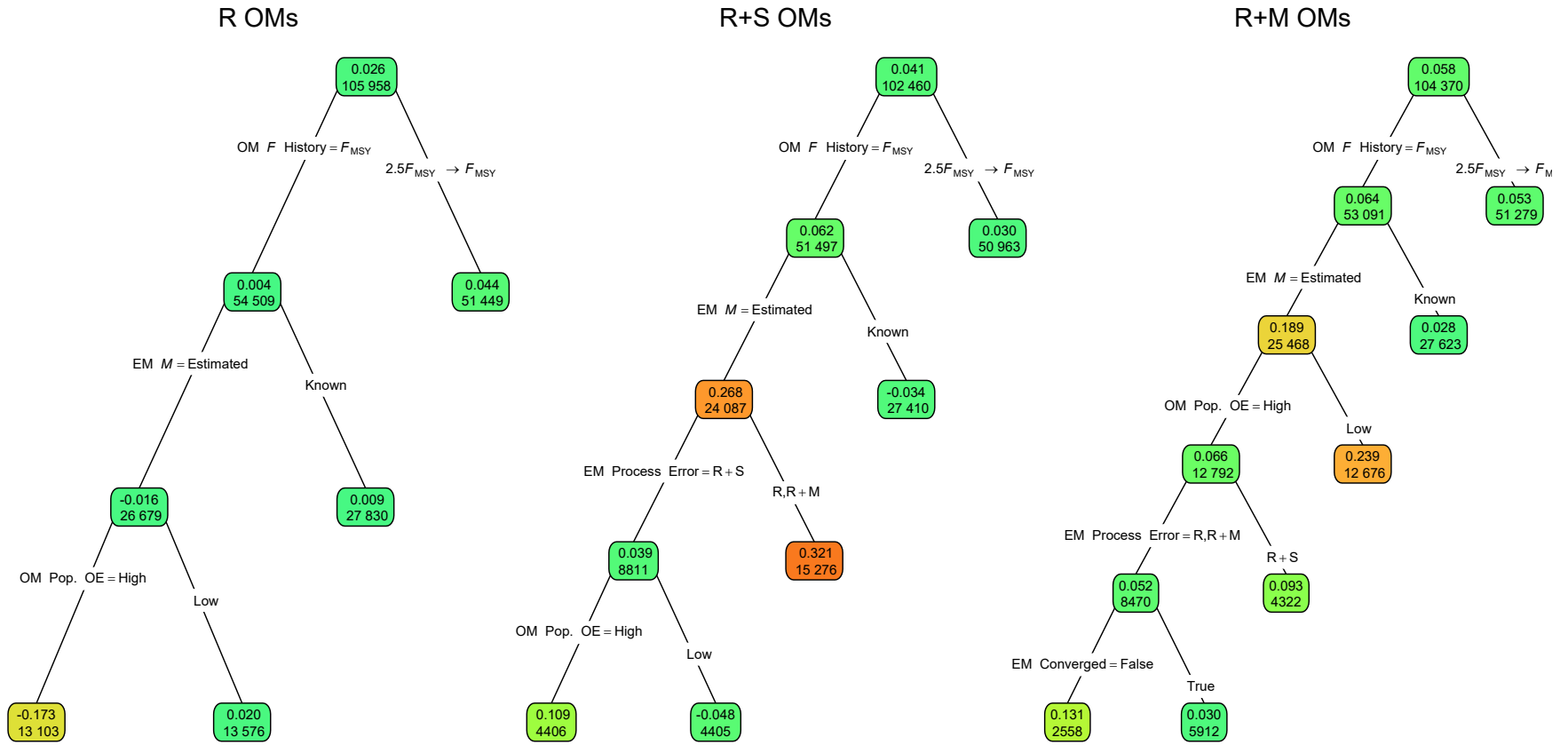


Fig. S53. Regression trees for errors terminal year fishing mortality rate as measured by Eq. 2. Each node shows the median error for the corresponding subset. Lower or higher absolute median errors are indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

Table S17. Percent reduction in deviance for linear regression models fit to errors in terminal year fishing mortality rate as measured by Eq. 2. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
Convergence	<0.01	0.19	0.01
OM F history	1.92	2.42	1.94
OM Obs. Error	0.74	0.12	1.12
OM σ_e	0.01	<0.01	<0.01
OM σ_E	<0.01	<0.01	<0.01
OM ρ_E	<0.01	<0.01	<0.01
OM β_E	<0.01	0.01	<0.01
EM Process Error	0.40	0.92	0.48
EM β_E assumption	0.01	0.03	0.02
EM M assumption	1.58	1.85	1.68
All factors	4.95	5.82	5.35
+ All Two Way	10.91	11.03	12.23
+ All Three Way	14.16	13.24	16.19

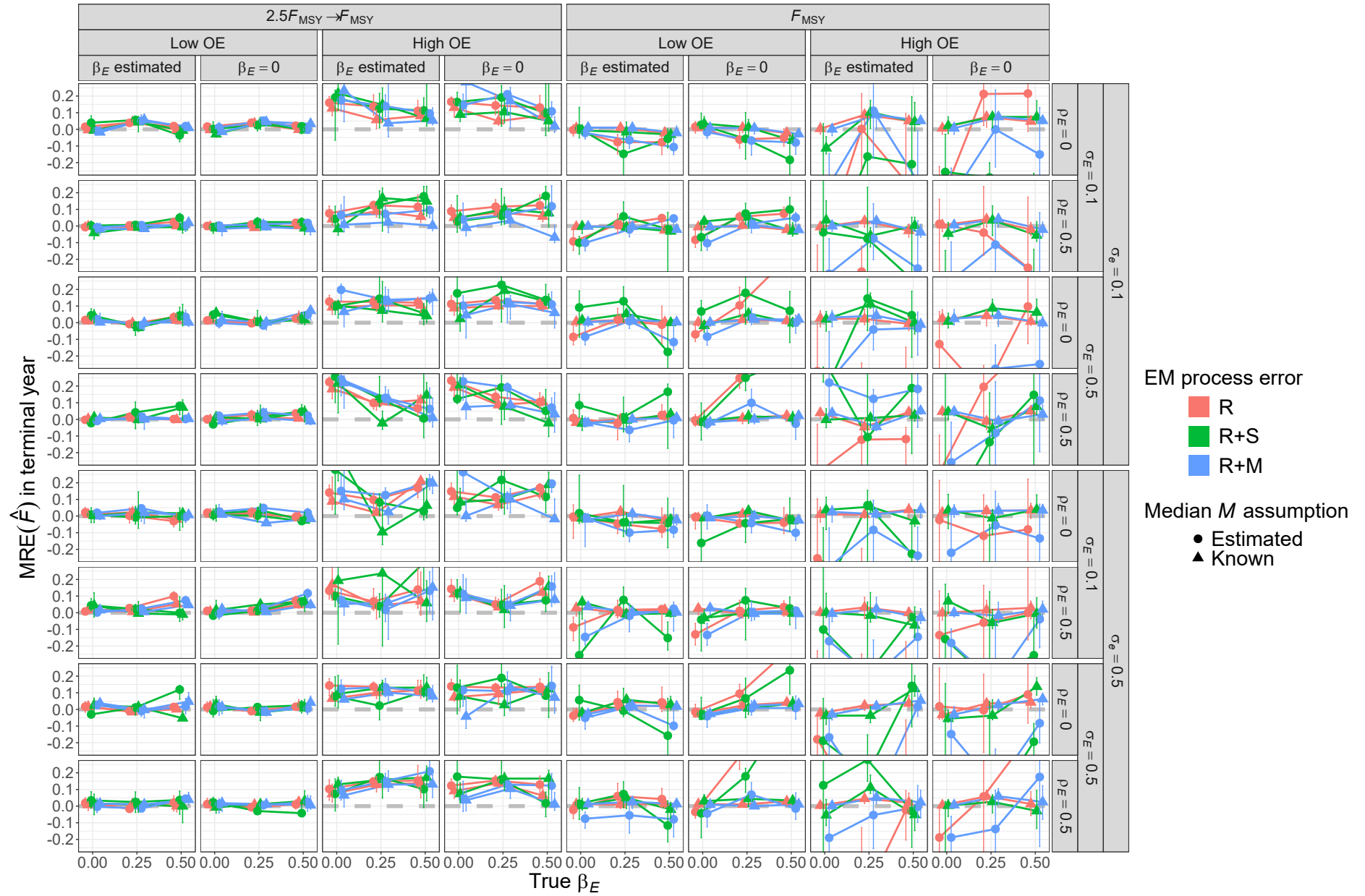


Fig. S54. For R operating models, median relative error (MRE) of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

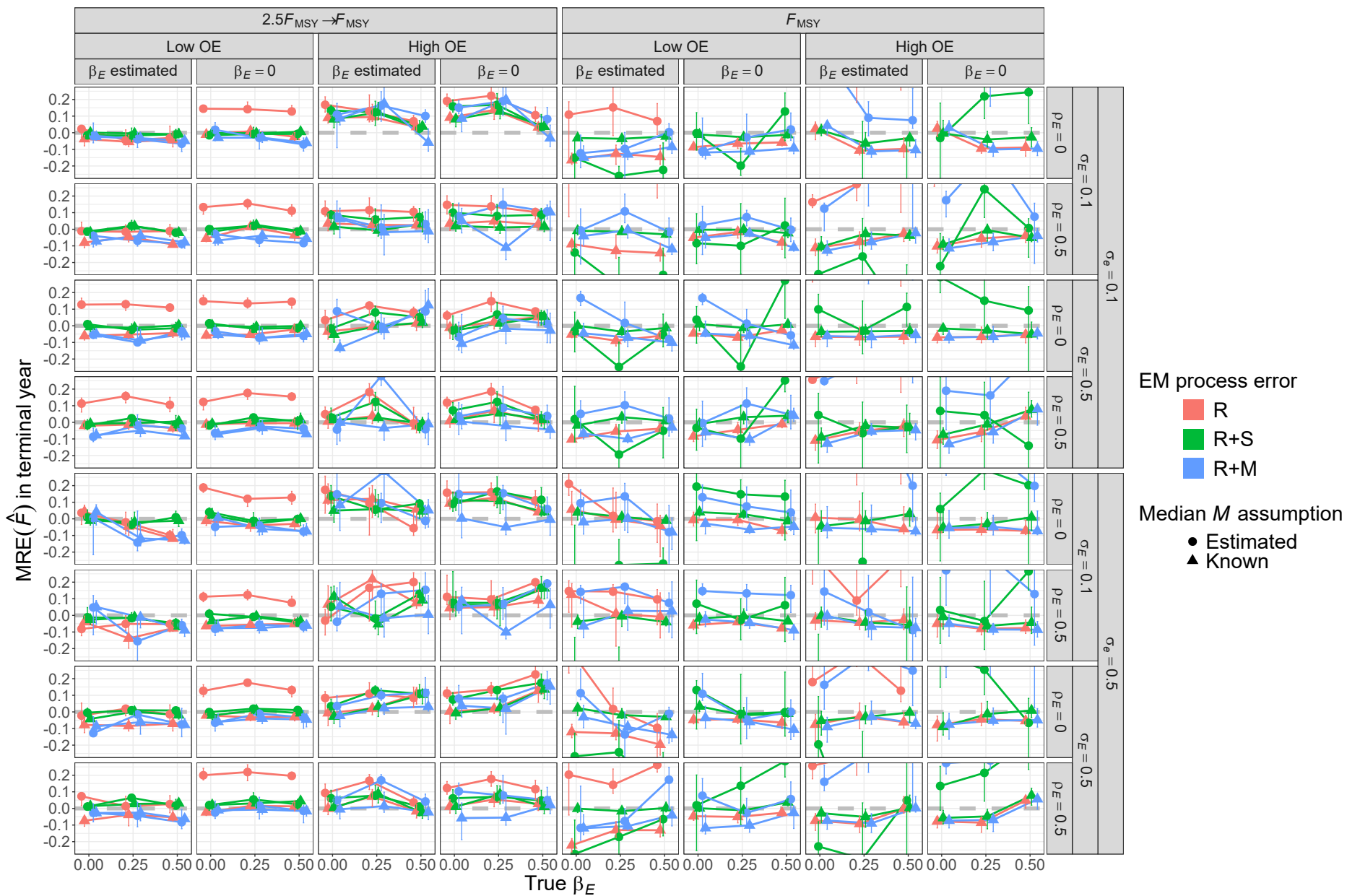


Fig. S55. For R+S operating models, median relative error (MRE) of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

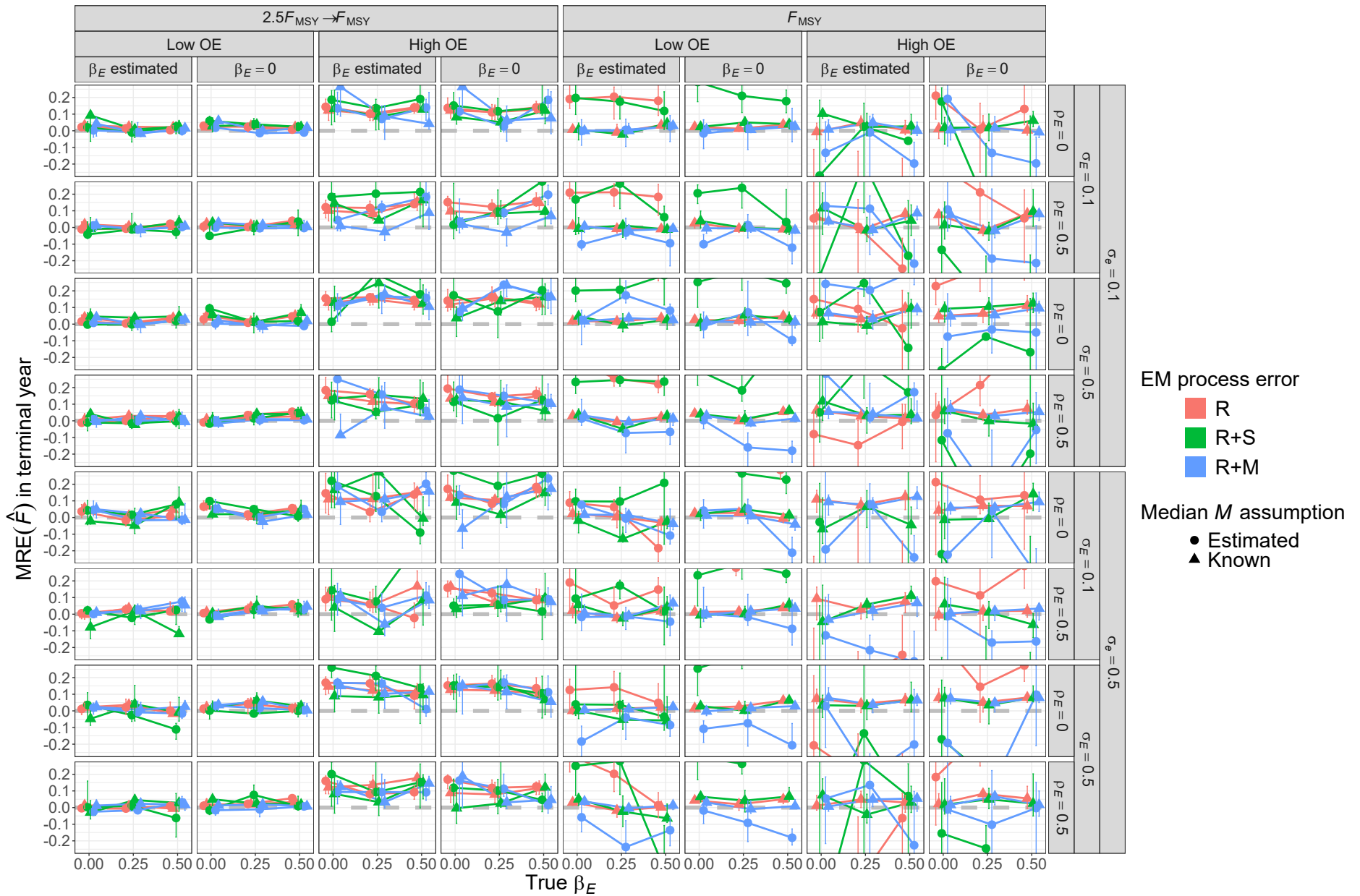


Fig. S56. For R+M operating models, median relative error (MRE) of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

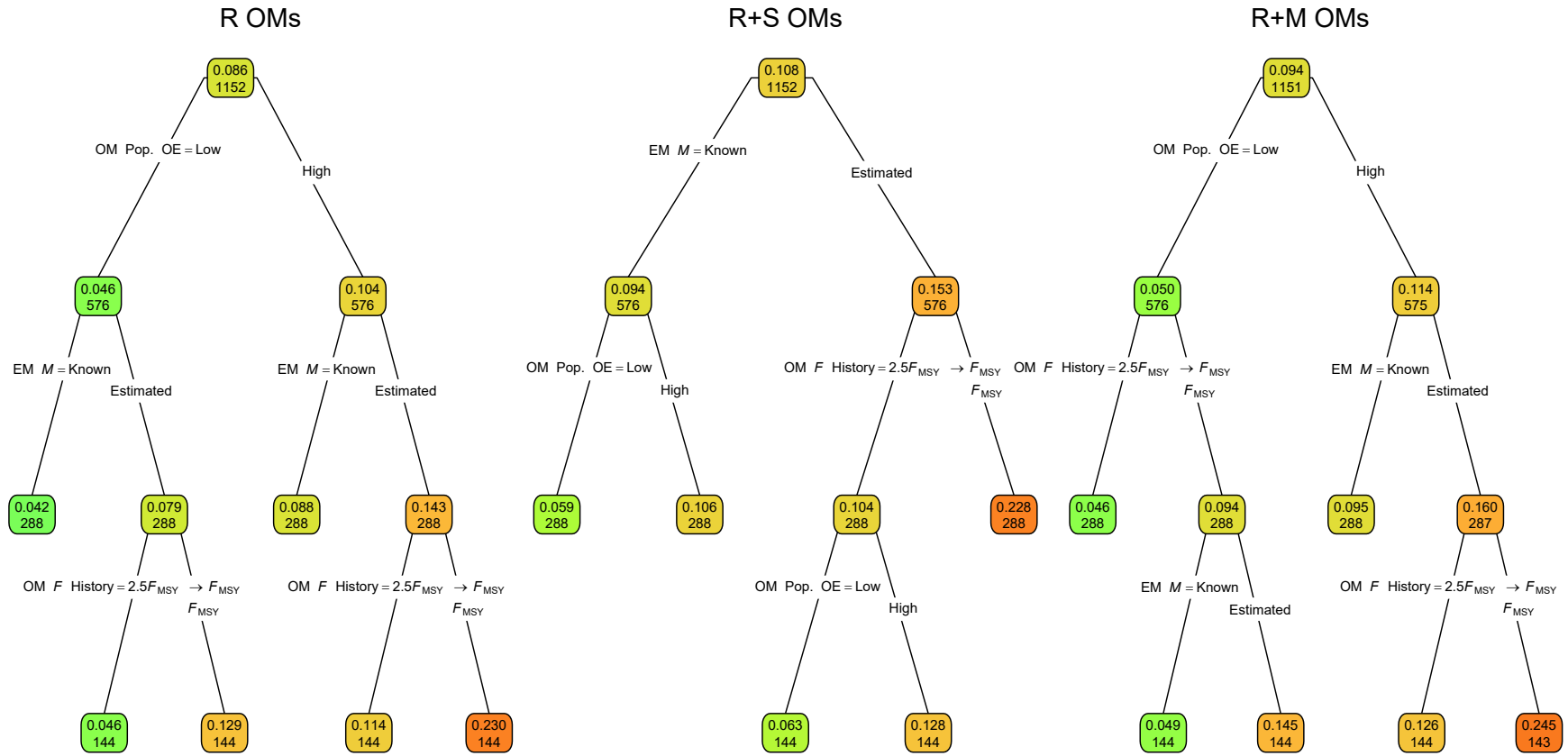


Fig. S57. Regression trees for the log-transformed RMSE of terminal year fishing mortality rate. Each node shows the median error for the corresponding subset. Lower or higher median RMSE is indicated by more green or red polygons, respectively. See Figure 1 caption and Table S1 for further details.

Table S18. Percent reduction in deviance for linear regression models fit to log-transformed RMSE of terminal year fishing mortality rate. See Tables S2 and S1 for further details.

Factor	R	R+S	R+M
OM F history	11.29	25.63	13.46
OM Obs. Error	37.27	21.42	35.47
OM σ_e	<0.01	0.02	0.02
OM σ_E	0.09	0.03	0.14
OM ρ_E	0.02	0.01	0.04
OM β_E	0.15	0.03	0.21
EM Process Error	0.42	0.75	0.63
$EM\beta_E$ assumption	0.04	<0.01	0.04
EM M assumption	25.38	26.10	24.68
All factors	74.65	73.99	74.76
+ All Two Way	92.28	91.29	91.64
+ All Three Way	94.31	95.51	93.73

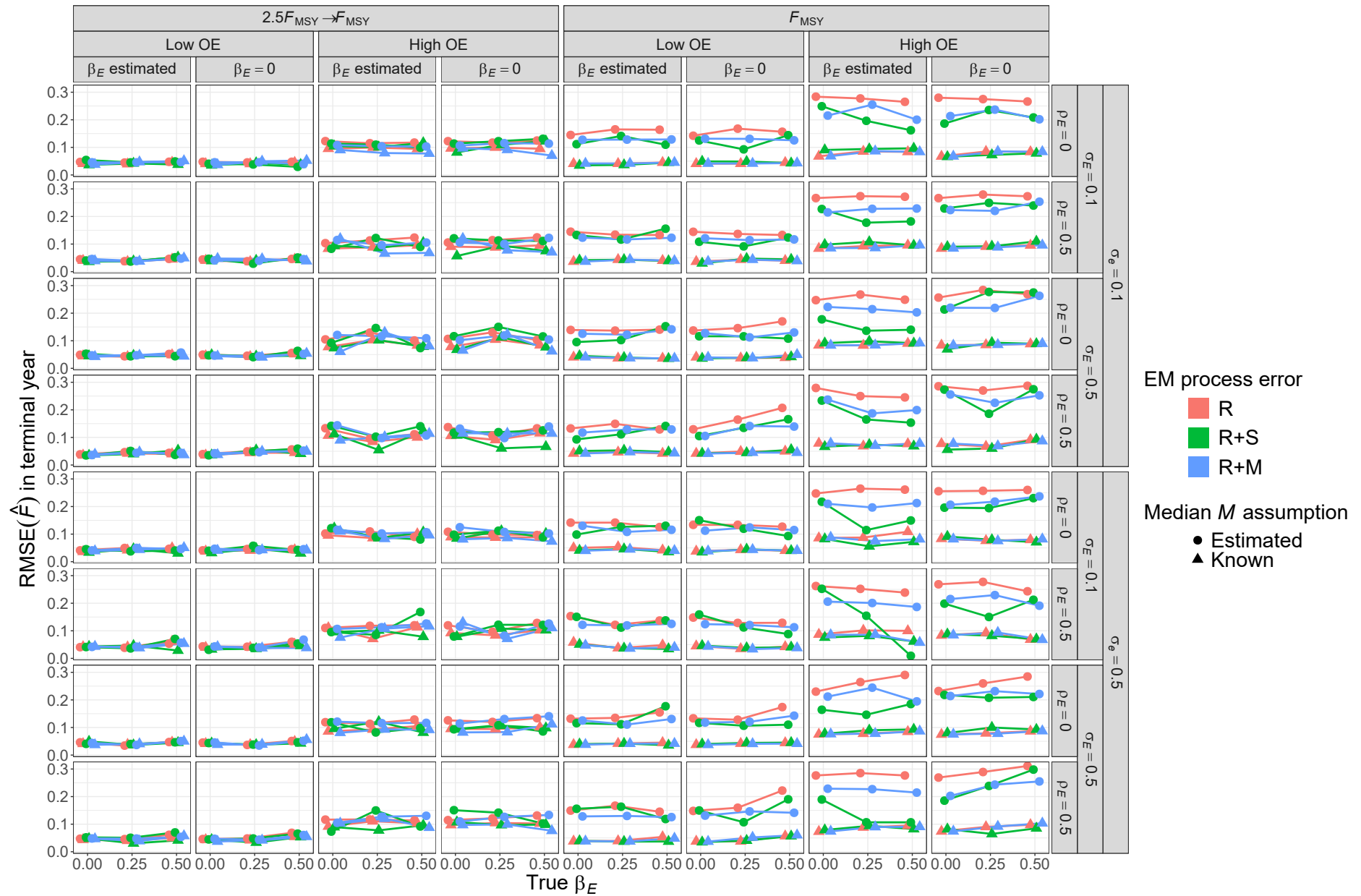


Fig. S58. For R operating models, RMSE of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

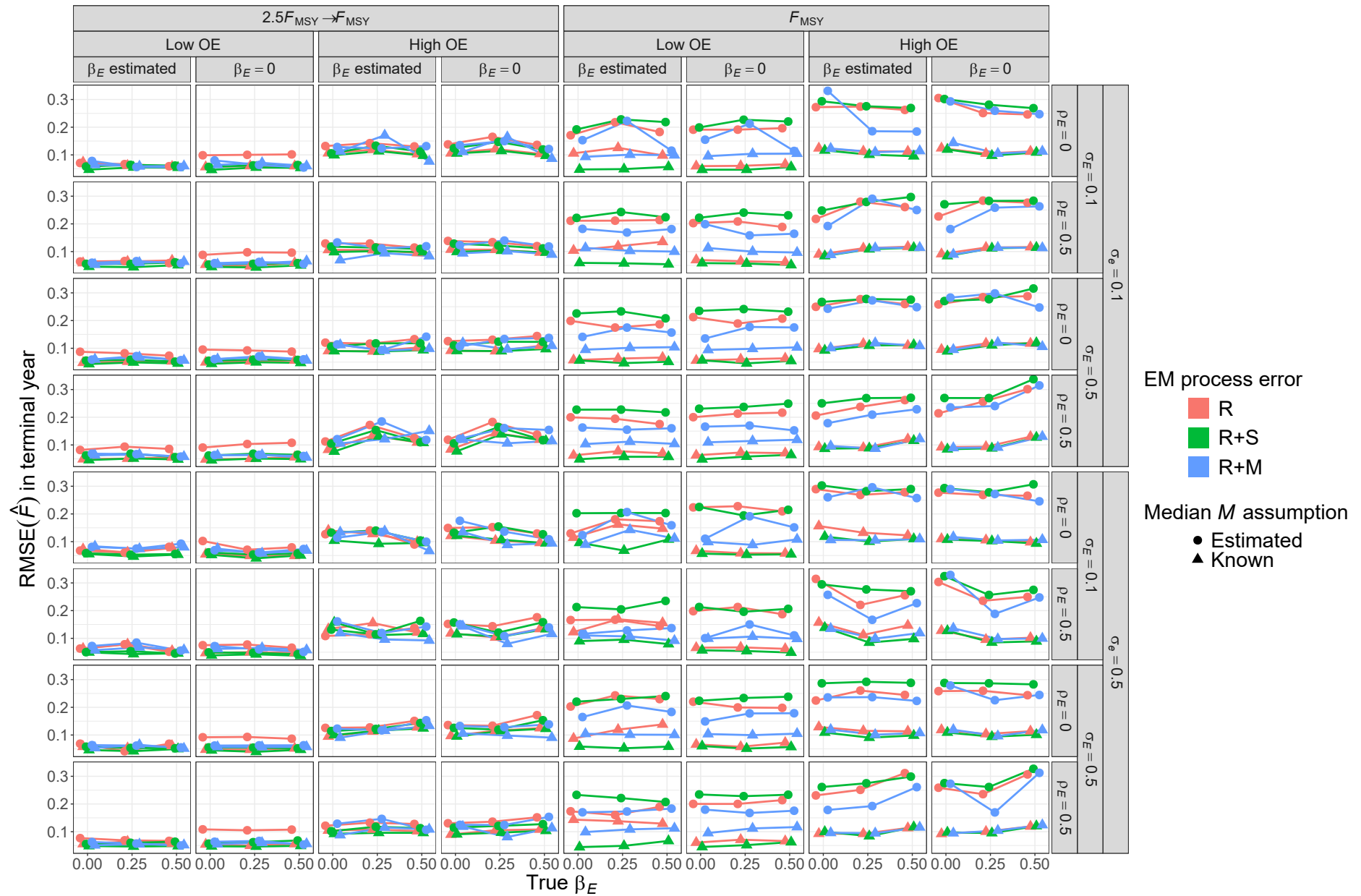


Fig. S59. For R+S operating models, RMSE of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).

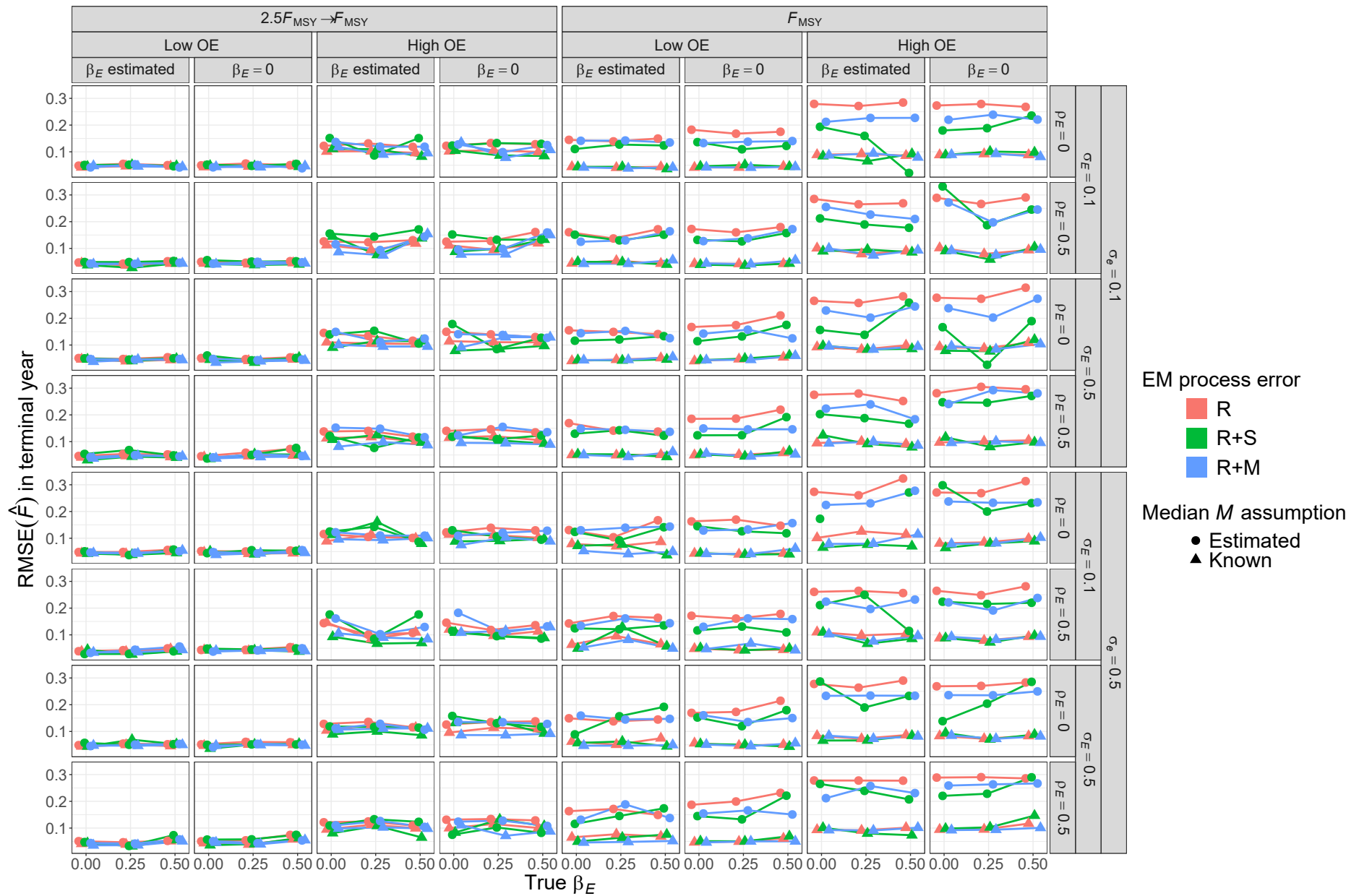


Fig. S60. For R+M operating models, RMSE of estimates of fully-selected fishing mortality (F) in the terminal year for estimating models with alternative process error assumptions, treatment of covariate effect ($\beta_E = 0$ or estimated), and treatment of median M parameter (β_M estimated or known).